DEPARTMENT OF MATHEMATICS

INTRODUCTION

1.0 HISTORY

The Department of Mathematics was first established in the School of Mathematical and Physical Sciences in October, 1964 and produced the first set of graduates in 1967. The Department was then composed of two subject areas, namely. Mathematics and Mathematical Physics, and could be taken as two subjects or as one subject. During that period, students in the School of Biological Sciences and the School of Mathematical and Physical Sciences were being examined in three subjects or, in exceptional cases, in one subject in Part III. The Department became part of the newly reconstituted Faculty of Science in 1973. As from 1974, Senate approved a new degree structure based on course units and in 1978, in addition to the honours degree programme in Mathematics, the Department started a combined honours degree Programme in Mathematics and Statistics and in 1983, the Department started a Four-Year Degree Programme. The Department started a B.Sc. (Hons) Degree Programme in Industrial Mathematics in 2010/2011. Recently, the Department of Mathematics gave birth to Department of Statistics and the Programme was approved in 2020/2021.

1.1 VISION AND MISSION STATEMENT OF THE DEPARTMENT

VISION

Our vision is to provide a first-class learning environment for the study of Mathematics in all its forms and produce mathematicians of all persuasion's-both theoretical and applied mathematicians in the fullest meanings of these terms. Our vision includes the desire to have a department that is always dynamically transformed.

MISSION

Our mission is to train students to acquire mathematical knowledge and skills to the highest attainable level; produce for society, scientists with unmatched competencies in the fields of mathematical and cognate sciences, and who can help to imbue society with discipline and positive values which are ingredients that propel and sustain national development.

Table 1

POST	NAMES & QUALIFICATIONS
Head of Department	Prof. Hudson AKEWE
Postgraduate Coordinator	Dr. Blessing I. AKINNUKAWE
Departmental Examination Officer	Dr. A. A. AYOADE
Chairman, Quality Assurance	Prof. A. A. MOGBADEMU

1.2 PROGRAMME: B.Sc. Mathematics

Overview
Mathematics programme develops students' self-confidence in handling problems with minimal or no supervision. Graduates of the programme will acquire sufficient knowledge to develop confidence in appreciating and solving problems in general.

Philosophy
The philosophy of the mathematics programme is to train students to acquire academic excellence and competence in Mathematical reasoning and problem–solving through the use of logic and computational skills with main purpose of meeting our national needs in the area of technological advancement which is currently a global trend.

Objectives
1. To instill in students a sense of enthusiasm for mathematics, an appreciation of its application in different areas and to involve them in an intellectually stimulating and satisfying experience of learning and studying.
2. To provide students with a broad and balanced foundation in mathematics and practical skills in statistics and computer science.
3. To develop in students the ability to apply their mathematics knowledge and skills to the solution of theoretical and practical problems in mathematics.
4. To develop in students, through an education in mathematics, a range of transferable skills of value in mathematical related and non-mathematical related employment.
5. To provide students with knowledge and skills base from which they can proceed to further studies in specialized areas of mathematics or multi-disciplinary areas involving mathematics.
6. To generate in students an appreciation of the importance of mathematics in an industrial, economic, environmental and social context.

Employability Skills
Mathematics is embodiment of employability skills and the graduates will be equipped with skills that include the following:
1. Learning and innovation skills
2. Life and career skills
3. Information, media and technology skills
4. Quantitative reasoning;
5. Ability to manipulate precise and intricate ideas;
6. Numeracy.

21st Century Skills
1. Creative and critical thinking;
2. Problem solving;
3. Analytical thinking;
4. Logical thinking;
5. Communication;
6. Time management;

7. Teamwork;
8. Independence.

Unique Features of the Programme

The unique features of the programme include:

1. Graduates will certainly possess the needed skills to bring to bear the applications of mathematics to address industrial and societal problems towards improvement of quality of life in both the developed and developing worlds taking advantage of current innovations in technology;
2. They will be well equipped to pursue careers in several other emerging areas that encompasses all the mathematics disciplines as well as many other areas of science, social science, business, etc. Examples include finance and cryptography, artificial intelligence, machine learning, actuarial science, climate change, energy and sustainable development, mathematical modelling, biomathematics; and
3. Traduates will be equipped to demonstrate anywhere they find themselves that mathematical skills propel a better world and enable one to excel in every other field

Admission and Graduation Requirements

Admission Requirements

The entry requirements shall be at least credit level passes in five subjects including English Language, Mathematics, and Physics to form the core subjects with credit in any other two relevant science subjects at the Senior Secondary Certificate (SSC) or its equivalent. In addition, an acceptable pass in the Unified Tertiary Matriculation Examination (UTME), with appropriate subject combination is required for admission into 100 Level.

Candidates with two A level passes (graded A-E) at the GCE/IJMB Advanced Level in relevant subjects (Mathematics, Further Mathematics, Physics and Chemistry) may be admitted into 200-level.

Graduation Requirements

Expected duration for UTME candidates shall be 4 years and students are required to pass a minimum of 120 units, while for Direct entry students, expected duration for graduation shall be 3 years and would be expected to pass a minimum of 90 units which must include all compulsory courses. Students in 5 years' programmes are expected to pass a minimum of 150 units which must include all compulsory courses.

Duration

A student will not be allowed to exceed an additional 50 per cent of the duration of the programme if he/she fails to graduate within the minimum number of years.

UTME

Four (4) academic sessions or eight (8) semesters

Direct Entry

Three academic sessions or six (6) semesters. In general, no student will be allowed to exceed an additional 50% of the normal duration of the programme.

Regime of Subject Knowledge
The programmes and their curricula should give students comprehensive education and training that equip them with knowledge, decision-making and problem-solving skills in a variety of areas.

Competencies and Skills
General skills should include competencies in computer literacy, quantitative skills, communication skills, interpersonal skills, organization skills, Information Technology skills and Entrepreneurship skills.

Administrative and Management related cognitive abilities and skills required are as follows:
1. ability to recognize and analyze management and administrative problems and evolve trategies for their solutions;
2. ability to recognize and implement good management and administrative policies.
3. computational and data processing skills, relating to administrative, financial and manpower data; and
4. ability to demonstrate knowledge and understanding of essential facts, concepts and principles, and apply theories to administration and management. Acquire knowledge in problem solving through industrial attachment, industrial seminars and student apprenticeship scheme.

Behavioural Attributes
Graduates on these programmes should:
1. understand the social-cultural environment in which they find themselves and how such environment conditions behaviour;
2. be able to understand, explain, predict and influence human behaviour in work organizations;
3. relate the knowledge of human behaviour to the ethics of their relevant professions; and
4. understand the relationship between culture and behaviour and why a unimodal system of behaviour may not work.

Course System
Credits are weights attached to a course. One credit is equivalent to one hour per week per semester of 15 weeks of lectures or three hours of laboratory/studio/workshop work per week per semester of 15 weeks.

Grading of Courses
Grading of courses shall be done by a combination of percentage marks and letter grades translated into a graduated system of Grade Point as shown in Table 2.

Table 2: Grade Point System

Marks %	Letter Grade	Grade Point
70 - 100	**A**	**5**
60 - 69	**B**	**4**
50 - 59	**C**	**3**
45 - 49	**D**	**2**
40 - 44	**E**	**1**
0 - 39	**F**	**0**

Grade Point Average and Cumulative Grade Point Average
For the purpose of determining a student's standing at the end of every semester, the Grade Point Average (GPA) system shall be used. The GPA is computed by dividing the total number of Units x Grade Point (TUGP) by the total number of units (TNU) for all the courses taken in the semester as illustrated in Table 3.

The Cumulative Grade Point Average (CGPA) over a period of semesters is calculated in the same manner as the GPA by using the grade points of all the courses taken during the period.

Table 3: Calculation of GPA and CGPA

Course	Units	Grade Point	Units × Grade Point (UGP)
C_1	U_1	GP_1	$U_1 \times GP_1$
C_2	U_2	GP_2	$U_2 \times GP_2$
-	-	-	-
-	-	-	-
C_i	U_i	GP_i	$U_i \times GP_i$
-	-	-	-
-	-	-	-
C_N	U_N	GP_N	$U_N \times GP_N$
TOTAL	TNU		TUGP

$$\textbf{TNU} = \sum_{i=1}^{N} U_i \text{ , } \textbf{TUGP} = \sum_{i=1}^{N} U_i \times GP_i \text{ , } \textbf{CGPA} = \frac{TUGP}{TNU}$$

Degree Classifications
Classes of degree are to be awarded depending on the cumulative GPA obtained. The classes of degrees that may be awarded are First Class Honours, Second Class Honours (Upper Division), Second Class Honours (Lower Division) and Third Class Honours (see Table 4).

Table 4: Degree Classification

CGPA	Class of Degree
4.50 – 5. 00	First Class
3.50 – 4.49	Second Class Upper
2.40 – 3.49	Second Class Lower
1.50 – 2.39	Third Class
1.00 – 1.49	Pass

Probation
Probation is a status granted to a student whose academic performance fall below an acceptable standard. A student whose Cumulative Grade Point Average is below 1.00 at the end of a particular year of study, earns a period of probation for one academic session.

Withdrawal
A candidate whose Cumulative Grade Point Average is below 1.00 at the end of a particular period of probation should be required to withdraw from the University. Where possible, consideration

may be given to a student who withdraws from a programme of study for transfer to any other programme within the same university.

Subject to the conditions for withdrawal and probation, a student may be allowed to repeat the failed course Unit(s) at the next available opportunity, provided that the total number of credit units carried during that semester does not exceed 24, and the Grade Points earned at all attempts shall count towards the CGPA.

Modes of Student Assessment
All courses taken must be evaluated and a final grade given at the end of the semester. To arrive at the final grade, the evaluation must be a continuous process consisting of some or all of the following where applicable:
(i) Continuous Assessment
(ii) Examinations

Continuous Assessment
Continuous assessment shall be done through essays and tests. Scores from continuous assessment shall normally constitute 30-40 per cent of the full marks for courses which are primarily theoretical.

Examinations
In addition to continuous assessment, final examinations should normally be given for every course at the end of each semester. All courses shall be graded out of a maximum of 100 marks comprising:
Final Examination: 60% - 70%
Continuous assessment (Quizzes, Homework, Tests and Practical): 30% - 40%

External Examiner System
The involvement of external examiners from other universities is a crucial quality assurance requirement for all courses in Nigerian University System. In this regard, external examiner should go beyond mere moderation of examination questions to examining of examination papers to scope and depth of examination questions vis a vis the curricular expectation.

SIWES Rating and Assessment
All students taking any degree in the Sciences must undergo industrial training to earn a minimum of 6 credit units. The minimum duration of the Students Industrial Work Experience Scheme (SIWES) should be 24 weeks. Students should be assessed using the Logbook, a report and a Seminar.

1.3 LIST OF ACADEMIC STAFF

S/N	Name	Rank	Phone Number	Email Address	Area of Specialization	Research Focus
1.	Prof. H. Akewe	Professor (HOD)	08036011788	hakewe@unilag.edu.ng	Functional Analysis	Fixed Point Theory and its Applications
2.	Prof. S. A. Okunuga	Professor	08023244422	sokunuga@unilag.edu.ng	Numerical and Computational Mathematics	Numerical and Computational Mathematics
3.	Prof. J. O. Olaleru	Professor	08037265755	jolaleru@unilag.edu.ng	Functional Analysis	Fixed Point Theory and its Applications
4.	Prof. I. O. Abiala	Professor	08056060214	iabiala@unilag.edu.ng	Analytical Dynamics (Solid and Fluid Mechanics)	Analytical Dynamics (Solid and Fluid Mechanics)
5.	Prof. O. J. Fenuga	Professor	08055060122	ofenuga@unilag.edu,ng	Fluid Mechanics	Fluid Mechanics
6.	Prof. A. A. Mogbademu	Professor	08030827141	amogbademu@unilag.edu.ng	Functional Analysis: Real Analysis, Iterative Method.	Functional Analysis: Real Analysis, Iterative Method.
7.	Prof. O. A. Akinfenwa	Professor	07057175195	0akinfenwa@unilag.edu.ng	Numerical and Computational Mathematics	Numerical and Computational Mathematics
8.	Dr. J. O. Ehigie	Assoc. Professor	08028490310	jehigie@unilag.edu.ng	Numerical and Computational Mathematics	Numerical and Computational Mathematics
9.	Dr. H. O. Olaoluwa	Assoc. Professor	08131981751	holaoluwa@unilag.edu.ng	Functional Analysis: Nonlinear Analysis	Functional Analysis: Nonlinear Analysis
10.	Dr. B. I. Akinnukawe	Senior Lecturer	07065338323	iahinnukawe@unilag.edu.ng	Numerical and Computational Mathematics	Numerical and Computational Mathematics
11	Dr. K. S. Eke	Senior Lecturer	08100577318	skanayo@unilag.edu.ng	Functional Analysis	Fixed point theory and applications.
12.	Dr. J. O. Hamzat	Senior Lecturer	08060085718	jhamzat@unilag.edu.ng	Complex Analysis	Complex Analysis
13.	Dr. S. A. Adebisi	Senior Lecturer	07041639013	sadebisi@unilag.edu.ng	Algebra	Algebra
14.	Dr. S. J. Aroloye	Senior Lecturer	07065714177	saroloye@unilag.edu,ng	Applied Mathematics	Applied Mathematics
15.	Dr. A. A. Ayoade	Lecturer 1	07033702969	aayoade@unilag.edu.ng	Mathematical Modelling	Mathematical Modelling
16.	Dr. O. O. George	Lecturer 1	08034295104	oogeorge@unilag.edu.ng	Algebra	Algebra
17.	Dr. A. P. Adetunji	Lecturer 1	08035735350	padetunji@unilag.edu.ng	Algebra	Algebra
18.	Dr. R. S. Olowe	Lecturer 11	08030773064	rolowe@unilag.edu.ng	Numerical and Computational Mathematics	Numerical and Computational Mathematics

1.4 LIST OF NON-TEACHING STAFF

S/N	Name	Date and Grade of First Appointment		Current Status	Date of Confirmation
		Date	Grade		
1.	ADETUNJI, Aina Beatrice	25/09/1992	Typist 1	Principal Pers. Assist 1	01/10/2022
2.	OLORUNTOBA, Theophilus	15/05/2000	Senior Clerical Officer	Assist. Chief Executive Officer	01/10/2022
3.	OMOSANYA, Kolawole	05/09/1995	Temp. Senior Driver II	Chief Transport Officer	01/10/2021
4.	IBIYEMI, Mary Abiona	31/10/2013	Assistant IV	Higher Executive Officer	Nov.12 2025

1.5 B.Sc. MATHEMATICS COURSE STRUCTURE

GLOBAL COURSE STRUCTURE

100 Level
FIRST SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
BIO 101	General Biology I	2	C	30	-
CHM 101	General Chemistry I	2	C	30	-
COS 101	Introduction to Computing Sciences	3	C	30	45
GST 111	Communication in English	2	C	15	45
MTH 101	Elementary Mathematics I	2	C	30	-
PHY 101	General Physics I	2	C	30	-
PHY 103	General Physics III	2	C	30	-
STA 121	Statistical Inference I	3	C	45	-
	Total Compulsory Units	**18**			

SECOND SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
BIO 102	General Biology II	2	C	30	
CHM 102	General Physics II	2	C	30	-
COS 102	Problem Solving	3	C	30	45
ECO 102	Principles of Economics II	2	C	30	-
GST 112	Nigerian People and Culture	2	C	30	-
MTH 102	Elementary Mathematics II	2	C	30	
MTH 103	Elementary Mathematics III	2	C	30	
PHY 102	General Physics II	2	C	30	
STA 112	Introduction to Statistics and Probability	3	C	45	
	Total Compulsory Units	**20**			

200 Level
FIRST SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
ENT 211	Entrepreneurship and Innovation	2	C	30	-
COS 201	Computer Programming I	3	C	30	45
MTH 201	Mathematical Methods I	2	C	30	-
MTH 203	Set Logic and Algebra I	2	C	30	-
MTH 204	Linear Algebra I	2	C	30	-
MTH 207	Real Analysis I	2	C	30	-
MTH 209	Introduction to Numerical Analysis	2	C	30	-
MTH 210	Vector Analysis	1	C	15	-
	Total Compulsory Units	**16**			

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 215	History of Mathematics	2	E	30	-
*MAT 217	Mathematics for Chemist I	2	E	30	-
*MAT 218	Electricity and Magnetism	2	E	30	-
*MAT 234	Mechanics II	3	C	45	-
PHY 205	Thermal Physics	2	E	30	-
	Total Compulsory Units	**3**			
	Total Elective Units	**8**			

SECOND SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
GST 212	Philosophy, Logic and Human Existence	2	C	30	
MTH 202	Elementary Differential Equations	2	C	30	-
MTH 205	Linear Algebra II	1	C	15	-
STA 202	Statistics for Physical Sciences and Engineering	3	C	45	-
	Total Compulsory Units	**8**			

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 224	Mechanics III	3	C	30	-
*MAT 229	Introduction to Scientific Computing	2	C	30	45
*MAT 248	Pure Mathematics II	2	E	30	-
*CSC 227	Introduction to Information Processing	3	E	30	45
*PHS 228	Theoretical Physics I	2	E	30	-
LAG-STA 214	Introduction to Biostatistics	3	E	45	-
LAG-STA 222	Linear Model I	2	E	30	-
	Total Compulsory Units	**5**			
	Total Elective Units	**12**			

NOTE: *MAT represents UNILAG Code (For example, *MAT 224 =LAG-MAT 224)

MTH 204: Linear Algebra I is a prerequisite of MTH 205: Linear Algebra II
***MAT 217 (CHM): Mathematics for Chemist I is offered by Chemistry students only**

300 Level
GLOBAL COURSE STRUCTURE
FIRST SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
MTH 300	Abstract Algebra I	2	C	30	-
MTH 301	Metric Space Topology	2	C	30	-
MTH 303	Vector and Tensor Analysis	2	C	30	-
MTH 304	Complex Analysis I	2	C	30	-
MTH 307	Real Analysis II	2	C	30	-
MTH 310	Mathematical Methods II	2	C	30	-
ENT 311	Enterprise Appreciation	2	C	30	-
	Total Compulsory Units	**14**			

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 325	Analytical Dynamics	3	E	45	-
*MAT 334	Numerical Analysis II	3	C	30	45
*MAT 332	Real Analysis III	2	E	30	-
IMT 301	Financial Mathematics	2	E	30	-
Total Compulsory Units		**3**			
Total Elective Units		**7**			

SECOND SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
MTH 399	Industrial Attachment II (24 Weeks)	6	C	-	270
	Total Compulsory Units	**6**			
	Total Elective Units	**0**			

400 Level
FIRST SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
MTH 401	Theory of Ordinary Differential Equations	2	C	30	45
MTH 403	Functional Analysis	2	C	30	-
MTH 405	General Topology	3	C	15	45
MTH 407	Mathematical Methods	2	C	30	-
MTH 305	Complex Analysis II	2	C	30	-
	Total Compulsory Units	**11**			

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 415	Fluid Mechanics I	3	E	45	-
*MAT 438	Numerical Analysis III	3	E	30	45
*MAT 439	Number Theory	3	E	45	-
*STA 423	Stochastic Processes	3	E	30	-
Total Compulsory Units		**0**			
Total Elective Units		**12**			

SECOND SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
MTH 402	Theory of Partial Differential Equations	2	C	30	-
MTH 404	Project	6	C	-	270
MTH 406	Lebesgue Measure and Integrals	2	C	30	-
MTH 408	Entrepreneurship in Mathematics	2	C	30	-
MTH 302	Ordinary Differential Equation	2	C	30	-
MTH 306	Abstract Algebra II	2	C	30	-
MTH 308	Introduction to Mathematical Modelling	2	C	30	-
GST 312	Peace and Conflict Resolution	2	C	30	-
	Total Compulsory Units	**20**			

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 426	Numerical Analysis IV	3	E	30	45
*MAT 444	Algebraic Topology	2	E	30	-
*MAT 445	Fluid Mechanics II	3	E	45	-
*MAT 446	Elasticity and Plasma Dynamics	2	E	30	-
*MAT 448	Electromagnetic Theory	3	E	45	-
*MAT 464	Differential Geometry	3	E	45	-
*MAT 316	Mechanics IV	3	E	45	-
*MAT 326	Analytical Geometry	3	E	45	-
*MAT 328	Discrete Mathematics	2	E	30	-
*IMT 429	Introduction to Stochastic Calculus for Finance	3	E	45	-
IMT 404	Optimization Theory	2	E	30	-
Total Compulsory Units		**0**			
Total Elective Units		**29**			

NOTE: * denotes UNILAG Courses

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SEMESTER

FIRST SEMESTER

Level	Compulsory	Elective
100	18	0
200	19	8
300	17	7
400	11	12
Grand Total (First Semester)	**65**	**27**

SECOND SEMESTER

Level	Compulsory	Elective
100	20	0
200	13	12
300	6	0
400	20	29
Grand Total (First Semester)	**59**	**41**

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SESSION

Level	Compulsory	Elective
100	**38**	**0**
200	**32**	**20**
300	**23**	**7**
400	**31**	**41**
Grand Total	**124**	**68**

1.5 B. Sc. Industrial Mathematics

Regime of Subject Knowledge
The programmes and their curricula should give students comprehensive education and training that equip them with knowledge, decision-making and problem-solving skills in a variety of areas.

Competencies and Skills
The general skills should include competencies in computer literacy, quantitative skills, communication skills, interpersonal skills, organization skills, Information Technology skills and Entrepreneurship skills.

Administrative and Management related cognitive abilities and skills required are as follows:
1. ability to recognize and analyze management and administrative problems and evolve strategies for their solutions;
2. ability to recognize and implement good management and administrative policies.
3. computational and data processing skills, relating to administrative, financial and manpower data; and
4. ability to demonstrate knowledge and understanding of essential facts, concepts and principles, and apply theories to administration and management. Acquire knowledge in problem solving through industrial attachment, industrial seminars and student apprenticeship scheme.

Behavioural Attributes
Graduates on these programmes should:
1. understand the social-cultural environment in which they find themselves and how such environment conditions behaviour;
2. be able to understand, explain, predict and influence human behaviour in work organizations;
3. relate the knowledge of human behaviour to the ethics of their relevant professions; and
4. understand the relationship between culture and behaviour and why a unimodal system of behaviour may not work.

Admission Requirements and Expected Duration of the Programmes
The entry requirements shall be at least passes at Credit level at Senior School Certificate (SSC) or its equivalents in five subjects at not more than two sittings. Such subjects must include English Language, Mathematics and three other relevant subjects. In addition, an acceptable pass in the Unified Tertiary Matriculation Examination in (UTME) is required for admission into 100 Level. Candidates with at least two A level passes in relevant subjects at the GCE Advanced Level/ IJMB/ JUPEB or equivalent, may be considered for admission into 200 Level, provided they satisfy the 'O' level qualifications above.

Duration
A student will not be allowed to exceed an additional 50 per cent of the duration of the programme if he/she fails to graduate within the minimum number of years.

UTME
Four (4) academic sessions or eight (8) semesters

Direct Entry
Three academic sessions or six (6) semesters. In general, no student will be allowed to exceed an additional 50% of the normal duration of the programme.

Graduation Requirements
Expected duration for UTME candidates shall be 4 years and students are required to pass a minimum of 120 units, while for Direct entry students, expected duration for graduation shall be 3 years and would be expected to pass a minimum of 90 units which must include all compulsory courses. Students in 5 years' programmes are expected to pass a minimum of 150 units which must include all compulsory courses.

Course System
Credits are weights attached to a course. One credit is equivalent to one hour per week per semester of 15 weeks of lectures or three hours of laboratory/studio/workshop work per week per semester of 15 weeks.

Definition of Course System
This should be understood to mean a quantitative system of organization of the curriculum in which subject areas are broken down into unit courses which are examinable and for which students earn credit(s) if passed. The courses are arranged in progressive order of complexity or in levels of academic progress. Level 1 courses are for example 100 and 101; Level II courses are for example 200 and 202. The second aspect of the system is that courses are assigned weights allied to Units.

Units
Consist of specified number of student-teacher contact hours per week per semester. Units are used in two complementary ways: one, as a measure of course weighting, and the other, as an indicator of student workload. As a measure of course weighting for each Unit course (e.g. HIS 105, ZOO 203, ARCH 504), the credit unit to be earned for satisfactorily completing the course is specified; e.g. a 2-credit unit course may mean two 1-hour lecture per week per semester or one 1-hour lecture plus 3-hour practical per week per semester.

As a measure of workload, "One Credit Unit" means one hour of lecture or one hour of tutorial per week per semester. For other forms of teaching requiring student teacher contact, the following equivalents may apply: two hours of seminar, three hours of laboratory or field work, Clinical practice/practicum, studio practice or stadium sporting activity, six hours of teaching practice; four weeks of industrial attachment where applicable.

Normally, in Course Credit System, courses are mounted all year round, thus enabling students to participate in examinations in which they are unsuccessful or unable to participate on account of ill health or for other genuine reasons. In such a system, no special provisions are made for re-sit examinations.

The minimum number of credit units for the award of a degree is 120 units, subject to the usual Department and Faculty requirements. A student shall therefore qualify for the award of a degree when he has met the conditions.

The minimum credit load per semester is 15 credit units. For the purpose of calculating a student's cumulative GPA(CGPA) in order to determine the class of Degree to be awarded, grades obtained in all the courses whether compulsory or optional and whether passed or failed must be included in the computation.

Even when a student repeats the same course once or more before passing it or substitutes another course for a failed optional course, grades scored at each and all attempts shall be included in the computation of the GPA. Pre - requisite courses must be taken and passed before a particular course at a higher level.

Grading of Courses

Grading of courses shall be done by a combination of percentage marks and letter grades translated into a graduated system of Grade Point as shown in Table 1.2.

Table 1.2: Grade Point System

Marks %	Letter Grade	Grade Point
70 - 100	**A**	**5**
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50 - 59	**C**	**3**
45 - 49	**D**	**2**
40 - 44	**E**	**1**
0 - 39	**F**	**0**

Grade Point Average and Cumulative Grade Point Average

For the purpose of determining a student's standing at the end of every semester, the Grade Point Average (GPA) system shall be used. The GPA is computed by dividing the total number of Units x Grade Point (TUGP) by the total number of units (TNU) for all the courses taken in the semester as illustrated in Table 1.3.

The Cumulative Grade Point Average (CGPA) over a period of semesters is calculated in the same manner as the GPA by using the grade points of all the courses taken during the period.

Table 1.3: Calculation of GPA and CGPA

Course	Units	Grade Point	Units × Grade Point (UGP)
C_1	U_1	GP_1	$U_1 \times GP_1$
C_2	U_2	GP_2	$U_2 \times GP_2$
-	-	-	-
-	-	-	-
C_i	U_i	GP_i	$U_i \times GP_i$
-	-	-	-
-	-	-	-
C_N	U_N	GP_N	$U_N \times GP_N$
TOTAL	**TNU**		**TUGP**

TNU = $\sum_{i=1}^{N} U_i$, **TUGP** = $\sum_{i=1}^{N} U_i \times GP_i$, **CGPA** = $\frac{TUGP}{TNU}$

Degree Classifications

Classes of degree are to be awarded depending on the cumulative GPA obtained. The classes of degrees that may be awarded are First Class Honours, Second Class Honours (Upper Division), Second Class Honours (Lower Division) and Third Class Honours (see Table 1.4).

Table 1.4: Degree Classification

CGPA	Class of Degree
4.50 – 5. 00	First Class
3.50 – 4.49	Second Class Upper
2.40 – 3.49	Second Class Lower
1.50 – 2.39	Third Class
1.00 – 1.49	Pass

Probation

Probation is a status granted to a student whose academic performance fall below an acceptable standard. A student whose Cumulative Grade Point Average is below 1.00 at the end of a particular year of study earns a period of probation for one academic session.

Withdrawal

A candidate whose Cumulative Grade Point Average is below 1.00 at the end of a particular period of probation should be required to withdraw from the University. Where possible, consideration may be given to a student who withdrawn from a programme of study for transfer to any other programme within the same university.

Subject to the conditions for withdrawal and probation, a student may be allowed to repeat the failed course Unit(s) at the next available opportunity, provided that the total number of credit units carried during that semester does not exceed 24, and the Grade Points earned at all attempts shall count towards the CGPA.

Modes of Student Assessment

All courses taken must be evaluated and a final grade given at the end of the semester. To arrive at the final grade, the evaluation must be a continuous process consisting of some or all of the following where applicable:

(i) Continuous Assessment

(ii) Examinations

Continuous Assessment

Continuous assessment shall be done through essays and tests. Scores from continuous assessment shall normally constitute 30-40 per cent of the full marks for courses which are primarily theoretical.

Examinations

In addition to continuous assessment, final examinations should normally be given for every course at the end of each semester. All courses shall be graded out of a maximum of 100 marks comprising:

Final Examination: 60% - 70%
Continuous assessment (Quizzes, Homework, Tests and Practical): 30% - 40%

External Examiner System

The involvement of external examiners from other universities is a crucial quality assurance requirement for all courses in Nigerian University System. In this regard, external examiner should go beyond mere moderation of examination questions to examining examination papers to scope and depth of examination questions vis a vis the curricular expectation.

Students' Evaluation of Courses

There should be an established mechanism to enable students to evaluate courses delivered to them at the end of each semester. This should be an integral component of the course credit system to serve as an opportunity for feedback on the effectiveness of course delivery. Such an evaluation which should be undertaken by students at the end of each course should capture, among others:

1. improvement in the effectiveness of course delivery;
2. continual update of lecture materials to incorporate emerging new concepts;
3. effective usage of teaching aids and tools to maximize impact of knowledge on students;
4. improvement in students' performance through effective delivery of tutorials, timely in; and
5. presentation of continuous assessment and high-quality examination.

It is very important that students' evaluation of courses be administered fairly and transparently using well-designed questionnaires. The completed questionnaires should be professionally analyzed and results discussed with the course lecturer(s) towards improvement in course delivery in all its ramifications.

SIWES Rating and Assessment

All students taking any degree in the Sciences must undergo industrial training to earn a minimum of 6 credit units. The minimum duration of the Students Industrial Work Experience Scheme (SIWES) should be 24 weeks. Students should be assessed using the Logbook, a report and a Seminar.

Overview

Industrial Mathematics is an area with potential beneficial innovations if only it is designed and applied by qualified and appropriate professionals in the industry. It is the application of mathematics and computer programming skills to solve solutions to certain problems in the industry. Its body of knowledge can be applied to solve some complex, industrial, finance and economic problems.

Philosophy

The philosophy of the programme is to prepare graduates for career as mathematicians in business and industry. It should produce graduates of high technical abilities in the applications of mathematical solutions to real-world problems.

Objectives

The objectives of the programme are to:

1. prepare the students to meet the demand of mathematical professionals in both existing and emerging industrial sectors;

2. develop in students a sense of enthusiasm for mathematics; and to involve them in an intellectually stimulating and satisfying experience of learning and studying;
3. provide students a broad and balanced foundation in general undergraduate mathematics knowledge and practical skills in its applications in industry;
4. develop in students the ability to apply their mathematics knowledge and skills to the solution of theoretical and practical problems in mathematics;
5. develop in students a range of transferable skills of value in mathematical and non-mathematical related employment;
6. provide students with knowledge and skills base from which they can proceed to further studies in specialized areas of applied mathematics or multi-disciplinary areas involving mathematics; and
7. generate in students an appreciation of the importance of mathematics in an industrial, economic, environmental and social context.

Unique Features of The Programme

The B. Sc. Industrial Mathematics programme is designed to train the needed skilled manpower to bring to bear the applications of Mathematics to address industrial problems: financial, stock, etc. towards development of the country, taking advantage of current innovations in technology, which pure and applied Mathematics have not really addressed directly.

Employability Skills

Graduates from the programme will be able to have the ability to apply knowledge and skills to solving theoretical and practical problems in Mathematics and other related areas in relation to national and societal needs. These skills include the following:

1. creative and critical thinking;
2. problem solving;
3. analytical thinking;
4. initiative;
5. quantitative reasoning;
6. technology;
7. ability to manipulate precise and intricate ideas;
8. logical thinking;
9. communication;
10. time management;
11. teamwork;
12. independence; and
13. numeracy.

21st Century Skills

1. Problem solving skills
2. Teamwork
3. Computer and digital literacy
4. Critical thinking; and
5. Communication skills.

Admission and Graduation Requirements

Admission Requirements
There are two different pathways by which candidates can be admitted into the programmes in the discipline: the indirect entry mode and the Direct Entry.

Indirect entry mode
Admission through the indirect mode shall take the student to 100 Level. To be eligible for admission, the candidate must have, a minimum of credit pass in five subjects at not more than two sittings in Senior Secondary Certificate or its equivalent. The credit passes are required in the following subjects: English language, Mathematics, Chemistry, Physics and Biology/Agric. Science or Economics. In addition, the candidate is expected to pass both the Unified Tertiary Matriculation Examination (UTME) and the University screening test. The UTME subjects are: English Language, Physics, Mathematics and Chemistry or Economics.

Direct Entry
Candidates with two 'A' level passes (graded A-E) at the GCE/JUPEB Advanced Level in two or more relevant subjects (Mathematics, Further Mathematics, Physics and Chemistry/Economics) or Good Diploma at minimum of Upper Credit in Mathematics, Physics and Engineering, may be admitted into 200-level.

Graduation Requirements
To be eligible for the award of B.Sc. in Industrial Mathematics, a student must have:
1. passed all the core courses, University/Faculty/School required courses and electives;
2. accumulated a **minimum of 120 course units** for students admitted through UTME and **90 course units** for students admitted through direct entry; and
3. pass all required courses.

PROPOSED THIRTY PERCENT (30%) CCMAS FOR UNDERGRADUATE PROGRAMME

1. OVERVIEW

Industrial Mathematics is an inherently interdisciplinary field. This programme combines topics from Mathematics, Computer Science, Economics, Actuarial Science and engineering disciplines to provide you with better understanding of contemporary and advanced mathematical and computational techniques, and their practical application across a wide spectrum of industrial disciplines such as business, computer science and engineering, and trains mathematics students how to apply mathematical analysis to problems arising mainly from Industry. Modern manufacturing and service industries have changed drastically in recent times due to the explosion in the knowledge economy. The creation of mathematical modelling and the development of numerical methods and/or algorithms for computers to obtain solutions for problems in industry has come to be called industrial mathematical sciences or mathematics in the industry.These

industries include:banks, manufacturing companies, academic institutions, health care institutions, research firms and government agencies. There is a current and growing demand for mathematically trained individuals who are able to move into business and industry. Numerous reports and studies carried out by professional organizations show that there is an increasing need in the work force for mathematics graduates with the practical skills to work with managers, engineers

2. PHILOSOPHY

The Department offer B.Sc. Industrial Mathematics with the following options:

i. B.Sc. (Hons) Industrial Mathematics (Applied And Computational Mathematics)

ii. B.Sc. (Hons) Industrial Mathematics (Mathematics And Computer Science)

iii. B.Sc. (Hons) Industrial Mathematics (Mathematics And Economics

iv. B.Sc. (Hons) Industrial Mathematics (Mathematics And Actuarial Science)

OPTION 1: B. Sc. (Hons) INDUSTRIAL MATHEMATICS (APPLIED AND COMPUTATIONAL MATHEMATICS OPTION)

Philosophy

Industrial Mathematics (Applied and Computational Mathematics option) aims to impact the fundamental principles of applied Mathematics on the students and the students at the same time acquire the skills needed for modern computing. A student is expected to specialize in an area of applied mathematics so as to be able to tackle real life problems in his or her area of specialization by using the tools of Mathematics. This programme is designed to meet the demand of the analytical and computational skills needed in Engineering.

OPTION 2: B.Sc. (Hons) INDUSTRIAL MATHEMATICS (MATHEMATICS WITH COMPUTER SCIENCE OPTION)

Philosophy

The aim of this programme is to train students in acquiring strong mathematical skills and at the same time acquire adequate skills in modern computing. A student may specialize in numerical or mathematical computation and on the other hand have interest in statistical analysis of data. Approximately two-third of this programme will consist of courses in Mathematics and Statistics. The remaining one – third of the courses will be from Computer Science.

OPTION 3: B.Sc. (Hons) INDUSTRIAL MATHEMATICS (MATHEMATICS WITH ECONOMICS OPTION)

Philosophy

This programme offers a strong mathematical background for students who have interest in Mathematics and may wish to become an Economist. The programme gives the connection between the mathematical theories and those in Economics. It has strong mathematical orientation

to encourage students to prepare for professional examination in Economics Science and to meet the requirements of the industries and the profession.

OPTION 4: B.Sc. (Hons) INDUSTRIAL MATHEMATICS (MATHEMATICS WITH ACTUARIAL SCIENCE OPTION)

Philosophy

This programme is intended to offer a strong mathematical background for students who have interest in Mathematics and may wish to become actuaries. The programme is an interphase between mathematical theories and managerial science. It has strong mathematical orientation to encourage students preparing for professional examination in Actuarial Science and to meet the requirements of the industries and the profession.

3. OBJECTIVES

In the spirit of our vision and mission, the objectives of the CCMAS (Thirty Percent (30%)) for undergraduate Industrial Mathematics programme are:

1. to provide training in the applications of Mathematics to a whole range of problems in different areas of Science and industry.
2. to impact the fundamental principles of Applied Mathematics on the students and the students at the same time acquire the skills needed for modern computing.
3. to provide a strong mathematical background for students who have interest in Mathematics and may wish to become actuaries
4. to train high level manpower in Mathematics for employment in industries and the public services.
5. to develop the ability to apply their mathematics knowledge and skills to the solution of theoretical and practical problems in mathematics.
6. to develop a range of transferable skills of value in mathematical related and non-mathematical related employment.
7. to generate an appreciation of the importance of mathematics in an industrial, economic, environmental and social context.
8. to develop high technical abilities in the applications of mathematical solutions to real-world and industry problems.
9. prepare the students to meet the demands of mathematical professionals in both existing and emerging industrial sectors;

10. develop in students a sense of enthusiasm for mathematics; and to involve them in an intellectually stimulating and satisfying experience of learning and studying;

11. provide students a broad and balanced foundation in general undergraduate mathematics knowledge and practical skills in its applications in industry;

12. develop in students a range of transferable skills of value in mathematical and non-mathematical related employment;

13. provide students with knowledge and skills base from which they can proceed to further studies in specialized areas of applied mathematics or multi-disciplinary areas involving mathematics; and

14. generate in students an appreciation of the importance of mathematics in an industrial, economic, environmental and social context.

4. UNIQUE FEATURES OF THE PROGRAMME

The B. Sc. Industrial Mathematics programme is designed to train the needed skilled manpower to bring to bear the applications of Mathematics to address industrial problems. It helps students acquire the skills needed for modern computing, financial institutions, risk management, economics, etc. towards development of the country, taking advantage of current innovations in technology, which pure and applied Mathematics have not really addressed directly.

5. RATIONALE/JUSTIFICATION

Our mission is to train students to acquire mathematical knowledge and skills to the highest attainable level; produce for society, scientists with unmatched competencies in the fields of mathematical and cognate sciences, and who can help to imbue society with discipline and positive values which are ingredients that propel and sustain national development.

6. ADMISSION REQUIREMENTS

UTME

i. Candidates for admission into the B.Sc. (Hons) degree in Industrial Mathematics must possess 5 (five) senior school certificate examination (SSCE) or its equivalent O – level credit passes in English Language, Mathematics, Further Mathematics, Physics and any of Chemistry or Biology or Economics or Geography in one sitting.

ii. The unified Tertiary Matriculation Examination (UTME) subjects are English, Mathematics, Physics and any one of Chemistry or Economics or Geography.

DIRECT ENTRY

Direct Entry candidates to 200 level must possess A – Level passes in Mathematics and one of Physics or Chemistry or Economics. In addition to O - level subjects indicated in (i) above, candidates with the same subjects as A-Level or its equivalence are also eligible.

7. GRADUATION REQUIREMENT

UTME STUDENTS

For a candidate to graduate under a four-year B.Sc. (Hons) degree programmes in Industrial Mathematics, he or she MUST pass a minimum of 120 units including all compulsory courses for the programme.

DIRECT ENTRY STUDENTS

For a candidate to graduate under a three-year B.Sc. (Hons) degree programme in Industrial Mathematics, he or she MUST pass a minimum of 90 units including all compulsory courses.

Candidates for Industrial Mathematics programme will have the opportunity to specialize in different areas of interest. The different areas of interest are in different options, namely:

i. Applied and Computational Mathematics
ii. Mathematics and Actuarial Science
iii. Mathematics and Computer Science
iv. Mathematics and Economics

8. CAREER OPPORTUNITIES FOR INDUSTRIAL MATHEMATICIANS

The following are the top career opportunities in Industrial Mathematics

1. Actuarial analyst	2. Actuary	3. Astronomer
4. Data scientist	5. Investment analyst	6. Research Analysts
7. Research scientist (maths)	8. Scientific programmer	9. Secondary school teacher
10. Software engineer	11. Sound engineer	12. Statistician
13. Systems Engineers Operations		14. Technical consultant

GLOBAL COURSE SRUCTURE

100 Level
FIRST SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
BIO 101	General Biology I	2	C	30	-
CHM 101	General Chemistry I	2	C	30	-
COS 101	Introduction to Computing Sciences	3	C	30	45
GST 111	Communication in English	2	C	15	45
MTH 101	Elementary Mathematics I	2	C	30	-
PHY 101	General Physics I	2	C	30	-
PHY 103	General Physics III	2	C	30	-
STA 121	Statistical Inference I	3	C	45	-
	Total Compulsory Units	**18**			

SECOND SEMESTER

NUC 70% NUC CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
BIO 102	General Biology II	2	C	30	
CHM 102	General Physics II	2	C	30	-
COS 102	Problem Solving	3	C	30	45
ECO 102	Principles of Economics II	2	C	30	-
GST 112	Nigerian People and Culture	2	C	30	-
MTH 102	Elementary Mathematics II	2	C	30	
MTH 103	Elementary Mathematics III	2	C	30	
PHY 102	General Physics II	2	C	30	
STA 112	Introduction to Statistics and Probability	3	C	45	
	Total Compulsory Units	**20**			

GLOBAL COURSE STRUCTION

200 LEVEL
FIRST SEMESTER

NUC 70% CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
ENT 211	Entrepreneurship and Innovation	2	C	30	-
COS 201	Computer Programming I	3	C	30	45
MTH 201	Mathematical Methods I	2	C	30	-
MTH 203	Set Logic and Algebra I	2	C	30	-
MTH 204	Linear Algebra I	2	C	30	-
MTH 207	Real Analysis I	2	C	30	-
MTH 209	Introduction to Numerical Analysis	2	C	30	-
MTH 210	Vector Analysis	2	C	30	-
	Total Compulsory Units	**17**			

SECOND SEMESTER

NUC 70% CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
GST 212	Philosophy, Logic and Human Existence	2	C	30	
MTH 202	Elementary Differential Equations	2	C	30	-
MTH 205	Linear Algebra II	1	C	15	-
IMT 206	Abstract Algebra	2	C	30	-
	Total Compulsory Units	**7**			

NOTE: The above 200L 70% courses are for all the options

OPTION 1: B. Sc. (Hons) Industrial Mathematics (Applied and Computational Mathematics)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 215	History of Mathematics	2	E	30	-
*MAT 218	Electricity & Magnetism	2	E	30	-
STA 211	Probability II	3	E	45	-
STA 231	Statistical Computing II	2	E	-	90
*STA 225	Statistical Quality Control	2	E	30	-
	Total Compulsory Units	**0**			
	Total Elective Units	**11**			

SECOND SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 224	Mechanics III	3	E	30	-
*MAT 229	Introduction to Scientific Computing	3	C	30	45
*CSC 227	Introduction to Information Processing	2	C	30	-
*PHS 229	Theoretical Physics I	2	E	30	-
PHY 205	Thermal Physics	3	E	45	-
LAG-STA 214	Introduction to Biostatistics	3	E	45	-
LAG-STA 222	Linear Model I	2	E	30	-
	Total Compulsory Units	**5**			
	Total Elective Units	**13**			

OPTION 2: B. Sc. (Hons) Industrial Mathematics (Mathematics and Computer Science)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*CSC 211	Software Workshop II	2	E	30	-
*CSC 213	Foundations of Sequential Programs	2	E	30	-
STA 231	Statistical Computing II	2	E	30	-
	Total Compulsory Units	**0**			
	Total Elective Units	**6**			

SECOND SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 224	Mechanics III	3	E	45	-
*MAT 229	Introduction to Scientific Computing	3	C	30	45
COS 202	Computer Programming II	3	C	30	-
*CSC 224	Introduction to Information Processing	3	C	30	-
*PHS 229	Theoretical Physics I	2	E	30	-
PHY 205	Thermal Physics	2	E	30	-
LAG-STA 214	Introduction to Biostatistics	3	E	45	-
LAG-STA 222	Linear Model I	2	E	30	-
	Total Compulsory Units	**9**			
	Total Elective Units	**12**			

NOTE: * denotes UNILAG courses (For example, *MAT 224 = LAG-MAT 224)

OPTION 3: B.Sc. (Hons) Industrial Mathematics (Mathematics and Economics)
FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
STA 231	Statistical Computing II	2	E	-	90
*STA 251	Introduction to Data Science I	3	E	45	-
*STA 223	Categorical Data Analysis	3	E	45	-
*STA 225	Statistical Quality Control	3	E	45	-
ACC 201	Financial Accounting I	3	E	30	45
ECO 201	Introduction to Microeconomics I	2	E	30	-
ECO 203	Introduction to Macroeconomics I	2	E	30	-
FIN 209	Elements of Banking	3	E	45	-
	Total Compulsory Units	**0**			
	Total Elective Units	**21**			

SECOND SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 229	Introduction to Scientific Computing	3	C	30	45
ACC 202	Financial Accounting II	3	E	30	45
ACC 204	Cost Accounting	3	E	30	45
ECO 202	Introduction to Microeconomics II	2	E	30	-
ECO 204	Introduction to Macroeconomics II	2	E	30	-
FIN 202	Principles of Insurance	2	E	30	-
	Total Compulsory Units	**3**			
	Total Elective Units	**12**			

OPTION 4: B. Sc. (Hons) Industrial Mathematics (Mathematics and Actuarial Science)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*STA 251	Introduction to Data Science I	3	E	45	-
STA 211	Probability II	3	E	45	-
STA 231	Statistical Computing II	2	E	-	90
*STA 223	Categorical Data Analysis	3	E	45	-
*STA 225	Statistical Quality Control	3	E	30	-
ACC 201	Financial Accounting I	3	E	30	45
ECO 201	Introduction to Microeconomics I	2	E	30	-
ECO 203	Introduction to Macroeconomics I	2	E	30	-
FIN 209	Elements of Banking	3	E	45	-
	Total Compulsory Units	**0**			
	Total Elective Units	**24**			

SECOND SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 229	Introduction to Scientific Computing	3	C	45	-
ACC 202	Financial Accounting II	3	E	30	45
ACC 204	Cost Accounting	3	E	30	45
ACS 206	Mathematics of Demography	2	E	30	-
ECO 202	Introduction to Microeconomics II	2	E	30	-
ECO 204	Introduction to Macroeconomics II	2	E	30	-
FIN 202	Principles of Insurance	2	E	30	-
	Total Compulsory Units	**3**			
	Total Elective Units	**14**			

NOTE: * denotes UNILAG courses

300 LEVEL

GLOBAL COURSE STRUCTURE

FIRST SEMESTER

NUC 70% CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
IMT 301	Financial Mathematics	2	C	30	-
IMT 303	Mathematical Computing	2	C	15	45
IMT 306	Introduction to Operations Research	2	C	30	-
IMT 312	Financial Derivatives	2	C	30	-
MTH 303	Vector and Tensor Analysis	2	C	30	-
ENT 312	Venture Creation	2	C	15	45
	Total Compulsory Units	**12**			

SECOND SEMESTER

NUC 70% CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
IMT 320	Industrial Attachment (24 Weeks)	6	C		270
	Total Compulsory Units	**6**			

NOTE: The above 300L 70% courses are for all the options

OPTION 1: B. Sc. (Hons) Industrial Mathematics (Applied and Computational Mathematics)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 316	Mechanics IV	2	E	30	-
*MAT 332	Real Analysis III	2	E	30	-
*MAT 334	Numerical Analysis II	3	C	30	45
	Total Compulsory Units	**3**			
	Total Elective Units	**4**			

OPTION 2: B. Sc. (Hons) Industrial Mathematics (Mathematics and Computer Science)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 334	Numerical Analysis II	3	E	30	45
*CSC 316	Introduction to System Analysis and Design	3	E	30	-
*CSC 331	Real-Time Systems and Programming	3	E	30	45
STA 321	Statistical Inference III	2	E	30	-
*MAT 332	Real Analysis III	2	E	30	-
	Total Compulsory Units	**0**			
	Total Elective Units	**13**			

OPTION 3: B.Sc. (Hons) Industrial Mathematics (Mathematics and Economics)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 334	Numerical Analysis II	3	E	30	45
MTH 307	Real Analysis II	2	E	30	-
STA 311	Probability III	3	E	45	-
ECO 306	Introduction to Econometrics	3	C	45	-
ECN 337	Introduction to Industrial Economics	2	E	30	-
ECN 339	Introduction to Monetary Economics	2	E	30	-
SSC 302	Research Methods I	3	E	45	-
FIN 303	Principles of Finance	3	E	45	-
*MAT 332	Real Analysis III	2	E	30	-
	Total Compulsory Units	**3**			
	Total Elective Units	**20**			

OPTION 4: B. Sc. (Hons) Industrial Mathematics (Mathematics and Actuarial Science)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 334	Numerical Analysis II	3	E	30	45
STA 321	Statistical Inference III	3	E	30	45
STA 312	Distribution Theory I	3	E	30	45
ACS 302	Life Contingencies	3	E	45	-
ACS 304	Financial Mathematics for Actuaries	3	E	45	-
ACS 306	Life and Health Insurance	2	E	30	-
ACS 308	Ratemaking and Insurance Pricing Models	2	E	30	-
ACS 310	Survival Models	2	E	30	-
FIN 303	Principles of Finance	3	E	45	-
FIN 313	Financial Management	2	E	30	-
INS 301	Risk Management I	3	E	45	-
*MAT 332	Real Analysis III	2	E	30	-
	Total Compulsory Units	**0**			
	Total Elective Units	**31**			

NOTE: * denote UNILAG courses

400 LEVEL

GLOBAL COURSE STRUCTURE

FIRST SEMESTER

NUC 70% CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
MTH 401	Theory of Ordinary Differential Equations	2	C	30	-
MTH 405	General Topology	2	C	15	45
IMT 411	Mathematical Economics I	2	C	30	-
MTH 305	Complex Analysis II	2	C	30	-
IMT 307	Functional Analysis	2	C	30	-
	Total Compulsory Units	**10**			

SECOND SEMESTER

NUC 70% CCMAS COURSES					
Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
MTH 402	Theory of Partial Differential Equations	2	C	30	-
IMT 404	Optimization Theory	2	C	30	-
IMT 412	Mathematical Economics II	2	C	30	-
MTH 302	Ordinary Differential Equations	2	C	30	-
MTH 308	Introduction to Mathematical Modelling	2	C	30	-
GST 312	Peace and Conflict Resolution	2	C	30	-
IMT 499	Project	6	C	-	270
	Total Compulsory Units	**18**			

NOTE: The above 400 L courses are compulsory for all the options

OPTION 1: B. Sc. (Hons) Industrial Mathematics (Applied and Computational Mathematics)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 415	Fluid Mechanics I	3	C	30	-
*MAT 438	Numerical Analysis III	3	C	30	45
*STA 423	Stochastic Processes	3	E	45	-
STA 415	Regression and Analysis of Variance II	3	E	45	-
1NS 401	Project Management	2	E	30	-
PHY 401	Quantum Mechanics	2	E	30	-
	Total Compulsory Units	**6**			
	Total Elective Units	**10**			

SECOND SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 426	Numerical Analysis IV	3	E	30	45
*MAT 445	Fluid Mechanics II	2	E	30	-
*MAT 446	Electricity and Plasma Dynamics	2	E	30	-
*IMT 429	Introduction to Stochastic Calculus for Finance	3	E	45	-
MTH 406	Lebesgue Measure and Integrals	2	E	30	-
*STA 432	Data Analysis	3	E	45	-
*MAT 325	Analytical Dynamics	2	E	30	-
*MAT 326	Analytical Geometry	3	E	45	-
*MAT 328	Discrete Mathematics	2	E	30	-
	Total Compulsory Units	**0**			
	Total Elective Units	**22**			

OPTION 2: B. Sc. (Hons) Industrial Mathematics (Mathematics and Computer Science)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 415	Fluid Mechanics I	3	E	45	-
*MAT438	Numerical Analysis III	3	C	30	45
*STA 423	Stochastic Processes	3	C	45	-
CSC 401	Algorithm and Complexity Analysis	2	E	30	-
ICT 418	Design and Installation of Electrical & ICT Equipment	3	E	15	45
	Total Compulsory Units	**6**			
	Total Elective Units	**8**			

SECOND SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 426	Numerical Analysis IV	3	E	30	45
*IMT 429	Introduction to Stochastic Calculus for Finance	3	E	45	-
*CSC 522	Principles of Programming Languages	3	E	30	45
MTH 406	Legesgue Measure and Integrals	2	E	30	-
	Total Compulsory Units	**0**			
	Total Elective Units	**11**			

OPTION 3: B.Sc. (Hons) Industrial Mathematics (Mathematics and Economics)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
STA 415	Regression and Analysis of Variance II	3	C	45	-
ECO 401	Advanced Microeconomics I	2	E	30	-
ECO 403	Advanced Macroeconomics I	2	E	30	-
	Total Compulsory Units	**3**			
	Total Elective Units	**4**			

SECOND SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*STA 432	Data Analysis	3	E	45	-
*IMT 429	Introduction to Stochastic Calculus for Finance	3	E	45	-
MTH 406	Legesgue Measure and Integrals	2	E	30	-
ECO 402	Advanced Microeconomics II	2	E	30	-
ECO 404	Advanced Macroeconomics II	2	E	30	-
	Total Compulsory Units	**0**			
	Total Elective Units	**12**			

OPTION 4: B. Sc. (Hons) Industrial Mathematics (Mathematics and Actuarial Science)

FIRST SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 438	Numerical Analysis III	3	C	30	45
*STA 423	Stochastic Processes	3	C	30	-
STA 415	Regression and Analysis of Variance II	3	C	45	-
ACS 401	Further Life Contingencies	3	E	45	-
ACS 402	Pensions and Social Insurance	3	E	45	-
ACS 404	Actuarial Valuation	3	E	45	-
ACS 405	Advanced Risk Management	3	E	45	-

ECO 401	Advanced Microeconomics I	2	E	30	-
ECO 403	Advanced Macroeconomics I	2	E	30	-
	Total Compulsory Units	**9**			
	Total Elective Units	**16**			

SECOND SEMESTER

INSTITUTIONAL 30% CCMAS COURSES					
COURSE CODE	**COURSE TITLE**	**UNITS**	**STATUS**	**LH**	**PH**
*MAT 426	Numerical Analysis IV	3	E	30	45
*IMT 429	Introduction to Stochastic Calculus for Finance	3	E	45	-
MTH 406	Legesgue Measure and Integrals	2	E	30	-
ECO 402	Advanced Microeconomics II	2	E	30	-
ECO 404	Advanced Macroeconomics II	2	E	30	-
	Total Compulsory Units	**0**			
	Total Elective Units	**12**			

NOTE: * denote UNILAG courses

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES

OPTION 1: B. Sc. (Hons) Industrial Mathematics (Applied and Computational Mathematics)

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SEMESTER

FIRST SEMESTER

Level	**Compulsory**	**Elective**
100	18	0
200	17	18
300	15	4
400	16	10
Grand Total (1st Semester)	**66**	**32**

SECOND SEMESTER

Level	**Compulsory**	**Elective**
100	20	0
200	12	13
300	6	0
400	18	22
Grand Total (2nd Semester)	**56**	**35**

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SESSION

Level	Compulsory	Elective
100	38	0
200	29	31
300	21	4
400	31	32
Grand Total	**119**	**67**

OPTION 2: B. Sc. (Hons) Industrial Mathematics (Mathematics and Computer Science)

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SEMESTER

FIRST SEMESTER

Level	Compulsory	Elective
100	18	0
200	17	6
300	12	19
400	16	8
Grand Total (1st Semester)	**63**	**33**

SECOND SEMESTER

Level	Compulsory	Elective
100	20	0
200	16	12
300	6	0
400	18	11
Grand Total (2nd Semester)	**60**	**23**

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SESSION

Level	Compulsory	Elective
100	**38**	**0**
200	**33**	**18**
300	**18**	**19**
400	**34**	**19**
Grand Total	**123**	**56**

OPTION 3: B.Sc. (Hons) Industrial Mathematics (Mathematics and Economics)
GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SEMESTER
FIRST SEMESTER

Level	Compulsory	Elective
100	18	0
200	17	21
300	15	20
400	13	4
Grand Total (1st Semester)	**57**	**43**

SECOND SEMESTER

Level	Compulsory	Elective
100	20	0
200	10	12
300	6	0
400	18	12
Grand Total (2nd Semester)	**54**	**24**

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SESSION

Level	Compulsory	Elective
100	38	0
200	27	33
300	21	20
400	31	16
Grand Total	**117**	**69**

OPTION 4: B. Sc. (Hons) Industrial Mathematics (Mathematics and Actuarial Science)

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SEMESTER
FIRST SEMESTER

Level	Compulsory	Elective
100	18	0
200	17	24
300	12	31
400	19	16
Grand Total (1st Semester)	**66**	**71**

SECOND SEMESTER

Level	Compulsory	Elective
100	20	0
200	10	14
300	6	0
400	18	12
Grand Total (2nd Semester)	**54**	**26**

GRAND TOTAL UNITS OF COMPULSORY AND ELECTIVE COURSES PER SESSION

Level	Compulsory	Elective
100	38	0
200	27	38
300	18	31
400	37	28
Grand Total	**120**	**97**

MATHEMATICS AND INDUSTRIAL MATHEMATICS 100 LEVEL COURSE CONTENTS AND LEARNING OUTCOMES (NUC 70% CCMAS)

100 LEVEL

BIO 101: General Biology I (2 Units C: LH 30)
Learning Outcomes
At the end of this course, students should be able to:
1. explain cell structure and organizations;
2. summarize functions of cellular organelles;
3. characterize living organisms and state their general reproduction;
4. describe the interrelationship that exists between organisms;
5. discuss the concept of heredity and evolution; and
6. enumerate habitat types and their characteristics.

Course Contents
Cell structure and organisation, functions of cellular organelles. Characteristics and classification of living things. Chromosomes, genes; their relationships and importance. General reproduction, interrelationships of organisms (competitions, parasitism, predation, symbiosis, commensalisms, mutualism, saprophytism). Heredity and evolution (introduction to Darwinism and Lamarkism, Mendelian laws, explanation of key genetic terms). Elements of ecology and types of habitats.

BIO 102: General Biology II (2 Units C: LH 30)
Learning Outcomes
At the end of the lectures, students should be able to:
1. list the characteristics, methods of identification and classification of viruses, bacteria and fungi;
2. state the unique characteristics of plant and animal kingdoms;
3. describe ecological adaptations in the plant and animal kingdoms;
4. explain nutrition, respiration, excretion and reproduction in plants and animals; and
5. describe growth and development in plants and animals.

Course Contents
Basic characteristics, identification and classification of viruses, bacteria and fungi.
A generalized survey of the plant and animal kingdoms based mainly on the study of similarities and differences in the external features. Ecological adaptations. Briefs on physiology include nutrition, respiration, circulatory systems, excretion, reproduction, growth and development.

CHM 101: General Chemistry I (2 Units C: LH 30)
Learning Outcomes
At the end of this course, students should be able to:
1. define atom, molecules and chemical reactions;
2. discuss the modern electronic theory of atoms;
3. write electronic configurations of elements on the periodic table;
4. justify the trends of atomic radii, ionization energies, electronegativity of the elements based on their position in the periodic table;
5. identify and balance oxidation – reduction equation and solve redox titration problems;
6. illustrate shapes of simple molecules and hybridized orbitals;

7. identify the characteristics of acids, bases and salts and solve problems based on their quantitative relationship;
8. apply the principles of equilibrium to aqueous systems using LeChatelier's principle to predict the effect of concentration, pressure and temperature changes on equilibrium mixtures;
9. analyse and perform calculations with the thermodynamic functions, enthalpy, entropy and free energy; and
10. determine rates of reactions and its dependence on concentration, time and temperature.

Course Contents
Atoms, molecules, elements, compounds and chemical reactions. Modern electronic theory of atoms. Electronic configuration, periodicity and building up of the periodic table. Hybridization and shapes of simple molecules. Valence forces and structure of solids. Chemical equations and stoichiometry, chemical bonding and intermolecular forces and kinetic theory of matter. Elementary thermochemistry, rates of reaction, equilibrium and thermodynamics. Acids, bases and salts. Properties of gases. Redox reactions and introduction to electrochemistry. Radioactivity.

CHM 102: General Chemistry II (2 Units C: LH 30)
Learning Outcomes
At the end of this course, students should be able to:
1. state the importance and development of organic chemistry;
2. define fullerenes and their applications;
3. discuss electronic theory;
4. determine the qualitative and quantitative of structures in organic chemistry;
5. describe rules guiding nomenclature and functional group classes of organic chemistry;
6. determine rate of reaction to predict mechanisms of reaction;
7. identify classes of organic functional group with brief description of their chemistry;
8. discuss comparative chemistry of group 1A, IIA and IVA elements; and
9. describe basic properties of transition metals.

Course Contents
Historical survey of the development and importance of organic chemistry, fullerenes as fourth allotrope of carbon, uses as nanotubules, nanostructures and nano chemistry. Electronic theory in organic chemistry. Isolation and purification of organic compounds. Determination of structures of organic compounds including qualitative and quantitative analysis in organic chemistry. Nomenclature and functional group classes of organic compounds. Introductory reaction mechanism and kinetics. Stereochemistry. The chemistry of alkanes, alkenes, alkynes, alcohols, ethers, amines, alkyl halides, nitriles, aldehydes, ketones, carboxylic acids and derivatives. The Chemistry of selected metals and non-metals. Comparative chemistry of group IA, IIA and IVA elements. Introduction to transition of metal chemistry.

COS 101: Introduction to Computing Sciences (3 Units C: LH 30; PH 45)
Learning Outcomes
At the end of the course, students should be able to:

1. explain basic components of computers and other computing devices;
2. describe the various applications of computers;
3. explain information processing and its roles in the society;
4. describe the Internet, its various applications and its impact;
5. explain the different areas of the computing discipline and its specializations; and
6. demonstrate practical skills on using computers and the internet.

Course Contents
Brief history of computing. Description of the basic components of a computer/computing device. Input/Output devices and peripherals. Hardware, software and human ware. Diverse and growing computer/digital applications. Information processing and its roles in society. The Internet, its applications and its impact on the world today. The different areas/programs of the computing discipline. The job specializations for computing professionals. The future of computing.

Lab Work: Practical demonstration of the basic parts of a computer. Illustration of different operating systems of different computing devices including desktops, laptops, tablets, smart boards and smart phones. Demonstration of commonly used applications such as word processors, spreadsheets, presentation software and graphics. Illustration of input and output devices including printers, scanners, projectors and smartboards. Practical demonstration of the Internet and its various applications. Illustration of browsers and search engines. How to access online resources.

COS 102: Problem Solving (3 Units C: LH 30; PH 45)
Learning Outcomes
At the end of this course, students should be able to:
1. explain problem solving processes;
2. demonstrate problem solving skills;
3. describe the concept of algorithms development and properties of algorithms;
4. discuss the solution techniques of solving problem;
5. solve computer problems using algorithms, flowcharts, pseudocode; etc.; and
6. solve problems using programming language using C, PYTHON, etc.

Course Contents
Introduction to the core concepts of computing. Problems and problem-solving. The identification of problems and types of problems (routine problems and non-routine problems). Method of solving computing problems (introduction to algorithms and heuristics). Solvable and unsolvable problems. Solution techniques of solving problems (abstraction, analogy, brainstorming, trial and error, hypothesis testing, reduction, literal thinking, meansend analysis, method of focal object, morphological analysis, research, root cause analy proof, divide and conquer). General Problem-solving process. Solution formulation and design: flowchart, pseudocode, decision table, decision tree. Implementation, evaluation and refinement. Programming in C, Python etc.
Lab Work: Use of simple tools for algorithms and flowcharts; writing pseudocode; writing assignment statements, input-output statements and condition statements; demonstrating simple programs using any programming language (Visual Basic, Python, C)
GST 111: Communication in English (2 Units C: LH 15; PH 45)
Learning Outcomes
At the end of this course, students should be able to:

1. identify possible sound patterns in English language;
2. list notable language skills;
3. classify word formation processes;
4. construct simple and fairly complex sentences in English;
5. apply logical and critical reasoning skills for meaningful presentations;
6. demonstrate an appreciable level of the art of public speaking and listening; and
7. write simple and technical reports.

Course Contents
Sound patterns in English language (vowels and consonants, phonetics and phonology). English word classes (lexical and grammatical words, definitions, forms, functions, usages and collocations). Sentence in English (types, structural and functional, simple and complex). Grammar and usage (tense, mood, modality, concord and aspects of language use in everyday life). Logical and critical thinking and reasoning methods (logic and syllogism, inductive and deductive argument and reasoning methods, analogy, generalisation and explanations). Ethical considerations, copyright rules and infringements. Writing activities: (Pre-writing, writing, post writing, editing and proofreading, brainstorming, outlining, paragraphing, types of writing, summary, essays, letter, curriculum vitae, report writing, note making and mechanics of writing). Comprehension strategies (reading and types of reading, comprehension skills, 3RsQ). Information and communication technology in modern language learning. Language skills for effective communication. Major word formation processes. Writing and reading comprehension strategies. Logical and critical reasoning for meaningful presentations. Art of public speaking and listening. Report writing.

GST 112: Nigerian Peoples and Culture (2 Units C: LH 30)

Learning Outcomes
At the end of the course, students should be able to:
1. analyse the historical foundation of the Nigerian culture and arts in pre-colonial times;
2. list and identify the major linguistic groups in Nigeria;
3. explain the gradual evolution of Nigeria as a political unit;
4. analyse the concepts of trade, economic and self-reliance status of the Nigerian peoples towards national development;
5. enumerate the challenges of the Nigerian State towards nation building;
6. analyse the role of the Judiciary in upholding people's fundamental rights;
7. identify acceptable norms and values of the major ethnic groups in Nigeria; and
8. list and suggest possible solutions to identifiable Nigerian environmental, moral and value problems.

Course Contents
Nigerian history, culture, and art up to 1800 (Yoruba, Hausa and Igbo peoples and culture; peoples and culture of the ethnic minority groups). Nigeria under colonial rule (advent of colonial rule in Nigeria and colonial administration of Nigeria). Evolution of Nigeria as a political unit (amalgamation of Nigeria in 1914, formation of political parties in Nigeria, nationalist movement and struggle for independence). Nigeria and challenges of nation building (military intervention in Nigerian politics and Nigerian Civil War). Concept of trade and economics of self-reliance

(indigenous trade and market system, indigenous apprenticeship system among Nigeria people, trade, skill acquisition and self-reliance). Social justices and national development (law definition and classification. judiciary and fundamental rights. Individual, norms and values (basic Nigeria norms and values, patterns of citizenship acquisition, citizenship and civic responsibilities, indigenous languages, usage and development, negative attitudes and conducts, cultism, kidnapping and other related social vices). Re-orientation, moral and national values (The 3R's – Reconstruction, Rehabilitation and Re-orientation; re-orientation strategies: Operation Feed the Nation (OFN), Green Revolution, Austerity Measures, War Against Indiscipline (WAI), War Against Indiscipline and Corruption (WAIC), Mass Mobilization for Self-Reliance, Social Justice and Economic Recovery (MAMSER), National Orientation Agency (NOA). Current socio-political and cultural developments in Nigeria.

MTH 101: Elementary Mathematics I (Algebra and Trigonometry)
(2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. explain basic definition of set, subset, union, intersection, complements and use of venn diagrams;
2. solve quadratic equations;
3. solve trigonometric functions;
4. identify the various types of numbers; and
5. solve some problems using binomial theorem.

Course Contents
Elementary set theory, subsets, union, intersection, complements and venn diagrams. Real numbers, integers, rational and irrational numbers, mathematical induction, real sequences and series, theory of quadratic equations and binomial theorem. Complex numbers, algebra of complex numbers and the argand diagram. De-Moivre's theorem, nth roots of unity. circular measure, trigonometric functions of angles of any magnitude, addition and factor formulae.

MTH 102: Elementary Mathematics II (Calculus) (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. identify the types of rules in differentiation and integration;
2. describe the meaning of function of a real variable, graphs, limits and continuity; and
3. solve some applications of definite integrals in areas and volumes.

Course Contents
Function of a real variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation. Extreme curve sketching, integration as an inverse of differentiation. Methods of integration and definite integrals. Application to areas and volumes.

MTH 103: Elementary Mathematics III (Vectors, Geometry and Dynamics)
(2 Units C: LH 30)
Learning Outcomes

At the end of the course, students should be able to:
1. solve some vectors in addition and multiplication;
2. calculate force and momentum; and
3. solve differentiation and integration of vectors.

Course Contents
Geometric representation of vectors in 1-3 dimensions, components, direction cosines. Addition, scalar, multiplication of vectors, linear independence. Scalar and vector products of two vectors. Differentiation and integration of vectors with respect to a scalar variable. Two-dimensional co-ordinate geometry. Straight lines, circles, parabola, ellipse, hyperbola. Tangents, normals. Kinematics of a particle. Components of velocity and acceleration of a particle moving in a plane. Force, momentum, laws of motion under gravity, projectiles and resisted vertical motion. Elastic string and simple pendulum. Impulse, impact of two smooth spheres and a sphere on a smooth surface.

PHY 101: General Physics I (Mechanics) (2 Units C: LH 30)
Learning Outcomes
On completion of the course, students should be able to:
1. identify and deduce the physical quantities and their units;
2. differentiate between vectors and scalars;
3. describe and evaluate motion of systems on the basis of the fundamental laws of mechanics;
4. apply Newton's laws to describe and solve simple problems of motion;
5. evaluate work, energy, velocity, momentum, acceleration, and torque of moving or rotating objects;
6. explain and apply the principles of conservation of energy, linear and angular momentum;
7. describe the laws governing motion under gravity; and
8. explain motion under gravity and quantitatively determine behaviour of objects moving under gravity.

Course Contents
Space and time. units and dimension. vectors and scalars. differentiation of vectors: displacement, velocity and acceleration. Kinematics. Newton laws of motion (Inertial frames, Impulse, force and action at a distance, momentum conservation). relative motion. Application of Newtonian mechanics. equations of motion. conservation principles in physics, conservative forces, conservation of linear momentum, Kinetic energy and work, Potential energy, System of particles, Centre of mass. Rotational motion. torque, vector product, moment, rotation of coordinate axes and angular momentum, polar coordinates. conservation of angular momentum; Circular motion. Moments of inertia, gyroscopes and precession. gravitation: Newton's Law of Gravitation, Kepler's Laws of Planetary Motion, Gravitational Potential Energy, Escape velocity, Satellites motion and orbits.

PHY 102: General Physics II (Electricity &Magnetism) (2 Units C: LH 30)
Learning Outcomes
On completion of the course, students should be able to:
1. describe the electric field and potential and related concepts, for stationary charges;
2. calculate electrostatic properties of simple charge distribution using Coulomb's law, Gauss's law and electric potential;

3. describe and determine the magnetic field for steady and moving charges;
4. determine the magnetic properties of simple current distributions using Biot-Savart and Ampere's law;
5. describe electromagnetic induction and related concepts and make calculations using Faraday and Lenz's laws;
6. explain the basic physical of Maxwell's equations in integral form;
7. evaluate DC circuits to determine the electrical parameters; and
8. determine the characteristics of AC voltages and currents in resistors, capacitors and inductors.

Course Contents
Forces in nature, electrostatics, electric charge and its properties, methods of charging, Coulomb's law and superposition, electric field and potential, Gauss's law, capacitance, electric dipoles, energy in electric fields, conductors and insulators, current, voltage and resistance, Ohm's law and analysis of DC circuits, magnetic fields, Lorentz force, Biot-Savart and Ampère's laws, magnetic dipoles, dielectrics, energy in magnetic fields, electromotive force, electromagnetic induction, self and mutual inductances, Faraday and Lenz's laws, step up and step down transformers. Maxwell's equations, electromagnetic oscillations and waves, AC voltages and currents applied to inductors, capacitors, resistance and combinations.

STA 121: Statistical Inference I (3 Units C: LH 45)
Learning Outcome
At the end of the course, students should be able to:
1. differentiate population from sample as well as point from interval estimate;
2. test for hypothesis concerning population mean and proportions for large and small samples;
3. compute regression and obtain the fitted line. Likewise, the computation for correlation coefficient well understood; and
4. describe the fundamentals of time series analysis.

Course Contents
Population and samples. Random sampling distributions. Estimation (point and interval) and tests of hypotheses concerning population mean and proportion (one and two large sample cases). Regression and correlation. Elementary time series analysis.

STA 112: Probability I (3 Units C: LH 45)
Learning Outcomes
At the end of the course students should be able to
1. explain the differences between permutation and combination;
2. explain the concept of random variables and relate it to probability and distribution functions;
3. describe the basic distribution functions; and
4. explain the concept of exploratory data analysis.

Course Contents

Permutation and combination. Concepts and principles of probability. Random variables. Probability and distribution functions. Basic distributions: Binomial, geometric, Poisson, normal and sampling distributions; exploratory data analysis.

MATHEMATICS COURSE CONTENT (ONLY)
200 LEVEL

NUC 70% CCMAS
GST 212: Philosophy, Logic and Human Existence (2 Units C: LH 30)

Learning Outcomes
A student who has successfully gone through this course should be able to:
1. know the basic features of philosophy as an academic discipline;
2. identify the main branches of philosophy & the centrality of logic in philosophical discourse;
3. know the elementary rules of reasoning;
4. distinguish between valid and invalid arguments;
5. think critically and assess arguments in texts, conversations and day-to-day discussions;
6. critically asses the rationality or otherwise of human conduct under different existential conditions;
7. develop the capacity to extrapolate and deploy expertise in logic to other areas of knowledge, and
8. guide his or her actions, using the knowledge and expertise acquired in philosophy and logic.

Course Contents
Scope of philosophy; notions, meanings, branches and problems of philosophy. Logic as an indispensable tool of philosophy. Elements of syllogism, symbolic logic— the first nine rules of inference. Informal fallacies, laws of thought, nature of arguments. Valid and invalid arguments, logic of form and logic of content — deduction, induction and inferences. Creative and critical thinking. Impact of philosophy on human existence. Philosophy and politics, philosophy and human conduct, philosophy and religion, philosophy and human values, philosophy and character molding, etc.

ENT 211: Entrepreneurship and Innovation (2 Units C: LH 15; PH 45)

Learning Outcomes
At the end of this course, students should be able to
1. explain the concepts and theories of entrepreneurship, intrapreneurship, opportunity seeking, new value creation, and risk-taking state the characteristics of an entrepreneur
2. analyse the importance of micro and small businesses in wealth creation, employment, and financial independence engage in entrepreneurial thinking;
3. identify key elements in innovation; describe stages in enterprise formation, partnership and networking including business planning;
4. describe contemporary entrepreneurial issues in Nigeria, Africa and the rest of the world; and
5. state the basic principles of e-commerce.

Course Contents
Concept of entrepreneurship (entrepreneurship, intrapreneurship/corporate entrepreneurship). Theories, rationale and relevance of entrepreneurship (Schumpeterian and other perspectives, risk-taking, necessity and opportunity-based entrepreneurship and creative destruction). Characteristics of entrepreneurs (opportunity seeker, risk taker, natural and nurtured, problem solver and change agent, innovator and creative thinker). Entrepreneurial thinking (critical thinking, reflective thinking, and creative thinking). innovation (concept of innovation, dimensions of innovation, change and innovation, knowledge and innovation); enterprise formation, partnership and networking (basics of business plan, forms of business ownership, business registration and forming alliances and join ventures). Contemporary entrepreneurship Issues (knowledge, skills and technology, intellectual property, virtual office, networking). Entrepreneurship in Nigeria (biography of inspirational entrepreneurs, youth and women entrepreneurship, entrepreneurship support institutions, youth enterprise networks and environmental and cultural barriers to entrepreneurship); basic principles of e-commerce.

COS 201: Computer Programming I (3 Units C: LH 30; PH 45)

Learning Outcomes
At the end of this course, students should be able to:
1. Explain the principles of good programming and structured programming concepts;
2. Explain the programming constructs, syntax and semantics of a higher-level language;
3. Describe the chosen programming language variables, types, expressions, statements and assignment; simple input and output;
4. Describe the programme control structures, functions and parameter passing, and structured decomposition; and
5. Develop simple programmes in the taught programming language as well as debug and test them.

Course Contents
Introduction to computer programming. Functional programming; Declarative programming; Logic programming; Scripting languages. Introduction to object-orientation as a technique for modelling computation. structured, and even some level of functional programming principles; Introduction of a typical object-oriented language, such as Java; Basic data types, variables, expressions, assignment statements and operators; Basic object-oriented concepts: abstraction; objects; classes; methods; parameter passing; encapsulation. Class hierarchies and programme organisation using packages/namespaces; Use of API – use of iterators/enumerators, List, Stack, Queue from API; Searching; sorting; Recursive algorithms; Event-driven programming: event-handling methods; event propagation; exception handling. Introduction to Strings and string processing; Simple I/O; control structures; Arrays; Simple recursive algorithms; inheritance; polymorphism.

Lab work: Programming assignments; design and implementation of simple algorithms, e.g., average, standard deviation, searching and sorting; Developing and tracing simple recursive algorithms. Inheritance and polymorphism.

MTH 201: Mathematical Methods 1 (2 Units C: LH 30)

Learning Outcomes
At the end of the course, students should be able to:
1. explain real-valued functions of a real variable;
2. solve some problems using mean value theorem and Taylor series expansion; and
3. evaluate line integral, surface integral and volume integrals.

Course Contents
Real-valued functions of a real variable. Review of differentiation and integration and their applications. Mean value theorem. Taylor series. Real-valued functions of two and three variables. Partial derivatives chain rule, extrema, lagrangian multipliers. Increments, differentials and linear approximations. Evaluation of line integrals. Multiple integrals.

MTH 202: Elementary Differential Equations (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. define the following: order and degree of a differential equation;
2. describe some techniques for solving first and second order linear and non-linear equations; and
3. solve some problems related to geometry and physics.

Course Contents
Derivation of differential equations from primitive geometry, physics etc. Order and degree of differential equation. Techniques for solving first and second order linear and non-linear equations. Solutions of systems of first order linear equations. Finite linear differential equations. Application to geometry and physics.

MTH 203: Sets, Logic and Algebra I (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. solve various problems using the concept of set theory;
2. recognize Algebraic structures; and
3. describe the meaning of logic in Mathematics.

Course Contents
Introduction to the language and concepts of modern mathematics. Topics include: basic set theory: mappings, relations, equivalence and other relations, Cartesian products, binary logic, methods of proof, binary operations. Algebraic structures, semi-groups, rings, integral domains, fields. Homeomorphics. Number systems; properties of integers, rationals, real and complex numbers.

MTH 204: Linear Algebra I (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. explain vector space;
2. describe linear transformations and their representation by matrices; and
3. calculate algebra of various matrices.

Course Contents

Vector space over the real field. Sub-spaces, linear independence, basis and dimension. Linear transformations and their representation by matrices – rings, null space, rank. Singular and non-singular transformation and matrices. Algebra of matrices.

MTH 205: Linear Algebra II (1 Unit C: LH 15)

Learning Outcomes

At the end of the course, students should be able to:

1. recognize systems of linear equations.
2. calculate the Eigen values and Eigen vectors.
3. describe the Cayley-Hamilton theorem and its uses.

Course Contents

Systems of linear equation, change of basis, equivalence and similarity. Eigen values and Eigen vectors. Minimum and characteristic polynomials of a linear transformation (matrix). Cayley-Hamilton theorem. Bi-linear and quadratic forms, orthogonal diagonalisation. Canonical forms.

MTH 207: Real Analysis I (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. describe convergence of sequence of numbers;
2. discuss the monotone, Cauchy sequences;
3. test for convergence of series; and
4. state roles and mean value theorem.

Course Contents

Bounds of real numbers, convergence of sequence of numbers. Monotone sequences, the theorem of nested intervals. Cauchy sequences, tests for convergence of series. Absolute and conditional convergence of series and re-arrangements. Completeness of reals and incompleteness of rationale. Continuity/and differentiability of functions. Rolles' mean and value theorems for differentiable functions, Taylor series.

MTH 209: Introduction to Numerical Analysis (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. solve some numerical solution of algebraic and transcendental equations;
2. describe curve fitting;
3. discuss error analysis;
4. calculate interpolation and approximation;
5. solve some numerical differentiation and numerical integration problems; and
6. solve some numerical problems in ordinary Differential equations with initial value problems;

Course Contents

Solution of algebraic and transcendental equations. Curve fitting. Error analysis. Interpolation and approximation. Zeros of non-linear equations 'in one variable'. Systems of linear equations. Numerical differentiation and integration. Initial value problems in ordinary differential equation.

MTH 210: Vector Analysis (1 Unit C: LH 15)

Learning Outcomes

At the end of the course, students should be able to:

1. describe vector algebra;
2. explain geometrical equation of lines and planes; and
3. outline problems in gradients, divergent and curl.

Course Contents

Elementary vector algebra, vector and vector triple, vector products (more application solution of vector equation, plain curves and space curves. Geometrical equation of lines and planes. Linear independence of vectors; components of vectors, direction cosines; position vector and scaler products; Frenet-Serret formulas; differential definition of gradients, divergent and simple multiplication)

INSTITUTIONAL 30% CCMAS

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 215: HISTORY OF MATHEMATICS (2 Units E: LH 30)**
SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching history of Mathematics is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

This course covers introductory and in-depth history of Mathematics starting from history of early Egyptian, Greek and Japanese Mathematics, emphasizing topics useful in other disciplines.

The course therefore, makes the beneficiary to appreciate and apply these foundation knowledge like the history of European Medieval mathematician from 500 AD and describe the development of plane pure geometry. The course is both helpful for further studies and practical use.

OBJECTIVES

The objectives of the course are to:

1. state the relevance of history of Mathematics;

2. demonstrate the Greek's method of counting numbers.
3. perform similar calculations using the Egyptian.
4. analyze the discovery of European Medieval mathematicians from 500 AD.
5. apply the Pythagorean principle to solve real life problems.

LEARNING OUTCOMES

At the end of the course, students should be able to:
1. discuss the contribution of mathematicians to the development of calculus.
2. explain the history of European Mathematics from 1500 AD - 1700 and the development of non-Euclidean and modern geometry.
3. discuss the History of European Medieval mathematicians from 500 AD; and
4. describe the development of plane pure geometry.
5. state some of the authors that made this history.

COURSE CONTENTS

Early Number Systems and Symbols. Primitive counting. Number recording of the Egyptians and Greeks, and the Babylonians. Mathematics in early civilizations. The Rhind Papyrus. Egyptian arithmetic. Egyptian geometry. Babylonian mathematics. Plimpton. The Beginnings of Greek Mathematics. The geometrical discoveries of Thales. Pythagorean Mathematics. Construction Problems of Antiquity. The quadratrix of Hippias. Archimedes. The Alexandrian School Euclid. Euclid and the Elements. Euclidean Geometry. Euclid's Number Theory. The Twilight of Greek Mathematics: Diophantus. The Decline of Alexandrian Mathematics. The Arithmetica. Diophantine Equations in Greece, India, and China. Mathematics in the Near and Far East. The First Awakening: Fibonacci. The Decline and Revival of Learning. The Liber Abaci and Liber Quadratorum. The Fibonacci Sequence. Fibonacci and the Pythagorean Number Triples. The Renaissance of Mathematics: Cardan and Tartaglia. Europe in the Fourteenth and Fifteenth Centuries. The Battle of the Scholars. Cardan's Ars Magna. Ferrari's Solution of the Quartic Equation.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 217: MATHEMATICS FOR CHEMISTRY I (2 Units C: LH 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Mathematics for Chemistry students is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is

seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
This course covers introductory and in-depth application of Calculus starting from Differential calculus: Differentiation of algebraic functions. Some applications are also considered in exponential growth and decay models.

The course therefore, makes the beneficiary appreciate Differential and Integral calculus.

The course is both helpful for further studies and practical use.

OBJECTIVES
The objectives of the course are to:
1. state the relevance of Calculus to Chemistry.
2. demonstrate the relation between differentiation and integration.
3. perform simple calculations on integration.
4. state the chain rules.
5. apply calculus to exponential growth and decay models.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. state the different integration techniques.
2. solve problems on areas and volumes.
3. state the formula for exponential growth.
4. analyze the decay models with the knowledge of calculus;
5. define a differential equation.

COURSE CONTENTS
Differential calculus: Differentiation of algebraic functions, trigonometric functions, composites function and chain rule, higher order derivatives. Integral calculus: Integration as inverse of differentiation, definite integral, techniques of integration. Application: Areas, volumes. Elementary differential equations: First order equations; Separable, homogeneous, exact, linear physical application: Exponential function, exponential growth and decay models.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 218: ELECTRICITY AND MAGNETISM (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Electricity ad Magnetism is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Electricity is basically the presence and motion of charged particles.On the other hand, magnetism refers to the force which the magnets exert when they attract or repel each other.A current gives rise to a magnetic field, and a magnetic field gives rise to a force on a current.

This course is an introduction to Electricity and Magnetism and their mathematical description connecting electric and magnetic phenomena via the special theory of relativity. Topic includes electrostatics, magnetic fields, electromagnetic induction, DC and AC circuits, and the electromagnetic properties of matter. The general topics included in this course are the principles and application of the said subject.

OBJECTIVES

The objectives of the course are to:

1. develop a basic understanding of electric and magnetic fields in free space using the integral forms of Maxwell's laws;
2. apply the knowledge of electricity and magnetism to explain natural and physical processes and related technological advances;
3. use an understanding of calculus along with physical principles to effectively solve problems encountered in everyday life, further study in Science, and in the professional world;
4. perform basic calculations using Coulomb's law and Gauss's law;
5. outline the significance of Maxwell's equations in integral form.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. describe the electric field and potential, and related concepts, for stationary charges;
2. describe and determine the magnetic field for steady and moving charges;
3. determine the magnetic properties of simple current distributions using Biot-Savart and Ampere's law;
4. describe electromagnetic induction and related concepts and make calculations using Faraday and Lenz's laws;
5. explain the basic physical significance of Maxwell's equations in integral form;
6. evaluate DC circuits to determine the electrical parameters; and
7. determine the characteristics of AC voltages and currents in resistors, capacitors, and Inductors.
8. develop a conception of space and time adequate for understanding electrodynamics.
9. develop and train problem-solving skills.

COURSE CONTENTS

Forces in nature. Electrostatics (electric charge and its properties, methods of charging). Coulomb's law and superposition. Electric field and potential. Gauss's law. Capacitance. Electric

dipoles. Energy in electric fields. Conductors and insulators. DC circuits (current, voltage and resistance). Ohm's law. Resistor combinations. Analysis of DC circuits. Magnetic fields. Lorentz force. Biot-Savart and Ampère's laws. Magnetic dipoles. Dielectrics. Energy in magnetic fields. Electromotive force. Electromagnetic induction. Self and mutual inductances. Faraday and Lenz's laws. Step up and step down transformers. Maxwell's equations in vacuum using vector differential and integral calculus. Electromagnetic oscillations and waves. AC voltages and currents applied to inductors, capacitors, and resistance. Relativity. Optics. Wave guides and Transmission lines. Radiation.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS

***MAT 234: MECHANICS II (3 Units C: LH 45)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Mechanics II is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Mechanics II is the branch of Mathematics dealing with the study of motion. No matter what students interest in science or engineering, mechanics will be important for them, motion is a fundamental idea in all of science.

Mechanics can be divided into two areas: kinematics, dealing with describing motions, and dynamics, dealing with the causes of motion.

This course covers introductory Newtonian Mechanics which include but not limited to force, momentum and laws of motion, different kinds of force, emphasizing topics useful in other disciplines.

The course therefore, makes the beneficiary to appreciate and apply the basic principles of dynamics of a rigid body. The course is both a preparation for further study and practical use.

OBJECTIVES

The objectives of the course are to:
1. define basic concepts of Newtonian Mechanics;
2. demonstrate the compound pendulum.

3. apply kinetic energy formula of a body rotating about a fixed axis to solve problems in appropriate situations.
4. demonstrate principle of dynamics of a rigid body and apply them to appropriate situations.
5. apply momentum and laws of motion to solving real life problems.
6. state and apply parallel axes and perpendicular axes theorems.
7. describe the Attwood's machine.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. define force ad momentum;
2. state the laws of motion;
3. state the different kinds of force;
4. state the equation of motion on an inclined plane;
5. define a projectile and describe the movement of a projectile motion;
6. define angular momentum of a rigid body.

COURSE CONTENTS
Newtonian mechanics: force, momentum and laws of motion. Different kinds of force (Gravitational, reactions, tension, thrust). Force and motion. Equation of motion. Motion on an inclined plane. Connected particles. Attwood's machine. Work, energy, power. Work-energy principle. Conversation laws of energy. Impulse and momentum. Impacts, direct and oblique collision. Impulsive tension. Projectiles: Range (on horizontal and inclined planes) trajectory. Dynamics of a rigid body. Moment of inertia. Radius of gyration. Parallel axes and perpendicular axes theorems. Kinetic energy of a body rotating about a fixed axis. Motion of a body about fixed axis. Angular momentum of a rigid body and principle of angular momentum. The compound pendulum. **Pre-requisite: MTH 103.**

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 224: MECHANICS III (3 Units C: LH 45)**

SENATE APPROVED RELEVANCE
Graduates who are highly skilled and knowledgeable in teaching Mechanics II is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
The aim of the course is to enhance students' knowledge and understanding of the mathematics and scientific principles related to mechanics and materials, and to develop their ability to apply this knowledge in a number of topics.

This course covers motion in 2-dimensions which include the use of polar coordinates, motion in a circle, emphasizing topics useful in other disciplines. The course is thus beneficial to those who wish to have a solid background in Mechanics. The course is both a preparation for further study and practical use.

OBJECTIVES
The objectives of the course are to:
1. demonstrate the motion in 2-dimensions;
2. describe constrained motion in a vertical circle and on a sphere;
3. use the simple harmonic motion to demonstrate real life problems;
4. illustrate the vertical motion under gravity, with resistance proportional to speed and to square of speed and apply them to appropriate situations.
5. apply projectiles in resisting medium to solving real life problems

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. define the canonical pendulum;
2. describe a simple harmonic motion;
3. describe simple harmonic motion problems;
4. solve some problems the dynamics of rigid body;
5. state D'Alembert's principle.

COURSE CONTENTS
Motion in 2-dimensions. Use of polar coordinates. Motion in a circle. The conical pendulum. The governor. Constrained motion in a vertical circle and on a sphere. Simple Harmonic Motion. Amplitude, period, frequency, the simple pendulum, elastic strings and springs. Second pendulum. Motion in a resisting medium. Vertical motion under gravity, with resistance proportional to speed and to square of speed. Damped, harmonic motion. Projectiles in resisting medium. Variable mass motion. Dynamics of a rigid body. Product of inertia,. Principal axes. The momenta ellipse. Virtual work. D'Alembert's principle.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 229: INTRODUCTION TO SCIENTIFIC COMPUTING (3 Units C: LH 30, PH 45)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Introduction to Scientific Computing is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Scientific Computing is the study of algorithms and their implementations for computations involving mathematical symbols as well as numbers. The course involves developing and applying a wide variety of scientific computing methods to many areas including physics and astronomy.

This course covers introduction to computer package Mathematica, Maple and Matlab. It also contain how to type in Mathematics, special characters, basic constructions. The course is both a preparation for further study and practical use.

OBJECTIVES

The objectives of the course are to:

1. describe the use of MATLAB;
2. demonstrate practically the Maple package;
3. use new commands which can perform more complicated tasks;
4. demonstrate the use of Java, C++, or Python;
5. apply the various packages to solving real life problems.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. use MATLAB in their own work;
2. illustrate their MATLAB programming skills and tackle other languages for computing, such as Java, C++, or Python;
3. define a computer package;
4. state standard functions;
5. define prime numbers.

COURSE CONTENTS

Introduction to computer package called Mathematica. Computer package called Mapleand Matlab. How to type in Mathematics. Special characters. Basic constructions. Numerical computations. Standard functions. Accuracy, use of variables. Working with whole numbers. Finding prime numbers. Handling algebraic expressions. Doing symbolic computations. Graphics. Calculus. Solving equations in Mathematica, Maple or Matlab. Introduction to programming (procedural vs functional) using various numerical and algebraic methods to solve equations and sets of equations. Defining new functions. Writing new commands which can perform more complicated tasks.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS

***CSC 227: INTRODUCTION TO INFORMATION PROCESSING (3 Units E: LH 30, PH 45)**

SENATE APPROVED RELEVANCE

Today, information system is at the core of processing data to obtain information in every aspect of the society, from technology companies like Google and Microsoft to banks, financial firms, to manufacturing companies and local supermarkets. These organisations use computer-based information systems to efficiently manage their businesses for improved productivity by providing high-quality goods and services as well as gain/sustain competitive advantages over their rivals. To produce graduates that are industry and future ready, this course was designed for students enrolled in non-computer science major courses in accordance with the University of Lagos' vision.

OVERVIEW

The purpose of information systems is to produce high-quality information that can be used to support the activities and decision making of managers in the organisation. To prepare non-computer science major students for the industry and the future, the students require a good understanding of information systems and its processes. This course would introduce the students to the various aspects of information processing such as the definition of data, information, systems, information system and their characteristics.

Introduction to information processing system a study of the acquisition, recording, organization, retrieval, display, and dissemination of information. In recent years, professionals in all fields and organizations use digital systems to enhance their productivity.

Any institution that would produce world class graduates must introduce information processing to their students so that the students would be prepared for the future. The goal of this course is to introduce non computer science majors to information processing and information systems.

OBJECTIVES

The objectives of the course are to:
1. introduce the general systems theory.
2. describe the basic concepts of information systems.
3. introduce the different types of information.
4. present the role of management information systems in a business organization.
5. give various ways of processing, storing, and dissemination of information.
6. apply the knowledge gained in the acquisition, processing, and presenting business information.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. Explain the concept and characteristics of a system.
2. Describe the basic components of an Information System.
3. State popular examples of Management Information Systems.
4. Describe the role of Information Processing in Information Systems.
5. Give detail explanation of Information Systems input, output, and storage.
6. Draw and explain Models of Data Communication Systems
7. Describe tools and techniques used in Data Security and Control
8. Apply Sorting and Searching techniques like quick sort and binary search procedures in Information Processing.

COURSE CONTENTS

Information Systems. Management Information Systems. Information Processing. Information Systems Output and Storage. Data Security and Control. Sorting and searching. Principles of Data Communication. Data communication system. Data communication channels. Twisted Pair Cable. Data Signal. Analog and Digital signals. Introduction to Computer Networks. Types of Computer Networks. Local Area Network Topologies. Bus Network Topology. Mesh Topology. Ring Network Topology. Star Network Topology. Hybrid Network Topology. Inter-Networking Model. The OSI Model. The Internet and Internet Services.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***PHS 228: THEORETICAL PHYSICS I (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Preparing graduates with broad knowledge of Physics and its utilization in theoretical physics, enabling them to be adequately prepared for a technologically-driven community, where dependence on foreign products would be drastically reduced is in tandem with the vision and mission of UNILAG. The students of the programme will be exposed to the relevant foundation physics courses that underpin an understanding of basic electronics courses, which form part of the curriculum aimed at providing them with the needed skills for job creation and innovations in the areas of applied physics and electronics.

OVERVIEW

Theoretical physics uses mathematical models and abstractions of physical objects and systems to rationalize, explain and predict natural phenomena. Ability to use mathematical models and physical abstractions is important in today's world. This underpins the importance of preparing students in Physics with Electronics with the requisite knowledge and skills on the far-reaching applications of theoretical physics and its application for everyday use.

This course will examine solutions to differential equations, partial differential equations, multiple integrals and their applications to moment of inertia, centre of mass etc and these shall be used to solve complex physical problems.

OBJECTIVES

The objectives of the course are to:

1. explain differential equations.
2. identify the different types of differential equations.
3. demonstrate multiple integration and its applications.
4. introduce matrices and its various applications.
5. introduce complex analysis and its use to solve physical problems.

LEARNING OUTCOMES

Upon the completion of the course, the students should be able to:

1. solve differential equations;
2. solve and apply partial differential equations;
3. apply multiple integration in moment of inertia and centre of mass;
4. solve some physical problems using the concept of matrices;
5. apply complex analysis to solve physical problems.

COURSE CONTENTS

Differential equations. Multiple integration: Line and double integrals. Triple integrals. Application of multiple integration in moment of inertia. Application of multiple integration in centre of mass and others. Transformation in 3-D. Partial differentiation. Matrices. Equality and addition of matrices. Multiplication of matrices. Product of matrices by portioning. Matrix solutions of linear equations. Complex analysis. Complex integration. Cauchy's integral. Taylor's expansion. Laurent's expansion.

MINIMUM ACADEMIC STANDARDS

Mathematical Simulations Laboratory with NUC-CCMAS requirement facilities and standard Lecture hall equipped with multimedia system.

STA 202: Statistics for Physical Sciences and Engineering (3 Units C: LH 45)

Learning Outcomes

At the end of the course, students should be able to:

1. highlight the scope for statistical methods in physical sciences and engineering;

2. define the measures of location, partition and dispersion;
3. explain the elements of probability; probability distribution: binomial Poisson, geometric, hypergeometric, negative-binomial, normal poisson, geometric, hypergeometric, negative-binomial, normal, Student's t and chi-square distribution;
4. differentiate point from internal estimation and could be able to tests for hypotheses concerning population means proportions and variances;
5. compute for regression and correlation as well as conduct some non–parametric tests with reference to contingency table analysis; and
6. explain the elements of design of experiments and analysis of variance (ANOVA).

Course Contents
Scope for statistical methods in physical sciences and engineering. Measures of location, partition and dispersion. Elements of probability. Probability distribution: binomial Poisson, geometric, hypergeometric, negative-binomial, normal poisson, geometric, hypergeometric, negative-binomial, normal, Student's t and chi-square distributions. Estimation (point and internal) and tests of hypotheses concerning population means proportions and variances. regression and correlation. non-parametric tests. contingency table analysis. introduction to design of experiments. analysis of variance.

300 Level
NUC 70% CCMAS

GST 312: Peace and Conflict Resolution (2 Unit C: LH 30)

Learning Outcomes
At the end of the course, students should be able to:
1. analyse the concepts of peace, conflict and security;
2. list major forms, types and root causes of conflict and violence;
3. differentiate between conflict and terrorism;
4. enumerate security and peace building strategies; and
5. describe roles of international organisations, media and traditional institutions in peace building.

Course Contents
Concepts of peace, conflict and security in a multi-ethnic nation. Types and theories of conflicts: ethnic, religious, economic and geo-political conflicts. Structural conflict theory, realist theory of conflict and frustration-aggression conflict theory. Root causes of conflict and violence in Africa: indigene and settlers' phenomenon, boundaries/boarder disputes, political disputes, ethnic disputes and rivalries, economic inequalities, social disputes, nationalist movements and agitations, selected conflict case studies – Tiv-Junkun, Zango Kartaf, chieftaincy and land disputes and many more. peace building, management of conflicts and security, peace & human development. Approaches to peace & conflict management: (religious, government, community leaders and many more.). Elements of peace studies and conflict resolution, conflict dynamics assessment scales: constructive & destructive. Justice and legal framework and concepts of social justice. The Nigeria legal system. Insurgency and terrorism. Peace Mediation and Peace Keeping. Peace and security council (international, national and local levels). Agents of conflict resolution:

conventions, treaties, community policing, evolution and imperatives. Alternative dispute resolution (ADR): a). dialogue b). arbitration c). negotiation d). collaboration and many more Roles of international organizations in conflict resolution: (a). The United Nations (UN) and its conflict resolution organs. (b). The African Union & Peace Security Council (c). ECOWAS in peace keeping. Media and traditional institutions in peace building. Managing post-conflict situations/crisis: refugees. internally displaced persons (IDPs). The role of NGOs in post-conflict situations/crisis.

ENT 312: Venture Creation (2 Units C: LH 15; PH 45)

Learning Outcomes

At the end of this course, students, through case study and practical approaches, would be able to:

1. describe the key steps in venture creation;
2. spot opportunities in problems and in high potential sectors regardless of geographical location;
3. state how original products, ideas, and concepts are developed;
4. develop business concept for further incubation or pitching for funding;
5. identify key sources of entrepreneurial finance;
6. implement the requirements for establishing and managing micro and small enterprises;
7. conduct entrepreneurial marketing and e-commerce; apply a wide variety of emerging technological solutions to entrepreneurship; and
8. appreciate why ventures fail due to lack of planning and poor implementation.

Course Contents

Opportunity Identification (Sources of business opportunities in Nigeria, Environmental scanning, Demand and supply gap/unmet needs/market gaps/Market Research, Unutilised resources, Social and climate conditions and technology adoption gap). New business development (business planning, market research). Entrepreneurial Finance (Venture capital, Equity finance, Micro finance, Personal savings, small business investment organisations and Business plan competition). Entrepreneurial marketing and e-commerce (Principles of marketing, customer acquisition & retention, B2B, C2C and B2C models of e-commerce, First mover advantage, E-commerce business models and successful E-Commerce Companies,). Small business management/family business: Leadership & management, Basic book keeping, Nature of family business and family business growth model. Negotiation and business communication (Strategy and tactics of negotiation/bargaining, traditional and modern business communication methods). Opportunity discovery demonstrations (business idea generation presentations, business idea contest, brainstorming sessions, idea pitching). Technological solutions (The concept of market/customer solution, customer solution and emerging technologies, business applications of new technologies-Artificial Intelligence (AI), Virtual/Mixed Reality (VR), Internet of Things (IoTs), Blockchain, Cloud computing, Renewable energy etc. Digital business and E-Commerce strategies).

MTH 300: Abstract Algebra I (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. state various groups, sub-groups, ring, field and Integral domain; and

2. calculate the H.C.F and L.C.M of Polynomials.

Course Contents
Group definition, examples including permutation groups. Sub-groups, Cosets. Lagrange's theorem and applications. Cyclic groups. Rings: definition and examples including Z, Zn, rings of polynomials and matrices. Integral domains, fields. Polynomial rings, factorisation. Euclidean algorithm for polynomials, H.C.F. and L.C.M. of polynomials.

MTH 301: Metric Space Topology (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1.describe the following: set, metric, open sphere, open set, closed sets, interior, exterior, neighbourhood, connectedness and compactness; and
2. discuss convergence in metric space.

Course Contents
Sets, metrics, and examples. Open spheres (or balls). Open sets and neighbourhoods. Closed sets. Interior, exterior, frontier, limit points and closure of a set. Dense sub-sets and separable space. Convergence in metric space homeomorphisms. Continuity and compactness, connectedness.

MTH 302: Ordinary Differential Equations (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. describe line dependence, Wronskian, reduction order, variation of parameters, series solution about ordinary and regular points; and
2. discuss orthogonal polynomials.

Course Contents
Ordinary differential equations: linear dependence, Wronskian, reduction order, variation of parameters, series solution about ordinary and regular points. Special functions: Gamma, Beta, Bessel, Legendre's theorem, hyper geometric. Laplace transform and applications to initial value problems

MTH 303: Vector and Tensor Analysis (2 Units C: LH 30)

Learning Outcomes
At the end of the course, students should be able to:
1. discuss vector differentiation and vector integration;
2. describe gradient, divergence and curl;
3. discuss Green's, Stoke's and divergence theorems;
4. solve some problems involving applications of vector differentiation and vector integration; and
5. discuss tensor and Cartesian tensor.

Course Contents

Vector differentiation and applications. Gradient, divergence and curl. Vector integration, line, surface and volume integrals, Greens, Stoke's and divergence theorems. Tensor products of vector spaces. Tensor algebra. Symmetry. Cartesian tensors.

MTH 304: Complex Analysis **(2 Units C: LH 30)**

Learning Outcomes

At the end of the course, students should be able to:

1. define functions of complex variable;
2. derive Cauchy-Riemann equations;
3. discuss conformal mapping;
4. solve some problems involving contour integrals, Power and Taylor series of function of a complex variable.

Course Contents

Functions of a complex variable. Limits and continuity of functions of a complex variable. Derivating the Cauchy-Riemann equations. Analytic functions. Bi-linear transformations, conformal mapping, contour integrals. Cauchy's theorems and its main consequences, convergence of sequences and series of functions of a complex variable. Power series. Taylor series.

MTH 305: Complex Analysis II **(2 Units C: LH 30)**

Learning Outcomes

At the end of the course, students should be able to:

1. discuss Laurent expansion, Isolated singularities and Residues; and
2. define Residue theorem and Rouche's theorem

Course Contents

Laurent expansions. Isolated singularities and residues. Residue theorem. Calculus of residue, and application to evaluation of integrals and to summation of series. Maximum modulus principle. Argument principle. Rouche's theorem. The fundamental theorem of algebra. Principle of analytic continuation. Multiple valued functions and Riemann surfaces.

MTH 306: Abstract Algebra II **(2 Units C: LH 30)**

Learning Outcomes

At the end of the course, students should be able to:

1. define normal subgroups and quotient groups;
2. state isomorphism theorem and Cayley's theorems;
3. discuss Sylow theorems; and
4. describe algebraic and transcendental extensions.

Course Contents

Normal subgroups and quotient groups. Monomorphism, isomorphism theorems. Cayley's theorems. Direct products. Groups of small order. Group acting on sets. Sylow theorems. Ideal

and quotient rings. P.I.D. &, U.F.D's Euclides rings. Irreducibility; Field extensions, degree of an extension, and minimum polynomial. Algebraic and transcendental extensions. Straight edged and compass constructions.

MTH 307: Real Analysis II (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. discuss Riemann integral of several functions;
2. define Riemann Stieltjes integral; and
3. describe continuous differentiable functions.

Course Contents

Riemann integral of functions Rn→R, continuous monopositive functions. Functions of bounded variation. The Riemann Stieltjes integral. Pointwise and uniform convergence of sequences and series of functions Rn→R. Effects on limits (sums) when the functions are continuous differentiable or Riemann integrable. Power series.

MTH 308: Introduction to Mathematical Modelling (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. develop some mathematical models in Biology, Physics, Chemistry, and Social Science;
2. describe flow Diagrams;
3. discuss method of Analysis of models formulated; and
4. describe the method of solutions to the models formulated.

Course Contents

Methodology of model building; identification, formulation and solution of problems, cause-effect diagrams. Equation types: algebraic, ordinary differential, partial differential, difference, integral and functional equations. Applications of mathematical models to physical, biological, social and behavioural sciences.

MTH 310: Mathematical Methods II (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. describe Sturm-Liouville problems.
2. discuss orthogonal polynomials and functions.
3. solve some problems using first and second order differential equations

Course Contents

Sturm – Liouville problems. Orthogonal polynomials and functions. Fourier series and integrals. Partial differential equations: general and particular solutions. Linear equations with constant coefficients, first and second order equations, solutions of the heat, wave and Laplace equations by the method of separation of variables. Eigen function expansions. Methods of variation of parameters. Fourier transforms.

MTH 399: Industrial Attachment II (12 Weeks) (3 Units C: PH 135)

Students should be attached to some industrial organizations for additional 12 weeks at their 300 Level preferably during the long vacation for more industrial experience. Students to be assessed based on seminar presentation, their reports and assessment by supervisors.

INSTITUTIONAL 30% CCMAS

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 316: MECHANICS IV (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE
Graduates who are highly skilled and knowledgeable in teaching Mechanics IV is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
This course contains the law of gravitation, Kepler's Laws. Central orbits; momentum and energy equations, determination of the orbits, apses, critical velocity, stability of circular orbit are some emphasizing topics useful in other disciplines.

The course therefore, makes the beneficiary to appreciate and apply the motion near the earth's surface to solve real life problems. The course is both a preparation for further study and practical use.

OBJECTIVES
The objectives of the course are to:
1. define the law of gravitation;
2. illustrate Kepler's Laws.
3. use motion near the earth's surface to demonstrate real life problem.
4. demonstrate principle of stability of circular orbit
5. illustrate rotating and translating frames.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. state the law of gravitation;
2. calculate the momentum and energy equations;
3. describe the motion near the earth's surface;
4. state Coriolis force;
5. describe the Foucault pendulum.

COURSE CONTENTS

The Newton's law of universal gravitation: definition, formula and facts. Kepler's first law of planetary motion: movement of planets in elliptical orbits with the Sun as a focus. Kepler's second law of planetary motion: a planet covers the same area of space in the same amount of time no matter where it is in its orbit. Kepler's third law of planetary motion: a planet's orbital period is proportional to the size of its orbit (its semi-major axis). Central orbits. Angular momentum about a pole. Energy equations. Determination of the orbits, apses, critical velocity, stability of circular orbit. Moving frames of reference; rotating and translating frames. Coriolis force (movement on rotating objects). The Coriolis effects. Earth's rotation and its effects on weather. Motion near the earth's surface. The Foucault pendulum. Definition, prediction and proof of Earth's rotation.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 325: ANALYTICAL DYNAMICS (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Analytical Dynamics is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Analytical dynamics develops Newtonian mechanics to the stage where powerful mathematical techniques can be used to determine the behaviour of many physical systems. The mathematical framework also plays a role in the formulation of modern quantum and relativity theories.

Analytical Dynamics is designed to teach you how mathematics could be used in solving problems in the contemporary Science/Technology and Engineering world. Therefore, the course is structured to expose you to the skills required to attain a level of proficiency in Analytical Dynamics. The course therefore, makes the beneficiary to appreciate and apply Langrange's equations to Holonomic and Non-Holonomic systems to solve real life problems.

OBJECTIVES

The objectives of the course are to:

1. define the term "constraint"
2. state the types of constraint;

3. state the difference between the various types of constraint;
4. apply D'Alambert's Principle and relevant theorems;
5. apply the Lagrange's equations to find the differential equations;
6. describe the Lagrange function of particle(s) moving in a conservation force field;
7. derive Lagrange's equations for holonomic and non-holonomicconstraint respectively;
8. define and explain simple harmonic motion;
9. state the forces causing simple harmonic motion;
10. explain the suspension by an elastic string;
11. describe the conical pendulum.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define the dynamics of a rigid body;
2. define Langrange's equations for Holonomic system;
3. discuss Kinetic and potential energies;
4. define nodes of oscillation;
5. state the difference between Holonomic and Non-Holonomic systems;
6. define degrees of freedom;
7. define the moment of a Force.

COURSE CONTENTS

Euler's dynamical equation for motion of a rigid body. Angular velocity. Angular momentum. Euler angles for transforming between the body-fixed and inertial frames of reference. The dynamics of rotational motion about a body-fixed point under the action of external force. The symmetrical top precessional motion. Force-free motion of a rigid symmetric top. Motion of spinning symmetric top with magnitudes. Degrees of freedom. Langrange's equations for Holonomic. Derivations and Lagrange multipliers for Holonomic constraints. Non-Holonomic systems. Impulsive Motion. Collation of smooth spheres. Small oscillation. Simple pendulum. Reduction of coplanar forces acting on a rigid body to a force and a couple. Moment of a force. The Hamiltonian. The Calculus of Variation. The Earth's gravitational field. Kinetic energy. Potential energy. Normal coordinates in a system of coupled oscillators and initial values. Nodes of oscillation.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS

***MAT 326: ANALYTICAL GEOMETRY (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Analytical Geometry is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Analytic geometry, also called coordinate geometry is a mathematical subject in which algebraic symbolism and methods are used to represent and solve problems in geometry. The importance of analytic geometry is that it establishes a correspondence between geometric curves and algebraic equations.

It is a branch of algebra, a great invention of Descartes and Fermat, which deals with the modelling of some geometrical objects, such as lines, points, curves, and so on. It is a mathematical subject that uses algebraic symbolism and methods to solve the problems. It establishes the correspondence between the algebraic equations and the geometric curves

The topics help students to build solid backgrounds Geometry which are applicable in different areas of Engineering and Physics. Some emphasizing topics useful in other disciplines are; conics, curves and surfaces. The course therefore, makes the beneficiary to appreciate and apply structures geometrically.

OBJECTIVES

The objectives of the course are to:

1. analyze the concepts of Geometry;
2. illustrates the use of Affine spaces ad sunspaces;
3. describe the construction of planar curves;
4. demonstrate projective Geometry;
5. state the use of convex hall.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define Affine spaces;
2. describe polar coordinates, spheres, spherical coordinates; conics;
3. discuss Projective Geometry;
4. define a linear subspace;
5. state the equation of an ellipse, hyperbola and parabola.

COURSE CONTENTS

Affine Geometry. Affine Spaces. Subspaces., Barycenter. Convexity. Convex hull. Affine Transformations. Euclidian Geometry: Planes, Points, Distances, Lines, Angles. Isometries. Shapes: Triangles, Circles, Polar Coordinates, Spheres. Spherical Coordinates. Conics: Cartesian and Polar Equations of ellipses, hyperbolas and parabolas. Curves and Surfaces. Vector Functions, Parametric Representation; Curves $f(x,y) = 0$, Tangents, Conics, Coordinates; Construction (plotting) of planar curves. Collinearity. Critical points. Branches.

Projective Geometry: Projective Spaces. Linear Subspaces. Projective Transformations. Duality, the Dual Projective Space. Projective Collinearity.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 328: DISCRETE MATHEMATICS (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Discrete Mathematics is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

This is an introductory course in discrete mathematics. The goal of this course is tointroduce students to ideas and techniques from discrete mathematics that are widely used in science and engineering. This course teaches the students techniques in how to think logically and mathematically and apply these techniques in solving problems.

To achieve this goal, students will learn logic and proof, sets, functions, as well as algorithms and mathematical reasoning. Key topics involving relations, graphs, trees, and formal languages and computability are covered in this course..The course therefore, makes the beneficiary to appreciate and apply turing machines.

OBJECTIVES

The objectives of the course are to:

1. describe Turing Machines;
2. relate logic and quantification to real life situations;
3. describe program correctness, graphs and its Applications;
4. demonstrate mathematical induction;
5. state the properties of matrices.
6. demonstrate how to use Finite-State Machines.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define Arithmetic Algorithms;
2. describe methods of proof;
3. define finite-state machines;

4. define logic and state its properties;
5. apply Turing Machines;
6. illustrate program information using a graph and state their applications.

COURSE CONTENTS
Propositional Logic. Predicate Logic. Quantification. Methods of Proof. Sets. Arithmetic Algorithms. Growth of Functions. Computational Complexity of Algorithms. Integer properties. Matrices. Mathematical Induction. Recursion. Sequences. Summations. Program Correctness. Graphs and its Applications. Trees and its Applications. Languages and Grammars. Finite-State Machines. Automata and Language Recognition. Turing Machines.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MTH 332: REAL ANALYSIS III (3 Units E: LH 45)**

SENATE APPROVED RELEVANCE
Graduates who are highly skilled and knowledgeable in teaching Real Analysis III is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
Mathematics relies on our ability to be able to solve equations, if not in explicit exact forms, then at least in being able to establish the existence of solutions. To do this requires a knowledge of "Analysis", which in many respects is just Calculus in very general settings. The foundations for this work are commenced in Real Analysis III.

This course deals with studies the behavior of improper integrals, their classifications and real life applications. It covers real numbers, sequences and series of real numbers, real functions and their topological concepts. It also covers the concept of open covering, Heine–Borel and Bolzano–Weierstrass theorems, which will seem mysterious at first.

The topics help students to build solid backgrounds in real analysis which are applicable in Economics, Engineering, Physics and Optimization. The course therefore, makes the students to appreciate and apply real analysis involving improper integrals to solve real life problems.

OBJECTIVES
The objectives of the course are to:

1. describe the topology of the real line;
2. illustrate the discontinuity of Improper Integral.
3. state some methods of proof.
4. describe how switching circuits work in logic;
5. describe open and closed sets topologically;
6. state the Weierstrass intermediate theorem.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. demonstrate skills in constructing rigorous mathematical arguments;
2. apply the theory in the course to solve a variety of problems at an appropriate level of difficulty;
3. define the convergence of an improper integral;
4. apply improper integrals to real life problem;
5. state and apply the Bolzano-Weierstrass theorem;
6. state and apply Heine Borel's theorem;
7. define a improper integral of type I;
8. define a metric space.

COURSE CONTENTS

Switching circuits in Logic. Connectives and validity of arguments. Some methods of proof. Extended real number system. Topology of the real line: open and closed sets. Interior, exterior, frontier. Closure of a set. Limit inferior and Limit superior criterion for convergence. Bolzano-Weierstrass theorem. Heine Borel's theorem. Weierstrass intermediate theorem. Enumerable and non-enumerable sets. Fundamental Theorem of Calculus. Improper integral of type I: Infinite limits of integration. Two infinite limits of integration. Discontinuity of Improper Integral. Convergence and divergence of improper integral. Comparison and quotient tests for improper integrals. Improper integral of type II. Real life applications of improper integral. Introduction to metric space topology. Sequence Spaces.

Pre-requisite: MTH 207

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 334: NUMERICAL ANALYSIS II (3Units C: LH 30, PH 45)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Numerical Analysis II is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in

Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Numerical Analysis II is concerned with approximation methods .Polynomials are very useful in the study of Numerical Analysis. Over the years polynomials have been used as approximation to several functions. Although polynomials are sometimes difficult to solve as equations yet they help in appreciating the value of certain functions. Polynomials are of paramount importance when it comes to approximation theory.

Orthogonal polynomials are of fundamental importance in many branches of mathematics in addition to approximation theory and their applications are numerous but we shall be mainly concerned with two special cases, the Legendre polynomials and the Chebyshev polynomials. More general applications are however easily worked out once the general principles have been understood. Legendre polynomial is known to possess some oscillatory property among which makes it of importance in the field of numerical analysis.

For approximation, or interpolation, of a function defined analytically a method due to Hermite is often useful. The method is superficially related to the spline method but in fact the two methods are very different because fitting of spline involves solving a system of linear equations to obtain numerical values for the second derivatives.

The course content is rich in topics useful in other disciplines Thus, the course is beneficial to those who wish to have a solid foundation in numerical analysis. The course is both a preparation for further study and practical use.

OBJECTIVES

The objectives of the course are to:

1. state the difference between discrete data and continuous function
2. define what orthogonal polynomials are.
3. use the least square methods and familiarize themselves with curve fitting;
4. state Simpson and trapezoidal rules and apply the rules to solve real life problems.
5. state the difference between initial value and boundary value problems in ordinary differential equations.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define simultaneous equations and solve problem using Gaussian elimination method;
2. explain how to fit a curve;
3. define least square method;
4. explain interpolation and approximation;
5. state some numerical differentiation and numerical integration problems; and
6. state some numerical problems in ordinary Differential equations with initial value problems.
7. state the necessary formulae for generating the Legendre polynomials.

COURSE CONTENTS

Solution of simultaneous equations. Direct method. Gaussian elimination method. Gauss Jordan method. L.U. decomposition method. Iterative schemes. Gauss Seidel approximation. Gauss Jacobi approximation. Curve fitting. Least square method. Orthogonal polynomials (Chebyshev and Legendry polynomials). Numerical integration. Simpson rule. Trapezoidal rule. Newton Cote formula. Gaussian quadrature. Initial value problems for ordinary differential equations. Cubic Spine Interpolation. Hermite Approximations.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

IMT 301: Financial Mathematics (2 Units C: LH 30)

Learning Outcomes

At the end of this course, the students should be able to:
1. demonstrate a firm understanding of modern theory of financial mathematics;
2. explain the basic concepts in Financial Mathematics; and
3. conduct transactions concerning annuities, actuarial, statistics and mortality analysis

Course Contents

Various topics in mathematics of finance are covered, including annuities. Actuarial. statistics and mortality analysis. principles and methods of actuarial treatment of statistical data, including the compilation and other rates. exposure to risk formula selection. multiple decrements. history and distinctive features of the principle's actuarial tables in common use. vital statistics, including censuses of births, deaths, marriages, and migration statistics. forecasting rates of mortality. General Theory of Projection.a short account of the population of Nigeria. construction of national life table, sickness and other rates. construction of tables. valuation of liabilities under life policies and special topics. multiple decrement (service) and associated single decrement tables. values of and contribution for sickness benefits. pension benefits, disability benefits and widows and orphans benefits.

MTH 399: Industrial Attachment II (24 Weeks) (6 Units C: PH 270)

Students should be attached to some industrial organizations for additional 24 weeks at their 300 Level preferably the whole of second semester and during the long vacation for more industrial experience. Students to be assessed based on seminar presentation, their reports and assessment by supervisors.

400 Level

NUC 70% CCMAS

MTH 401: Theory of Ordinary Differential Equations (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. define existence and uniqueness theorems;
2. describe Volterra and Fredholm types; and
3. solve some problems of reduction of ordinary differential equations to integral equations.

Course Contents
Existence and uniqueness theorems, dependence of solutions on initial data and parameters. Properties of solutions. Sturm comparison and Sonin-Polya theorems. Linear and non-linear systems. Floquet's theory and stability theory. Integral equations: classification, Volterra and Fredholm types, Neumann series. Fredholm alternative for degenerate Hilbert – Schmidt kernels. Reduction of ordinary differential equations to integral equations. Symmetric kernels, eigenfunction expansion with application.

MTH 402: Theory of Partial Differential Equations (2 Units C: LH 30)

Learning Outcomes
At the end of the course, students should be able to:
1. define first and second order linear equations;
2. describe Cauchy problems; and
3. solve some problems in parabolic, hyperbolic and elliptic equations.

Course Contents
Theory and solutions of first-order and second order linear equations. Classification, characteristics and canonical forms. Cauchy problems. Elliptic equations: Laplace's and Poisson's formulas, properties of harmonic functions. Hyperbolic equations; wave equations, retarded potential: transmission line equation, Riemann method. Parabolic equation: diffusion equation, singularity function, boundary and initial – value problems.

MTH 403: Function Analysis (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. define Hilbert spaces, Banach spaces and vector spaces;
2. describe Banach algebra.

Course Contents
Hilbert Spaces bounded linear functionals, operators on Banach spaces, topological vector spaces, Banach algebra.

MTH 404: Project: (6 Units C: PH 270)
A research project and dissertation to be undertaken on any topic of mathematical interest.

MTH 405: General Topology (2 Units C: LH 30)

Learning Outcomes
At the end of the course, students should be able to:
1. define the following: topological spaces, continuous functions and topological invariants;
2. describe pointwise and uniform convergence.

Course Contents
Topological spaces, definition, open and closed sets neighbourhoods. Coarser, and finer topologies. Basis and sub- basis. Separation axioms, compactness, local compactness, connectedness. Construction of new topological spaces from given ones. Sub-spaces and quotient spaces. Continuous functions, homeomorphisms and topological invariants. Spaces of continuous functions: Pointwise and uniform convergence.

MTH 406: Lebesgue Measure and Integrals (2 Units C: LH 30)

Learning Outcomes
At the end of the course, students should be able to:
1. define Lesbesgue measure; and
2. describe Measurable functions and Integral convergence theorems.

Course Contents
Lebesgue measure; measurable and non-measurable sets. Measurable functions. Lebesgue integral: Integration of non-negative functions, the general integral convergence theorems.

MTH 407: Mathematical Methods (2 Units C: LH 30)

Learning Outcomes
At the end of the course, students should be able to:
1. define calculus of variation;
2. describe isoperimetric problems; and
3. solve some problems using integral transforms like: Laplace, Fourier, Hankel, Mellin, Z, Hilbert transforms.

Course Contents
Calculus of variation. Lagrange's functional and associated density. Necessary condition for a weak relative extremum. Hamilton's principles. Lagrange's equations and geodesic problems. The Du Bois-Raymond equation and corner conditions. Variable end-points and related theorems. Sufficient conditions for a minimum. Isoperimetric problems. Variational integral transforms. Laplace, Fourier and Hankel transforms. Complex variable methods, convolution theorems. Application to solution of differential equations.

MTH 408: Entrepreneurship in Mathematics (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. read, write, execute, run and analyse prevailing programming languages
2. prepare documents using LaTeX
3. engage in web designing, printing technology and other related assignments

Course Contents
Student should be exposed to programming languages and how to write projects using latex, web designing, printing technology and mathematics improvement projects.
INSTITUTIONAL 30% CCMAS

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 415: FLUID MECHANICS I (3 Units C: LH 30))**
SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Fluid Mechanics I is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Fluid Mechanics I is an introductory part of this course. It is concerned with pressure, hydrostatics, and buoyancy; open systems and control volume analysis; mass conservation and momentum conservation for moving fluids; viscous fluid flows, flow through pipes; dimensional analysis; boundary layers, and lift and drag on objects.

Fluid flows are important in many scientific and technological problems including atmospheric and oceanic circulation, energy production by chemical or nuclear combustion in engines and stars, energy utilisation in vehicles, buildings and industrial processes, and biological processes such as the flow of blood. Considerable progress has been made in the mathematical modelling of fluid flows and this has greatly improved our understanding of these problems, but there is still much to discover. This course introduces students to the mathematical description of fluid flows and the solution of some important flow problems. Topics covered are: the mathematical description of fluid flow in terms of Lagrangian and Eulerian coordinates; the derivation of the Navier-Stokes equations from the fundamental physical principles of mass and momentum conservation; use of the stream function, velocity potential and complex potential are introduced to find solutions of the governing equations for inviscid, irrotational flow past bodies and the forces acting on those bodies; analytic and numerical solutions of the Navier-Stokes equation.

OBJECTIVES

The objectives of the course are to:

1. describe Real and Ideal fluid;
2. illustrate motion of fluids particles;
3. solve simplified examples of fluid mechanics,
4. demonstrate Stokes's stream function with applications;
5. apply the acquired knowledge and skills in professional and specialist courses

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define basic terms, values and laws in the areas of fluids properties, statics, kinematics and dynamics of fluids, and hydraulic design of pipes,
2. describe methods of implementing fluid mechanics laws and phenomena while analysing the operational parameters of hydraulic problems, systems and machines,
3. apply tables and diagrams, and equations that define the associated laws,

4. calculate and optimise operational parameters of hydraulic problems, systems and machines,
5. explain the correlation between different operational parameters,
6. select engineering approach to problem solving based on the acquired physics and mathematical knowledge

COURSE CONTENTS

Introduction: Basic concepts of fluid mechanics. Fundamental terms. Physical values. Fluids and their properties. Forces inside fluid. Fluid Statistics: Pascal's law. Euler's equation of fluid statics. Measurement of pressure. Relative statics of fluid constant acceleration, rotation. Forces of hydrostatic pressure. Buoyancy. Flotation. Stability. Fluid Kinematics: Euler and Lagrangian specification of fluid flow. Streamlines. Path lines. Stream surface. Stream tube. Mass/volume flow. Control volume. Fluid Dynamics: Continuity equation. Basic laws of fluid dynamics conservation of mass, conservation of linear momentum, conservation of energy. Ideal fluid flow. Application of Bernoulli's equation. Real fluid flow. Viscosity. Determination of losses. Reynolds experiment. Laminar and turbulent flow. Boundary layer. Velocity profile. Losses in pipes. Frictional losses. Nikuradse experiments. Moody's diagram. Local losses. Coefficients of resistance. Hydraulic design of pipeline: Different approaches in designing the pipeline – pressure drop. Mass/volume low, diameter of pipeline. Graphical view. Energy properties of pumps and hydraulic machines. Dimensional analysis. Theory of similarity. Flow of fluid in open channels. Non- stationary flow and hydraulic shock.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 421: ADVANCED FUNCTIONAL ANALYSIS II (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Advanced Functional Analysis is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Functional Analysis II deals with spaces, points, functions , distances and the relationship between them. It introduces locally convex vector spaces. As an example, the space of test functions and the space of distributions is taught. In a second part of the course, differential calculus in Banach spaces and some elements from nonlinear functional analysis are treated.

This course contains relevant topics which will help the students to build solid background in Functional Analysis. The topics are widely used in other disciplines to describe their spaces, distances and functions. This course will be highly beneficial to those who which to further their career in Pure Mathematics.

OBJECTIVES

The objectives of the course are to:

1. describe distances in selected spaces;
2. apply Wiener-Ikehara and the prime number theorem;
3. apply Fourier inversion formula;
4. demonstrate Fréchet-derivative;
5. state the consequences Pontryagin duality.

LEARNING OUTCOMES

By the end of the course, the student must be able to:

1. formulate the definitions and results of the lectures,
2. apply the concepts learned in class to concrete problems,
3. analyze problems related to the topics treated in the course,
4. choose an appropriate method to solve a given problem,
5. prove some elementary statements about the topics of the course,
6. solve exercises on the topics.

COURSE CONTENTS

Locally convex vector spaces. Test functions and the dual space of distributions. Fréchet-derivative. Implicit function theorem and consequences on Banach spaces. Fixed point theorems. Introduction to degree theory (if time permits). Trace class operators. Spectral Theorem. Baire category theorem. Guelfand's theory of Abelian C* Algebra. Positive definite functions of Bochner's theorem. Fourier Inversion formula. Plancherel's theorem. Pontryagin duality and consequences. Regular Abelian Banach Algebras. Minimal Ideals and Wiener theorem. Wiener-Ikehara and the prime number theorem. Applications of Wiener-Ikehara and the prime number theorem.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 426: NUMERICAL ANALYSIS IV (3 Units E: LH 30, PH 45)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Numerical Analysis IV is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in

Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Numerical Analysis IV concerned with the story of how functions, derivatives, integrals, and differential equations are handled as strings of numbers in the computer. At the heart of numerical analysis is an understanding of approximation of the solution of PDE, classification of PDE and approximation of derivatives by finite difference.

Most scientists and engineers are sooner or later faced with computing tasks that require some knowledge of numerical analysis. This course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable majorly in Engineering. Topics like the Crank Nicolson method. Introduction to finite element method, variation formulations are very important in other fields. This course makes the beneficiary to appreciate Engineering problems.

OBJECTIVES

The objectives of the course are to:

1. Learn how to describe solutions of difference equations;
2. illustrate the classification of PDEs;
3. differentiate the Crank Nicolson and Galerkin methods;
4. demonstrate a simple parabolic differential equation;
5. state explicit form of the difference equation and its convergence.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state the classical finite difference methods.;
2. solve problems on high-order compact difference methods;
3. state the explicit form of the difference equation and its convergence;
4. solve problems on finite difference methods for 1-D parabolic, hyperbolic, and elliptic equations;
5. state the classification of PDEs.

COURSE CONTENTS

Brief overview of PDEs and applicable areas. Models and quick review of methods. Finite difference methods for 1-D parabolic, hyperbolic, and elliptic equations. Classical finite difference methods. Applications finite difference schemes such as Crank-Nicholson, Leap-Frog, Dufort-Frankel, Douglas, Newark, Upwind schemes, Lax-Friedrich, Lax-Wendroff, Box, Beam-Warming to PDE model. Method of Lines for time-dependent problems. Analysis of finite difference methods. High-Order compact difference methods for 1-D problems. Finite difference methods for higher dimensional PDEs. ADI and splitting methods. Peaceman-Rachford, Douglas-Rachford. Mitchell-Fairweather schemes. Mixed Equations. Finite element methods. Basic theory. Variational and weak formulation. Weighted residual methods. Rayleigh-Ritz, collocation and Galerkin methods. Engineer's point of view of finite element methods with application to

Boundary value problems 1-D PDEs. Finite element methods of Laplace (Poisson) and time-dependent PDEs.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 438: NUMERICAL ANALYSIS III (3 Units C: LH 30, PH 45)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Numerical Analysis III is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Numerical Analysis III is the study of algorithms that use numerical approximation (as opposed to symbolic manipulations) for the problems of mathematical analysis. It is the study of numerical methods that attempt at finding approximate solutions of problems rather than the exact ones.

Numerical analysis finds application in all fields of engineering and the physical sciences, and in the 21st century also the life and social sciences, medicine, business and even the arts. Current growth in computing power has enabled the use of more complex numerical analysis, providing detailed and realistic mathematical models in science and engineering. The course is also concerned with existence of solution which include one step schemes and linear multistep methods, their theory of convergence and stability.

OBJECTIVES

The objectives of the course are to:

1. describe the existence of solutions;
2. illustrate one step schemes;
3. outline the difference between one step and multistep methods;
4. demonstrate extrapolation processes;
5. state integral equation and boundary value problems.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define a one step scheme;
2. state the criteria for convergence of one step scheme;
3. state the stability properties of multistep methods.
4. solve integral equation and boundary value problems.
5. apply linear multistep methods.
6. state the formula for Taylor and Runge-Kutta methods.
7. describe problems with singularities.

COURSE CONTENTS

Functions approximations: Tchebyshef polynomials, Rational functions and Pade approximations, least square approximations, min-max approximation. Linear Difference equations with constant coefficients. Numerical solutions of second order linear boundary value problems in ODE (Shooting methods). Approximate Analytic methods for solving BVPs and Integral Equations (Decomposition methods, Variational Iteration Methods, Homotopy Pertubation methods). Solution of Ordinary differential equations. One step methods. Taylor and Runge-Kutta methods, Rooted trees and general order conditions of Runge-Kutta methods, Variable-stepsize methods. Error bound for Runge-Kutta Methods. Multistep methods. Order conditions. Stability, Local Truncation Error. Error estimates. Stability and Weak-Stability theory. Predictor-Corrector methods and their Error Estimates. Stability theory for Predictor-Corrector algorithms. Methods for special problem: stiffness, Oscillatory, Multirate, Problems with singularities, etc. Extrapolation. Geometric integration of Symplectic (Hamiltonian) Reversible systems. General Linear Methods.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 439: NUMBER THEORY (2 Units E: LH 30)**
SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Number Theory is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Number theory is a rich and abstract branch of mathematics which explores the fundamental properties of our number system. It is the study of the set of positive whole numbers which are usually called the set of natural numbers.

Number theory has always fascinated amateurs as well as professional mathematicians. In contrast to other branches of mathematics, many of the problems and theorems of number theory can be understood by laypersons, although solutions to the problems and proofs of the theorems often require a sophisticated mathematical background.

In the mid-20th century, number theory was considered the purest branch of mathematics, with no direct applications to the real world. The advent of digital computers and digital communications revealed that number theory could provide unexpected answers to real-world problems. At the same time, improvements in computer technology enabled number theorists to make remarkable advances in factoring large numbers, determining primes, testing conjectures, and solving numerical problems once considered out of reach.

This course covers broad topics on divisibility, congruence and quadratic residues. This course contains relevant topics which will help the students to build solid background in other fields.

OBJECTIVES

The objectives of the course are to:

1. describe Euclidian Algorithm and Greatest Common Divisor;
2. state elementary properties Chinese theorem, Euler's function, Fermat's little theorem;
3. define the Euler's criterion;
4. demonstrate the Law of quadratic reciprocity;
5. describe infinite continued fractions, Irrational numbers;
6. describe the Jacobi symbol.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define greatest common divisor;
2. state Fermat's little theorem;
3. state the Law of quadratic reciprocity;
4. state the Pythagorean Triples, Pell's equation;
5. illustrate multiplicative arithmetic functions;
6. state the Chinese remainder's theorem;
7. state Morbius inversion formula;
8. illustrate the Pythagorean Triples.

COURSE CONTENTS

Divisibility. Euclidian Algorithm. Greatest Common Divisor. Unique factorization theorem. Least Common Multiple. Congruence and it elementary properties. Euler's function. Fermat's little theorem. Wilson's theorem. The Chinese remainder theorem. Quadratic residues. Euler's criterion. The Jacobi symbol. Gauss lemma. The Law of quadratic reciprocity. Arithmetic functions. Multiplicative arithmetic functions. Perfect numbers. Morbius inversion formula. Simple continued fraction. Infinite continued fractions. Irrational numbers. Approximation to

irrational numbers. Diophantine equations. Linear Diophantine equations. Pythagorean Triples. Pell's equation.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS

***MAT 444: ALGEBRAIC TOPOLOGY (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE
Graduates who are highly skilled and knowledgeable in teaching Algebraic Topology is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
Algebraic topology is a branch of mathematics that uses tools from abstract algebra to study topological spaces. The basic goal is to find algebraic invariants that classify topological spaces up to homeomorphism, though usually most classify up to homotopy equivalence.This course also deals with the singular homology of topological spaces.

The course includes material on the construction of covering maps over locally simply-connected topological spaces. In particular, it was shown that, given any of subgroup of the fundamental group of a locally simply-connected topological space, one can construct a corresponding covering space whose fundamental group is isomorphic to that subgroup.

This course isconcerned with two-dimensional manifolds and fundamental group.This course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable in other fields. This course makes the beneficiary to appreciate the use of covering spaces.

OBJECTIVES
The objectives of the course are to:
1. Learn how to describe two-dimensional manifolds;
2. illustrate the Fundamental Group;
3. state free groups and free product of groups;
4. demonstrate theBrowder fixed point theorem in dimension 2;
5. demonstrate the use of Covering Spaces.
6. describe implicial complexes.
7. define modules.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state two-dimensional manifolds ;
2. state Browder fixed point theorem in dimension 2;
3. explain free groups and free product of groups.
4. define singular homology.
5. define covering maps.
6. describe exact sequences of homology groups.

COURSE CONTENTS

Two-dimensional manifolds. Fundamental Group. Browder fixed point theorem in dimension 2. Free groups and free product of groups. Topological Space. Covering Spaces. Covering Maps and the Monodromy Theorem. Covering Maps and Discontinous Group Actions. Implicial Complexes. Singular homology. CW complexes, Homological algebra, Cohomology, and Poincare duality. Homology Calculations. Homotopies and the Fundamental Group. Modules. Exact Sequences of Homology Groups. The Hopf fibration shows how the three-sphere can be built by a collection of circles arranged like points on a two-sphere.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS

***MAT 445: FLUID MECHANICS II (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Fluid Mechanics II is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Fluid Mechanics II is a survey of principal concepts and methods of fluid dynamic. It is a vast area of Mathematics that contains governing equations of viscous flow, exact solutions, low Reynold's number solutions. Boundary layers, Compressible flows.

This course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable majorly in engineering. This course makes the beneficiary to appreciate engineering problems.

OBJECTIVES

The objectives of the course are to:

1. describe the governing equations of viscous flow;
2. define Reynold's number;
3. interpret Reynold's number solutions;
4. demonstrate compressible flows;
5. illustrate boundary layers;
6. describe the differential equations for conservation of mass.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state fluid kinematics,
2. demonstrate an ideal fluid flow,
3. illustrate basic viscous flows,
4. define laminar and turbulent boundary layers,
5. demonstrate momentum-integral methods,
6. demonstrate adiabatic and isentropic compressible flow,
7. describe normal shock waves,
8. state the governing equations of viscous flow,
9. define momentum,
10. state the boundary layer equations.

COURSE CONTENTS

Governing equations of viscous flow. exact solutions. low Reynold's number solutions. Boundary layers. Compressible flows. Review of the kinematics of the flow field. The differential equations for conservation of mass, momentum and energy. The stream function, vorticity and irrotationality. Simple viscous flow fields. The boundary layer equations and the laminar flat plate boundary layer. Momentum-integral estimates. The turbulent boundary layer. Boundary layers with pressure gradient. Experimental external flows. Review of Thermodynamics and the speed of sound. Compressible adiabatic and isentropic steady flow. Isentropic flow with area change. The normal shock wave. Converging and diverging nozzles. Compressible duct flow with friction. Frictionless duct flow with heat transfer.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***MAT 446: ELASTICITY AND PLASMA DYNAMICS (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Elasticity and Plasma Dynamics is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

This course deals with Elasticity and Plasma Dynamics. Elasticity is the tendency of solid objects and materials to return to their original shape after the external forces (load) causing a deformation are removed. Physical reasons for elastic behavior vary among materials and depend on the microscopic structure of the material.

This also course deals with the study of plasmas and particular emphasis on the orbital theory of plasmas and on magneto hydrodynamics (MHD) will be devoted. The course also treats some basic concepts on Elasticity together with some applications to Physics and Engineering.

This course describes the analysis of stress and strain, the elastic solid, Saint-Venant principle, generalized Hooke's law and two-dimensional elasticity. The course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable in other fields.

OBJECTIVES

The objectives of the course are to:

1. Learn how to describe Saint-Venant Principle in elastic solid;
2. illustrate the generalized Hooke's law;
3. state two-dimensional elasticity methods;
4. demonstrate the dynamics of plasma, motion of a particle in a uniform magnetic field;
5. demonstrate physical basis of relativity theory.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. describe the physical basis of relativity theory and its mathematical aids;
2. state Generalized Hooke's law;
3. state Saint-Venant principle;
4. explain the relativity electrodynamics of empty space.
5. state a number of phenomena associated with the behaviour of neutral fluids;
6. describe the behaviour of plasmas and some of the phenomena associated with them.
7. describe tensions and deformations in a solid and to relate them.

COURSE CONTENTS

Analysis of deformations. Displacement vector. Infinitesimal transformations (commutativity). Rotation (antissi metric part of the strain tensor). Values and eigenvectors of the deformation tensor. Relative variation on length, area and volume. Trace and invariants. Saint-Venant principle and the strain ellipsoid. Analysis of tensions. Equations of balance and motion. The stress tensor. Values and principal directions of tension. The Mohr diagram and its application. The quadric of

tensions. Utility; calculation of stress and its components. Extremes.Hooke's generalized law. The elastic tensor coefficients as a 4th order tensor. Isotropic media. Lamé parameters. Generalized Hooke's law for an isotropic media. Calculation of deformations given the tensions and vice-versa. Relations between Young modules, Poisson's ratio and compressibility and Lamé parameters. Energy density of elastic deformation. elástica. Clapeyron Theorem. Plasma orbital theory, dynamics of multiple charged particles and processes without collisions in plasmas. Approaches used in the theory of plasmas Orbits of motion of particles in a plasma. Effect of a perpendicular force, gradient and curvature. Larmor radius and frequency. Magnetic mirrors. Particle acceleration in astrophysics. Van Allen Belt. Cosmic rays. Basic properties of plasmas. Different types of plasma. Electrical neutrality in a plasma. Shielding and the Debye length. The plasma parameter. Oscillations in plasmas. The plasma frequency. Electromagnetic waves in hot and cold plasmas. Magneto hydrodynamics (MHD) Fundamental equations. The plasma equations. The equation of induction and its consequences. Magneto hydrostatics. Applications to Fusion and Tokamaks. Applications to the solar atmosphere. Magneto hydrodynamic waves. Alfvén waves and magneto acoustic waves. Magneto convection and sunspots. The buoyancy of magnetic flux tubes. The Parker instability of the magnetic field. The magnetic field as a carrier of angular momentum, magnetic braking and magnetized winds.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS

***MAT 448: ELECTROMAGNETIC THEORY (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Electromagnetic Theory is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Electromagnetic theory is primarily concerned with the magneto static field, equation of electromagnetic field, radiation. The course covers static and dynamic electric and magnetic fields and their interaction. Major topics include Electromagnetic Waves, Transmission Lines, Waveguides, and Antenna fundamentals. This course establish a correlation between the theoretical foundation of electromagnetism and its multifaceted practical application as well as its permeating influence on virtually everything we do every day of our lives.

This course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable in other fields. This course makes the beneficiary to appreciate the use of vector-field theory.

OBJECTIVES

The objectives of the course are to:

1. state the four Maxwell's equations of electromagnetism.
2. describe the Lorentz force law
3. explain why parallel currents attract and why anti-parallel currents repel.
4. relate Maxwell's equations to the electromagnetic properties of a material.
5. analyse each of Maxwell's equations in detail.
6. explain the continuity equation.
7. distinguish between rotation-free vector field and source-free vector field.
8. establish the relationship between electromagnetic wave and light
9. state the wave equation for electric field vector.
10. describe the wave equation for the magnetic field vector.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. apply vector calculus operations.
2. develop a knowledge of vector fields and scalar fields.
3. describe the fundamental nature of static fields, including steady current, static electric and magnetic fields.
4. apply Maxwell's equations and their application to time-harmonic fields, boundary conditions, wave equations, and Poynting's power-balance theorem.
5. describe the properties of plane waves in unbounded space, and understand such concepts as wavelength, phase velocity, and attenuation.
6. solve problems involving lossless transmission lines with time-harmonic excitation.

COURSE CONTENTS

Introduction to Maxwell's Equations. Force, energy and power. Guided waves. Electromagnetic Wave. Equation and Theory of Light. Lights as Transverse Waves. Reflections and Refraction of Plane. Boundary of Plane Waves. Energy Theorem in Maxwell's Theory. Radiation from Extended Sources Review of vector-field theory. Review of electric field theory. The magneto static field. Equation of electromagnetic field. Radiation. Ttransient and time-harmonic transmission lines. Smith charts.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS

B. Sc. MATHEMATICS

***MAT 464: DIFFERENTIAL GEOMETRY (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching Differential Geometry is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

This course is concerned with curves in R^3, Arclength, Curvature, Torsion, Frenet-Serret equations, surfaces in Space, Parametrized surfaces, First and Second Fundamental Forms, Gauss and Codazzi equations.

This course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable in other fields. Topics like Manifolds, Fibers, Tangent and cotangent Bundles are needed and applicable in Engineering. This course makes the beneficiary to appreciate the use of Curves in R^3.

OBJECTIVES

The objectives of the course are to:

1. define differential geometry,
2. state connection and significance of differential geometry to other areas of mathematics.
3. illustrate many of the basic concepts and tools of multivariable calculus and linear algebra to the contexts of calculus on Riemannian geometry,
4. develop and demonstrate a level of expertise in mathematical reasoning appropriate to a challenging upper-level mathematics course.
5. describe parametrized surfaces.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. describe Curves in R^3, Arclength, Curvature, Torsion;
2. state first and second fundamental forms, Gauss and Codazzi equations;
3. explain geometry of surfaces, Geodesics, Gauss-Bonnet Theorem.
4. demonstrate the Frenet-Serret Equations;
5. demonstrate the use of Tangent and cotangent Bundles.

COURSE CONTENTS

Historical background. Motivation and applications. Index notation and summation convention. Space curves. The tangent vector field. Parameterization. Curves in R^3. Arc length. Curvature. Torsion, Frenet-Serret Equations. Principal normal. Binormal, Torsion, Osculating, Normaland Rectifying planes. Frenet-Serret Theorem. Spherical images. Sphere curves. Spherical contacts. Fundamental theorem of space curves. Line integrals and Green's theorem. Local surface theory. Coordinate transformations. Tangent and the Normal planes. Parametric curves. First fundamental form and the metric tensor. Normal and geodesic curvatures. Gauss's

formulae. Christoffel symbols of first and second kinds. Parallel vector fields along a curve and parallelism. Second fundamental form and the Weingarten map. Principal, Gaussian, Mean and Normal curvatures, Dupin indicatrices. Conjugate and asymptotic directions. Isometries and the fundamental theorem of surfaces .Euler's Theorem. Geometry of Surfaces, Geodesics, Gauss-Bonnet Theorem, Hyperbolic Geometry. Manifolds, Fibers, Tangent and cotangent Bundles.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS

***IMT 429: INTRODUCTION TO STOCHASTIC CALCULUS FOR FINANCE (3 Units E: LH 45)**

SENATE APPROVED RELEVANCE
Graduates who are highly skilled and knowledgeable in teaching Introduction to Stochastic Calculus for Finance is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education

OVERVIEW
Stochastic calculus is a branch of mathematics that operates on stochastic processes. It allows a consistent theory of integration to be defined for integrals of stochastic processes with respect to stochastic processes. This field was created and started by the Japanese mathematician Kiyoshi Itô during World War II.

This course deals with processes containing a stochastic component and thus allows the modeling of random systems. Many stochastic processes are based on functions which are continuous, but nowhere differentiable. This rules out differential equations that require the use of derivative terms, since they are unable to be defined on non-smooth functions. Instead, a theory of integration is required where integral equations do not need the direct definition of derivative terms. In quantitative finance, the theory is known as Ito Calculus.

The main use of stochastic calculus in finance is through modeling the random motion of an asset price in the Black-Scholes model. The physical process of Brownian motion (in particular, a geometric Brownian motion) is used as a model of asset prices, through the Weiner Process. This

process is represented by a stochastic differential equation, which despite its name is in fact an integral equation.

OBJECTIVES

The objectives of the course are to:

1 learn how to construct Brownian motion as a continuous martingale;
2. state the Martingale convergence theorem;
3. perform Integrals with respect to a random walk;
4. state the multivariate normal distribution;
5. describe limits of sums of independent variables.
6. illustrate multivariate normal distribution.

LEARNING OUTCOMES

At the end of this course, the student should be able to:

1. define a Martingale, its properties and applications.
2. define a Brownian motion, and its properties.
3. construct a Brownian motion and apply it.
4. perform simple integration of simple and continuous processes, and Ito's formula.
5. define the Markov chain.
6. define Riemann integral,
7. state Ito's formula.

COURSE CONTENTS

Martingale conditional expectation. Martingale optional sampling theorem. Martingale convergence theorem. Square integrable martingales. Integrals with respect to a random walk. Brownian motion. Limits of sums of independent variables. Multivariate normal distribution. Limits of random walks. Construction of Brownian motion. Brownian motion as a continuous martingale. Brownian motion as a Markov. Brownian motion as a Gaussian. Computations for Brownian motion. Quadratic variation. Expected value at a future time. Stochastic integration. Understanding the concept of stochastic calculus. Review of Riemann integration. Integration of simple processes. Integration of continuous processes. Ito's formula.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. MATHEMATICS
***STA 423: STOCHASTIC PROCESSES (3 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduates who are highly skilled and knowledgeable in teaching stochastic processes is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of Academy, Industry, community and fostering development in education.

OVERVIEW

A stochastic process is a set of random variables indexed by time or space. Stochastic modelling is an interesting and challenging area of probability and statistics that is widely used in the applied sciences. In this course the students will gain the theoretical knowledge and practical skills necessary for the analysis of stochastic systems. The basic concepts of the theory of stochastic processes and explore different types of stochastic processes including Markov chains, Poisson processes and birth and death processes will be studied.

Stochastic processes are widely used as mathematical models of systems and phenomena that appear to vary in a random manner. Examples include the growth of a bacterial population, an electrical current fluctuating due to thermal noise, or the movement of a gas molecule. Stochastic processes have applications in many disciplines such as Biology, Chemistry, Ecology, neuroscience, physics, computer science, signal processing, image processing and telecommunication. Furthermore, seemingly random changes in financial markets have motivated the extensive use of stochastic processes in finance.

OBJECTIVES

The objectives of the course are to:

1. describe the defining properties of various stochastic process models, especially for Markov processes.
2. explain standard concepts and methods of stochastic modeling
3. Identify the best stochastic process for specific situations
4. apply stochastic analysis to realistic problems
5. recognize Markov chain process and apply it to the Birth and Death processes.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state three properties of various stochastic process models, specifically for Markov processes.
2. compute probabilities of transition between states and return to the initial state after long time intervals in Markov chains.
3. apply the most appropriate stochastic process technique for a given applied problem.
4. solve stochastic process problems mathematically and using software.
5. define Markov chain process and apply it to the Birth and Death processes.

COURSE CONTENTS
Discrete and continuous time Markov chains. General Random Walk. Simple Random Walk. Random walk: Gamblers ruin problem. gambler's ruin and successful runs. Chapman-Kolmogorov Equation. Classification of States. Long run behavior. Limit Theorems, Branching processes. Poisson processes. Counting Process. Interarrival and Waiting Time Distributions. Conditional Distribution of Arrival Times. Birth-and-death processes. Queuing processes: M/M/I, M/M/s, M/a/I queues and their waiting time distributions. Renewal Process. Some Limits Theorems. Wald's Equation. Elementary Renewal Theorem, Regenerative Process. Symmetric Random Walk and Arc Sine Laws. Introduction to Markov Chain Monte Carlo (MCMC) methods.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

IMT 404: OPTIMIZATION THEORY (2 Units E: LH 30)

LEARNING OUTCOMES
At the end of the of this course, students should be able:
1. develop a fundamental understanding of the role of optimization technique;
2. describe and explain complex systems ;
3. enumerate the theoretical and computational tools needed to solve equations that arise as models.

COURSE CONTENTS
Linear programming models. the simplex method, formulation and theory. quality integer programming. transportation problem. Two-person zero-sum games. nonlinear programming. quadratic programming Kuhn-tucker methods. optimality criteria. simple variable optimization. multivariable techniques. gradient methods.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

NUC 70% CCMAS

Minimum Academic Standards
Staffing
Academic Staff
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply. To start any programme in science, there should be a minimum of six academic staff. There is need to have a reasonable number of staff with PhD degrees accounting for at least 70% of the total number and having adequate teaching experience for every programme in the discipline.

Administrative support staff
The services of the administrative support staff are indispensable in the proper administration of departments and faculty offices. It is important to recruit very competent, computer literate senior staff.

Technical support personnel
The services of technical support staff, which are indispensable in the proper running of laboratories and workshops, are required. It is important to recruit very competent senior technical staff to maintain teaching and research equipment. They are also to undergo regular training developments in equipment operation and maintenance.

Classrooms, Laboratories, Workshops and Office Space
The NUC recommends the following physical space requirements:

Description		**Size m^2**
Professor's Office	-	18.50
Head of Department's Office	-	18.50
Tutorial Teaching Staff's Office	-	13.50
Other Teaching Staff Space	-	7.00
Technical Staff Space	-	7.00
Secretarial Space	-	7.00
Research Laboratory	-	16.50
Seminar Space/per student	-	1.85
Laboratory Space per FTE	-	7.50
Conference Room	-	37.0

Adequate space should be provided for all departments in the sciences. Effort must be made to provide each department with at least:

1. Four (4) large laboratories calculated according to specifications of 7.5 m2 per FTE; a minimum of four (4) preparatory rooms for each laboratory at the NUC specifications of 7 m2 each.
2. At least two lecture rooms capable of sitting at least sixty students at the specification of 1 m2 per FTE.
3. A departmental conference room.
4. A staff common room.

Library and Information Resources
Universities should leverage on available technology to put in place rich databases and other electronic/digital library and information resources. In addition, well stock and current hardcopies of reference and other textual materials should be provided centrally at the level of the faculty.
A well network digital library should serve the entire university community. Availability of wireless facilities (Wifi) with adequate bandwidth should enhance access to these electronic resources.
In any case, there should be internet-ready workstations available in the library for at least 25% of the total students enrolled in each academic programme. The funding of the library should be in line with NUC guidelines.

Equipment

S/N	Description	Quantity Required
1	Computers	60
2	Computer Software	6
3	Flash Drive	10
4	Printers	10
5	Projectors	10
6	Smart Board	5
7	Smart TV	5
8	Video Machine	5
9	Screen Board	10
10	Pen Drive	10
11	Loud Speakers with Microphone	5

Equipment S/N	Description	Quantity Required
1	Computers	60
2	Computer software	6
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4	Printers	10
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7	Smart TV	5
8	Video machine	5
9	Screen board	10
10	Pen drive	10
11	Loud speaker with microphones	5

INDUSTRIAL MATHEMATICS COURSE CONTENTS AND LEARNING OUTCOMES

200 LEVEL

70% CCMAS COURSE CONTENT

GST 212: Philosophy, Logic and Human Existence (2 Units C: LH 30)

Learning Outcomes

A student who has successfully gone through this course should be able to:

1. know the basic features of philosophy as an academic discipline;
2. identify the main branches of philosophy & the centrality of logic in philosophical discourse;
3. know the elementary rules of reasoning;
4. distinguish between valid and invalid arguments;
5. think critically and assess arguments in texts, conversations and day-to-day discussions;
6. critically asses the rationality or otherwise of human conduct under different existential conditions;
7. develop the capacity to extrapolate and deploy expertise in logic to other areas of knowledge, and
8. guide his or her actions, using the knowledge and expertise acquired in philosophy and logic.

Course Contents

Scope of philosophy; notions, meanings, branches and problems of philosophy. Logic as an indispensable tool of philosophy. Elements of syllogism, symbolic logic— the first nine rules of inference. Informal fallacies, laws of thought, nature of arguments. Valid and invalid arguments, logic of form and logic of content — deduction, induction and inferences. Creative and critical thinking. Impact of philosophy on human existence. Philosophy and politics, philosophy and human conduct, philosophy and religion, philosophy and human values, philosophy and character molding, etc.

ENT 211: Entrepreneurship and innovation (2 Units C: LH 15; PH 45)

Learning Outcomes

At the end of this course, students should be able to

1. explain the concepts and theories of entrepreneurship, intrapreneurship, opportunity seeking, new value creation, and risk-taking state the characteristics of an entrepreneur
2. analyse the importance of micro and small businesses in wealth creation, employment, and financial independence engage in entrepreneurial thinking;
3. identify key elements in innovation; describe stages in enterprise formation, partnership and networking including business planning;
4. describe contemporary entrepreneurial issues in Nigeria, Africa and the rest of the world; and
5. state the basic principles of e-commerce.

Course Contents

Concept of entrepreneurship (entrepreneurship, intrapreneurship/corporate entrepreneurship). Theories, rationale and relevance of entrepreneurship (Schumpeterian and other perspectives,

risk-taking, necessity and opportunity-based entrepreneurship and creative destruction). Characteristics of entrepreneurs (opportunity seeker, risk taker, natural and nurtured, problem solver and change agent, innovator and creative thinker). Entrepreneurial thinking (critical thinking, reflective thinking, and creative thinking). Innovation (concept of innovation, dimensions of innovation, change and innovation, knowledge and innovation); enterprise formation, partnership and networking (basics of business plan, forms of business ownership, business registration and forming alliances and join ventures). Contemporary entrepreneurship Issues (knowledge, skills and technology, intellectual property, virtual office, networking). Entrepreneurship in Nigeria (biography of inspirational entrepreneurs, youth and women entrepreneurship, entrepreneurship support institutions, youth enterprise networks and environmental and cultural barriers to entrepreneurship); basic principles of e-commerce.

COS 201: Computer Programming I (3 Units C: LH 30; PH 45)

Learning Outcomes

At the end of this course, students should be able to:

1. explain the principles of good programming and structured programming concepts;
2. explain the programming constructs, syntax and semantics of a higher-level language;
3. describe the chosen programming language variables, types, expressions, statements and assignment; simple input and output;
4. describe the programme control structures, functions and parameter passing, and structured decomposition; and
5. develop simple programmes in the taught programming language as well as debug and test them.

Course Contents

Introduction to computer programming. Functional programming; Declarative programming; Logic programming; Scripting languages. Introduction to object-orientation as a technique for modelling computation. structured, and even some level of functional programming principles; Introduction of a typical object-oriented language, such as Java; Basic data types, variables, expressions, assignment statements and operators; Basic object-oriented concepts: abstraction; objects; classes; methods; parameter passing; encapsulation. Class hierarchies and programme organisation using packages/namespaces; Use of API – use of iterators/enumerators, List, Stack, Queue from API; Searching; sorting; Recursive algorithms; Event-driven programming: event-handling methods; event propagation; exception handling. Introduction to Strings and string processing; Simple I/O; control structures; Arrays; Simple recursive algorithms; inheritance; polymorphism.

Lab work: Programming assignments; design and implementation of simple algorithms, e.g., average, standard deviation, searching and sorting; Developing and tracing simple recursive algorithms. Inheritance and polymorphism.

COS 202: Computer Programming II (3 Units C: LH 30; PH 45)

Learning Outcomes

At the end of this course, students should be able to:

1. demonstrate the principles of good programming and structured programming concepts;

2. demonstrate string processing, internal searching, sorting, and recursion;
3. demonstrate the basic use of OOP concepts: classes, objects, inheritance, polymorphism, data
abstraction;
4. apply the tools for developing, compiling, interpreting and debugging programmes; and
5. demonstrate the use of syntax and data objects, operators. Central flow constructs, objects and
classes programming, Arrays, methods, Exceptions, Applets and the Abstract, OLE, Persistence,
Window Toolkit.

Course Contents

Review and coverage of advanced object-oriented programming - polymorphism, abstract classes and
interfaces. Class hierarchies and programme organisation using packages/namespaces. Use of API use of iterators/enumerators. List. Stack. Queue from API. Searching. Sorting. Recursive algorithms. Event-driven programming: event-handling methods; event propagation; exception handling. Applications in Graphical User Interface Lab work: Programming assignments leading to extensive practice in problem-solving and programme development with emphasis on object-orientation. Solving basic problems using static and dynamic data structures. Solving various searching and sorting algorithms using iterative and recursive approaches. GUI programming.

MTH 201: Mathematical Methods 1 (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:
1. explain real-valued functions of a real variable;
2. solve some problems using mean value theorem and Taylor series expansion; and
3. evaluate line integral, surface integral and volume integrals.

Course Contents

Real-valued functions of a real variable. Review of differentiation and integration and their applications. Mean value theorem. Taylor series. Real-valued functions of two and three variables. Partial derivatives chain rule, extrema, lagrangian multipliers. Increments, differentials and linear approximations. Evaluation of line integrals. Multiple integrals.

MTH 202: Elementary Differential Equations (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:
1. define the following: order and degree of a differential equation;
2. describe some techniques for solving first and second order linear and non-linear equations; and
3. solve some problems related to geometry and physics.

Course Contents

Derivation of differential equations from primitive geometry, physics etc. Order and degree of differential equation. Techniques for solving first and second order linear and non-linear equations. Solutions of systems of first order linear equations. Finite linear differential equations. Application to geometry and physics.

MTH 203: Sets, Logic and Algebra I (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. solve various problems using the concept of set theory;
2. recognise Algebraic structures; and
3. describe the meaning of logic in Mathematics.

Course Contents

Introduction to the language and concepts of modern mathematics. Topics include: basic set theory: mappings, relations, equivalence and other relations, Cartesian products, binary logic, methods of proof, binary operations. Algebraic structures, semi-groups, rings, integral domains, fields. Homeomorphics. Number systems; properties of integers, rationals, real and complex numbers.

MTH 204: Linear Algebra I (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. explain vector space;
2. describe linear transformations and their representation by matrices; and
3. calculate algebra of various matrices.

Course Contents

Vector space over the real field. Sub-spaces, linear independence, basis and dimension. Linear transformations and their representation by matrices – rings, null space, rank. Singular and non-singular transformation and matrices. Algebra of matrices.

MTH 205: Linear Algebra II (1 Unit C: LH 15)

Learning Outcomes

At the end of the course, students should be able to:

1. recognise systems of linear equations.
2. calculate the Eigen values and Eigen vectors.
3. describe the Cayley-Hamilton theorem and its uses.

Course Contents

Systems of linear equation, change of basis, equivalence and similarity. Eigen values and Eigen vectors. Minimum and characteristic polynomials of a linear transformation (matrix). Cayley-Hamilton theorem. Bi-linear and quadratic forms, orthogonal diagonalisation. Canonical forms.

MTH 207: Real Analysis I (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. describe convergence of sequence of numbers;

2. discuss the monotone, cauchy sequences;
3. test for convergence of series; and
4. state roles and mean value theorem.

Course Contents
Bounds of real numbers, convergence of sequence of numbers. Monotone sequences, the theorem of nested intervals. Cauchy sequences, tests for convergence of series. Absolute and conditional convergence of series and re-arrangements. Completeness of reals and incompleteness of rationals. Continuity/and differentiability of functions. Rolles' mean and value theorems for differentiable functions, Taylor series.

MTH 209: Introduction to Numerical Analysis (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. solve some numerical solution of algebraic and transcendental equations;
2. describe curve fitting;
3. discuss error analysis;
4. calculate interpolation and approximation;
5. solve some numerical differentiation and numerical integration problems; and
6. solve some numerical problems in ordinary Differential equations with initial value problems;

Course Contents
Solution of algebraic and transcendental equations. Curve fitting. Error analysis. Interpolation and approximation. Zeros of non-linear equations 'in one variable'. Systems of linear equations. Numerical differentiation and integration. Initial value problems in ordinary differential equation.

MTH 210: Vector Analysis (1 Unit C: LH 15)
Learning Outcomes
At the end of the course, students should be able to:
1. describe vector algebra;
2. explain geometrical equation of lines and planes; and
3. outline problems in gradients, divergent and curl.

Course Contents
Elementary vector algebra, vector and vector triple, vector products (more application solution of vector equation, plain curves and space curves. Geometrical equation of lines and planes. Linear independence of vectors; components of vectors, direction cosines; position vector and scaler products; Frenet-Serret formulas; differential definition of gradients, divergent and simple multiplication)

IMT 206: Abstract Algebra (2 Units C: LH 30)
Learning Outcomes
At the end of this course, the students should be able to:
1. state various group, subgroup, ring, field and Integral Domain;
2. calculate the H.C.F and L.C.M of Polynomials;

3. elucidate the concept of groups, subgroups, normal subgroup, rings, fields;
4. apply the concept of groups, subgroups, normal subgroup, rings, fields to integral domain, sub-domain, direct sum and linear transformations.

Course Contents
Introduction to the language and concepts of Modern Mathematics: Basic Sets Theory, mappings, equivalence and other relations, Cartesian Products, Binary Logic, Method of Proof, Binary Operations, Algebraic Structures, Semi-groups, Rings, Integral Domain Fields, Homomorphism, Number Systems, Properties of Integers, Rational, Real and Complex Numbers; Euclidean algorithm for polynomials H.C.F. and L.C.M. of polynomials..

INSTITUTIONAL 30% CCMAS

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 215: HISTORY OF MATHEMATICS (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE
Graduate who are highly skilled and knowledgeable in teaching history of Mathematics is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
This course covers introductory and in-depth history of Mathematics starting from history of early Egyptian, Greek and Japanese Mathematics, emphasizing topics useful in other disciplines.

The course therefore, makes the beneficiary to appreciate and apply these foundation knowledge like the history of European Medieval mathematician from 500 AD and describe the development of plane pure geometry. The course is both helpful for further studies and practical use.

OBJECTIVES
The objectives of the course are to:
1. state the relevance of history of Mathematics;
2. demonstrate the Greek's method of counting numbers.
3. perform similar calculations using the Egyptian .
4. narrate the discovery of European Medieval mathematicians from 500 AD.
5. apply the Pythagorean principle to solve real life problems.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. discuss the contribution of mathematicians to the development of calculus.

2. explain the history of European Mathematics from 1500 AD - 1700 and the development of non-Euclidean and modern geometry.
3. discuss the History of European Medieval mathematician from 500 AD; and
4. describe the development of plane pure geometry;
5. state some of the authors that made these history.

COURSE CONTENTS

Early Number Systems and Symbols. Primitive counting. Number recording of the Egyptians and Greeks, and the Babylonians. Mathematics in early civilizations. The Rhind Papyrus. Egyptian arithmetic. Egyptian geometry. Babylonian mathematics. Plimpton. The Beginnings of Greek Mathematics. The geometrical discoveries of Thales. Pythagorean Mathematics. Construction Problems of Antiquity. The quadratrix of Hippias. Archimedes. The Alexandrian School Euclid. Euclid and the Elements. Euclidean Geometry. Euclid's Number Theory. The Twilight of Greek Mathematics: Diophantus. The Decline of Alexandrian Mathematics. The Arithmetica. Diophantine Equations in Greece, India, and China. Mathematics in the Near and Far East. The First Awakening: Fibonacci. The Decline and Revival of Learning. The Liber Abaci and Liber Quadratorum. The Fibonacci Sequence. Fibonacci and the Pythagorean Number Triples. The Renaissance of Mathematics: Cardan and Tartaglia. Europe in the Fourteenth and Fifteenth Centuries. The Battle of the Scholars. Cardan's Ars Magna. Ferrari's Solution of the Quartic Equation.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 218: ELECTRICITY AND MAGNETISM (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Electricity ad Magnetism is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Electricity is basically the presence and motion of charged particles. On the other hand, magnetism refers to the force which the magnets exert when they attract or repel each other. A current gives rise to a magnetic field, and a magnetic field gives rise to a force on a current.

This course is an introduction to Electricity and Magnetism and their mathematical description connecting electric and magnetic phenomena via the special theory of relativity. Topic includes

electrostatics, magnetic fields, electromagnetic induction, DC and AC circuits, and the electromagnetic properties of matter. The general topics included in this course are the principles and application of the said subject.

OBJECTIVES

The objectives of the course are to:

1. develop a basic understanding of electric and magnetic fields in free space using the integral forms of Maxwell's laws;
2. apply the knowledge of electricity and magnetism to explain natural and physical processes and related technological advances;
3. use an understanding of calculus along with physical principles to effectively solve problems encountered in everyday life, further study in Science, and in the professional world;
4. perform basic calculations using Coulomb's law and Gauss's law;
5. outline the significance of Maxwell's equations in integral form.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. describe the electric field and potential, and related concepts, for stationary charges;
2. describe and determine the magnetic field for steady and moving charges;
3. determine the magnetic properties of simple current distributions using Biot-Savart and Ampere's law;
4. describe electromagnetic induction and related concepts and make calculations using Faraday and Lenz's laws;
5. explain the basic physical significance of Maxwell's equations in integral form;
6. evaluate DC circuits to determine the electrical parameters; and
7. determine the characteristics of AC voltages and currents in resistors, capacitors, and Inductors.
8. develop a conception of space and time adequate for understanding electrodynamics.
9. develop and train problem-solving skills.

COURSE CONTENTS

Forces in nature. Electrostatics (electric charge and its properties, methods of charging). Coulomb's law and superposition. Electric field and potential. Gauss's law. Capacitance. Electric dipoles. Energy in electric fields. Conductors and insulators. DC circuits (current, voltage and resistance). Ohm's law. Resistor combinations. Analysis of DC circuits. Magnetic fields. Lorentz force. Biot-Savart and Ampère's laws. Magnetic dipoles. Dielectrics. Energy in magnetic fields. Electromotive force. Electromagnetic induction. Self and mutual inductances. Faraday and Lenz's laws. Step up and step down transformers. Maxwell's equations in vacuum using vector differential and integral calculus. Electromagnetic oscillations and waves. AC voltages and currents applied to inductors, capacitors, and resistance. Relativity. Optics. Wave guides and Transmission lines. Radiation.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 224: MECHANICS III (3 Units C: LH 45)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Mechanics II is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

The aim of the course is to enhance students' knowledge and understanding of the mathematics and scientific principles related to mechanics and materials, and to develop their ability to apply this knowledge in a number of topics.

This course covers motion in 2-dimensions which include the use of polar coordinates, motion in a circle, emphasizing topics useful in other disciplines. The course is thus beneficial to those who wish to have a solid background in Mechanics. The course is both a preparation for further study and practical use.

OBJECTIVES

The objectives of the course are to:

1. demonstrate the motion in 2-dimensions;
2. describe constrained motion in a vertical circle and on a sphere;
3. use the simple harmonic motion to demonstrate real life problems;
4. illustrate the vertical motion under gravity, with resistance proportional to speed and to square of speed and apply them to appropriate situations.
5. apply projectiles in resisting medium to solving real life problems

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define the canonical pendulum;
2. describe a simple harmonic motion;
3. describe simple harmonic motion problems;
4. solve some problems the dynamics of rigid body;
5. state D'Alembert's principle.

COURSE CONTENTS

Motion in 2-dimensions. Use of polar coordinates. Motion in a circle. The conical pendulum. The governor. Constrained motion in a vertical circle and on a sphere. Simple Harmonic Motion.

Amplitude, period, frequency, the simple pendulum, elastic strings and springs. Second pendulum. Motion in a resisting medium. Vertical motion under gravity, with resistance proportional to speed and to square of speed. Damped, harmonic motion. Projectiles in resisting medium. Variable mass motion. Dynamics of a rigid body. Product of inertia,. Principal axes. The momenta ellipse. Virtual work. D'Alembert's principle.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 229: INTRODUCTION TO SCIENTIFIC COMPUTING (3 Units C: LH 30, PH 45)**

SENATE APPROVED RELEVANCE
Graduates who are highly skilled and knowledgeable in teaching Introduction to Scientific Computing is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
Scientific Computing is the study of algorithms and their implementations for computations involving mathematical symbols as well as numbers. The course involves developing and applying a wide variety of scientific computing methods to many areas including physics and astronomy.

This course covers introduction to computer package Mathematica, Maple and Matlab. It also contain how to type in Mathematics, special characters, basic constructions. The course is both a preparation for further study and practical use.

OBJECTIVES
The objectives of the course are to:
1. describe the use of MATLAB;
2. demonstrate practically the Maple package;
3. use new commands which can perform more complicated tasks;
4. demonstrate the use of Java, C++, or Python;
5. apply the various packages to solving real life problems.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. use MATLAB in their own work;

2. illustrate their MATLAB programming skills and tackle other languages for computing, such as
   Java, C++, or Python;
3. define a computer package;
4. state standard functions;
5. define prime numbers.

COURSE CONTENTS

Introduction to computer package called Mathematica. Computer package called Maple and Matlab. How to type in Mathematics. Special characters. Basic constructions. Numerical computations. Standard functions. Accuracy, use of variables. Working with whole numbers. Finding prime numbers. Handling algebraic expressions. Doing symbolic computations. Graphics. Calculus. Solving equations in Mathematica, Maple or Matlab. Introduction to programming (procedural vs functional) using various numerical and algebraic methods to solve equations and sets of equations. Defining new functions. Writing new commands which can perform more complicated tasks.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 234: MECHANICS II (3 Units C: LH 45)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Mechanics II is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Mechanics II is the branch of Mathematics dealing with the study of motion. No matter what students interest in science or engineering, mechanics will be important for them, motion is a fundamental idea in all of science.

Mechanics can be divided into two areas: kinematics, dealing with describing motions, and dynamics, dealing with the causes of motion.

This course covers introductory Newtonian Mechanics which include but not limited to force, momentum and laws of motion, different kinds of force, emphasizing topics useful in other disciplines.

The course therefore, makes the beneficiary to appreciate and apply the basic principles of dynamics of a rigid body. The course is both a preparation for further study and practical use.

OBJECTIVES

The objectives of the course are to:

1. define basic concepts of Newtonian Mechanics;
2. demonstrate the compound pendulum.
3. apply kinetic energy formula of a body rotating about a fixed axis to solve problems in appropriate situations.
4. demonstrate principle of dynamics of a rigid body and apply them to appropriate situations.
5. apply momentum and laws of motion to solving real life problems.
6. state and apply parallel axes and perpendicular axes theorems.
7. describe the Attwood's machine.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define force ad momentum;
2. state the laws of motion;
3. state the different kinds of force;
4. state the equation of motion on an inclined plane;
5. define a projectile and describe the movement of a projectile motion;
6. define angular momentum of a rigid body.

COURSE CONTENTS

Newtonian mechanics: force, momentum and laws of motion. Different kinds of force (Gravitational, reactions, tension, thrust). Force and motion. Equation of motion. Motion on an inclined plane. Connected particles. Attwood's machine. Work, energy, power. Work-energy principle. Conversation laws of energy. Impulse and momentum. Impacts, direct and oblique collision. Impulsive tension. Projectiles: Range (on horizontal and inclined planes) trajectory. Dynamics of a rigid body. Moment of inertia. Radius of gyration. Parallel axes and perpendicular axes theorems. Kinetic energy of a body rotating about a fixed axis. Motion of a body about fixed axis. Angular momentum of a rigid body and principle of angular momentum. The compound pendulum.

Pre-requisite: MTH 103.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS

FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***CSC 224: INTRODUCTION TO INFORMATION PROCESSING (3 Units E: LH 30, PH 45)**

SENATE APPROVED RELEVANCE
In accordance with the University's vision, introduction to information processing was introduced to cater for students whose major study is not computer science. The University of Lagos has a vision to train information age graduates.

OVERVIEW
Introduction to information processing is a study of the acquisition, recording, organization, retrieval, display, and dissemination of information.

In recent years, several professionals use digital systems to enhance their productivity. Any institution that would produce world class graduates must introduce information processing to their students so that the students would be equipped for the future.

OBJECTIVES
The objectives of the course are to:
1. introduce the general systems theory.
2. describe the basic concepts of information systems.
3. introduce the different types of information.
4. present the role of management information systems in a business organization.
5. give the various ways of processing, storing, and dissemination of information.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. Explain the concept and characteristics of a system.
2. Describe the basic components of an Information System.
3. State popular examples of Management Information Systems.
4. Describe the role of Information Processing in Information Systems.
5. Understand Information Systems output and storage.
6. Draw and explain Models of Data Communication System
7. Describe tools and techniques used in Data Security and Control
8. Apply Sorting and Searching procedures in Information Processing.

COURSE CONTENTS
Information Systems. Management Information Systems. Information Processing. Information Systems Output and Storage. Data Security and Control. Sorting and searching. Principles of Data Communication. Data communication system. Data communication channels. Twisted Pair Cable. Data Signal. Analog and Digital signals. Introduction to Computer Networks. Types of Computer Networks. Local Area Network Topologies. Bus Network Topology. Mesh Topology. Ring Network Topology. Star Network Topology. Hybrid Network Topology. Inter-Networking Model. The OSI Model. The Internet and Internet Services.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***PHS 228: THEORETICAL PHYSICS I (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE
Preparing graduates with broad knowledge of Physics and its utilization in theoretical physics, enabling them to be adequately prepared for a technologically-driven community, where dependence on foreign products would be drastically reduced is in tandem with the vision and mission of UNILAG. The students of the programme will be exposed to the relevant foundation physics courses that underpin an understanding of basic electronics courses, which form part of the curriculum aimed at providing them with the needed skills for job creation and innovations in the areas of applied physics and electronics.

OVERVIEW
Theoretical physics uses mathematical models and abstractions of physical objects and systems to rationalize, explain and predict natural phenomena. Ability to use mathematical models and physical abstractions is important in today's world. This underpins the importance of preparing students in Physics with Electronics with the requisite knowledge and skills on the far-reaching applications of theoretical physics and its application for everyday use.
This course will examine solutions to differential equations, partial differential equations, multiple integrals and their applications to moment of inertia, centre of mass etc and these shall be used to solve complex physical problems.

OBJECTIVES
The objectives of the course are to:
1. explain differential equations.
2. identify the different types of differential equations.
3. demonstrate multiple integration and its applications.
4. introduce matrices and its various applications.
5. introduce complex analysis and its use to solve physical problems.

LEARNING OUTCOMES
Upon the completion of the course, the students should be able to:

1. solve differential equations;
2. solve and apply partial differential equations;
3. apply multiple integration in moment of inertia and centre of mass;
4. solve some physical problems using the concept of matrices;
5. apply complex analysis to solve physical problems.

COURSE CONTENTS

Differential equations. Multiple integration: Line and double integrals. Triple integrals. Application of multiple integration in moment of inertia. Application of multiple integration in centre of mass and others. Transformation in 3-D. Partial differentiation. Matrices. Equality and addition of matrices. Multiplication of matrices. Product of matrices by portioning. Matrix solutions of linear equations. Complex analysis. Complex integration. Cauchy's integral. Taylor's expansion. Laurent's expansion.

MINIMUM ACADEMIC STANDARDS

Mathematical Simulations Laboratory with NUC-CCMAS requirement facilities and standard Lecture hall equipped with multimedia system.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***STA 223 CATEGORICAL DATA ANALYSIS (3 Units E: LH = 45)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching categorical data analysis is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of Academy, Industry, community and fostering development in education.

OVERVIEW

Categorical data analysis is a field of statistical analysis dedicated to the analysis of dependent variables that are categorical in nature. A great advantage of studying categorical data analysis is that many concepts in statistics are very transparent when discussed in a categorical data context and at many places. Therefore, Categorical data can be counts of individuals, objects, or entities in different categories emerge frequently from many areas of study, such as geology, sociology, medicine and education. They provide important statistical information that can lead to real-life final decision making, discovery of new idea(s) and fresh knowledge. Meanwhile, ability to control, understand, and interpret categorical data becomes of interest to professionals and students across several disciplines.

However, some of the topics the course will focus on are as follows: The role of categorical data analysis like levels of measurement, categorical or continuous data, interaction in statistical analysis. Sampling distribution: sampling without and with replacement, the multinomial distribution, contingency tables etc. Although t-tests, linear regression, and analysis of variance are useful, valid methods for analysis of measurement data, categorical data requires a different methodology and techniques typically not encountered or dig deep in introductory statistics courses. In the aspect of practical, EXCEL, SPSS and R software package will be used for analysis to develop student's skills and more practical oriented.

OBJECTIVES

The objectives of the course are to:

1. establish the levels of measurement; Nominal, Ordinal, Ratio and interval
2. familiarize students with different study designs and data types in categorical data
3. equip with different descriptive and test statistics, generalized linear models for different types of outcomes, and their applications
4. utilize software packages and programs to conduct categorical data analysis and fit models where necessary
5. interpret model results properly and draw conclusions from the analysis
6. implement taught categorical data to analyze real-life datasets/phenomenon.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. differentiate between the **four** measurement levels with examples on each.
2. sate three different definitions of categorical data with examples
3. apply at least three appropriate statistics tools to solve given problems of categorical data.
4. produce good results from categorical data analysis conducted using two different software packages.
5. draw valid conclusion in two ways and make good decision from the results obtained
6. establish good usage of three software packages with any real-life (categorical data) datasets

COURSE CONTENTS

The role of categorical data analysis. Levels of measurement. Categorical and continuous data. Interaction in statistical analysis. Structure and stochastics in statistics. Sampling distributions. The Binomial distribution. The multinomial distribution. Contingency Tables. Conditional and marginal distribution. The Poisson distribution. Sampling with unequal selection Probabilities. Normal Approximation. Convergence in distribution. Normal approximation to the Binomial. Normal approximation to the Multinomial. Simple Estimation for Categorical Data. Introduction to maximum likelihood estimation. Standard Errors of Quantities Reported in Surveys. Basic Testing for Categorical Data. Test of $p = p_0$. Basic properties of the Pearson and the likelihood Ratio statistics. Test of $p \leq p_0$. Test of two-Way Tables. Chi-Square Test of Independence. Testing Independence for Ordinal Data. Association in Three-Way Tables.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***STA 225: STATISTICAL QUALITY CONTROL (3 Units E: LH = 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Statistical Quality Control is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of Academy, Industry, community and fostering development in education.

OVERVIEW

This course will present the theory and methods of quality monitoring including process capability, control charts, acceptance sampling, quality engineering, and quality design. Statistical Quality Control (SQC) is simply a statistical method for determining the extent. to which quality goals are being met without necessarily checking every item produced and. S.Q.C. also enables us to decide whether to reject or accept a particular product.

Statistical analysis in quality control is where statistical methods are used to measure, monitor and maintain the overall quality of products. Over time, the results help processes, such as manufacturing, ensure that the procedures will produce more specification-conforming products, therefore creating less waste.

OBJECTIVES

The objectives of the course are to:

1. define basic concepts of quality monitoring.
2. explain the statistical underpinnings of quality monitoring.
3. describe various available statistical tools of quality monitoring.
4. discuss the statistical and economical design issues associated with the monitoring tools
5. demonstrate the ability to design and implement these tools.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. explain the philosophy and basic concepts of quality improvement.
2. describe the DMAIC processes (define, measure, analyze, improve, and control).
3. demonstrate the ability to use the methods of statistical process control.
4. demonstrate the ability to design, use, and interpret control charts for variables.
5. demonstrate the ability to design, use, and interpret control charts for attributes.
6. perform analysis of process capability and measurement system capability.

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COURSE CONTENTS

Definitions of Quality and Quality Improvement, Statistical Methods and Management Aspects for Quality Control and Improvement. Basic concepts, Standardization and Specifications. Sources and detection of process variation. Control charts for attributes and variables and their

properties: d, p, x and charts. Process capacity studies. Cumulative sum charts and their properties. Sampling inspection for attributes and variables and their properties: single, double, multiple and sequential plans. Continuous sampling plans.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
*** STA 251: INTRODUCTION TO DATA SCIENCE I (3 Units E: LH = 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Introduction to Data Science I is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of Academy, Industry, community and fostering development in education.

OVERVIEW

Data Science is a combination of multiple disciplines that uses statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it. Data Science is about data gathering, analysis and decision-making. Data Science is about finding patterns in data, through analysis, and make future predictions. By using Data Science, companies are able to make: Better decisions, Predictive analysis and Pattern discoveries.

Data Science is used in many industries in the world today, e.g. banking, consultancy, healthcare, and manufacturing. Examples of where Data Science is needed are For route planning: To discover the best routes to ship, To foresee delays for flight/ship/train etc. (through predictive analysis), To create promotional offers, To find the best suited time to deliver goods, To forecast the next years revenue for a company, To analyze health benefit of training and To predict who will win elections.

OBJECTIVES

The objectives of the course are to:

1. describe Data Blending, Business Intelligence and Data Warehousing.
2. explain the process for data collection and data cleaning
3. identify the best data analysis tools in Python/R
4. apply Data Science analysis to realistic problems
5. recognize Ethics of big data.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. describe the tools for acquiring, cleaning data.
2. outline clean and reshape messy datasets.
3. apply data through web-scraping and data APIs.
4. define Introductory hypothesis testing and statistical inference.
5. illustrate ethics of big data.

COURSE CONTENTS

The major goals of this course are to learn how to use tools for acquiring, cleaning data. Data Collection and Data Blending, Business Intelligence and Data Warehousing, Data Science Project Life Cycle. Introduction to data analysis tools in Python/R, Descriptive statistics, Data structures, Introductory hypothesis testing and statistical inference and Web scraping and data acquisition. Reformatting messy datasets, Application to real life cases, Ethics of big data.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

STA 211: PROBABILITY II (3 Units E: LH 45)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. explain further permutation and combination;
2. define probability laws, conditional probability, and independence;
3. describe Bayes' theorem and explain some of the basic probability distribution for discrete and continuous random variables;
4. compute expectations and moments of random variables;
5. explain Chebyshev's inequality and apply it to real life situations;
6. explain joint marginal and conditional distributions and moments as well as Limiting distributions;
7. describe standard distributions, moments and moment-generating functions; and
8. explain laws of large numbers and the central limit theorem.

COURSE CONTENTS

Further permutation and combination. probability laws. conditional probability, independence. Bayes' theorem. probability distribution of discrete and continuous random variables: binomial, Poisson, geometric, hypergeometric, rectangular (uniform), negative exponential, binomial. Expectations and moments of random variables. Chebyshev's inequality. joint marginal and conditional distributions and moments. limiting distributions. discrete and continuous random variables, standard distributions, moments and moment-generating functions. laws of large numbers and the central limit theorem.

STA 231: STATISTICAL COMPUTING II (2 Units E: PH 90)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. explain the uses of computers in statistical computing;
2. demonstrate various statistical package;
3. use some statistical packages in solving problems in statistical methodology;
4. to demonstrate the use of spread sheet in application software; and
5. use packages such as SPSS, STATA, MINITAB etc, to demonstrate their abilities in statistical methodology.

COURSE CONTENTS

Uses of computers in statistical computing. introduction to various statistical packages. use of statistical packages in solving problems in statistics. spread sheet applications. Such as SPSS, STATA, MINITAB etc.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

PHY 205: THERMAL PHYSICS (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. discuss the concept of heat and temperature;
2. explain and determine thermodynamic processes;
3. explain and evaluate properties of real and ideal gases;
4. evaluate the consequences of the thermodynamic laws;
5. describe the basis of the kinetic theory; and
6. describe the statistical behaviour of gases with applications.

COURSE CONTENTS

The foundations of classical thermodynamics including the definition of temperature. The first law. Work, heat and internal energy. The second law. Carnot cycles and Carnot engines. Zeroth law. Entropy and irreversibility. Thermodynamic potentials and the Maxwell relations. Ideal gas equation. Internal energy and internal molecular modes. Qualitative discussion of phase transitions. Gibbs free energy. Clausius-Clapeyron equation. Examples of phase transitions. Van der Waals gas. Kinetic theory. Mean free path. Equi-partition of energy. Heat transfer. Diffusion rate.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained.

ACC 201: FINANCIAL ACCOUNTING I (3 Units C: LH 30; PH 45)

LEARNING OUTCOMES

At the end of this course, students should be able to:

1. appreciate the framework for the preparation and presentation of financial statements;
2. prepare simple financial statements of a limited liability company;
3. account for borrowing cost;
4. discuss accounting for government grants;
5. define relevant terms under IFRS 15 – Revenue;
6. account for inventory;
7. explain accounting policies; and
8. define relevant terms under IFRS 15 – Revenue.

COURSE CONTENTS

IASB framework for the preparation and presentation of financial statements. Introduction to IFRS 15 – Revenue and IAS 1. Presentation of Financial Statements. IAS 2. Accounting for inventories. Introduction to IAS 8 – Accounting policies, IAS 16 – Property, plant and equipment, IAS 20 – Government Grants and IAS 23 – Borrowing Costs.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained

Laboratory

Equipment and furniture needed in the traditional Accounting Laboratory

1. computers at least 50.

2. hands-on application of accounting software such as Peachtree complete accounting (best software), QuickBooks online (intuit), audit commander SPSS and other statistical software Packages.

ACC 202: FINANCIAL ACCOUNTING II (3 Units E: LH 30; PH 45)

LEARNING OUTCOMES

At the end of this course, students should be able to:

1. explain the conceptual framework for financial reporting;
2. prepare simple Partnership account;
3. explain Joint arrangement under IFRS 11;
4. account for goods on sale or return;
5. describe Revenue and Grants in accordance with the provision of the IFRS;
6. develop simple accounting policies; and
7. prepare final accounts .

COURSE CONTENTS
Conceptual framework for financial reporting. The principles of Small and Medium-sized Entities' Guidelines on Accounting (SMEGA) and Level 3 guidance. Introduction to partnership – definitions, types of partners, deeds, partners' capital and current accounts, profit sharing ratio and appropriation account. Introduction to IFRS 11 – Joint arrangement, and account for substance of transactions under goods on sale or return.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained

Laboratory
Equipment and furniture needed in the traditional Accounting Laboratory
1. computers at least 50.
2. hands-on application of accounting software such as Peachtree complete accounting (best software), QuickBooks online (intuit), audit commander SPSS and other statistical software Packages.

ACC 204: COST ACCOUNTING (3 Units E: LH 30; PH 45)

LEARNING OUTCOMES
At the end of this course, students should be able to:
1. explain the nature and objective of a cost accounting system;
2. describe the organisation of cost accounting department and its relationship with other departments;
3. explain elements of cost, cost classification, estimation and behaviour;
4. account for materials, labour and overhead in a cost accounting system;
5. explain costing methods - specific order and process costing;
6. familiarise with the treatment of process gains and losses;
7. prepare integrated and interlocking accounts;
8. discuss the concept of marginal and standard costing with simple variance analysis; and
9. explain functional budgets and cash budget.

COURSE CONTENTS
Definition of cost and cost accounting. The need for and installation of a cost accounting system, objectives of a cost accounting system, organisation of a cost accounting department and its relationship with other departments, advantages and limitations of a cost accounting system. Differences between cost and financial accounting, and cost and management accounting. Classification and estimation of cost into variable and fixed elements. Account for materials, labour and overhead. Preparation of specific order costing (job, batch and contract), process costing, including treatment of process gains and losses in line with IAS 2, integrated and interlocking accounts, explain and apply costing techniques marginal costing, (break-even and CVP analysis), standard costing. Prepare and explain simple variance analysis, and explain and prepare functional budget and cash budget.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained

Laboratory
Equipment and furniture needed in the traditional Accounting Laboratory
1. computers at least 50.
2. hands-on application of accounting software such as Peachtree complete accounting (best software), QuickBooks online (intuit), audit commander SPSS and other statistical software Packages.

ACS 206: MATHEMATICS OF DEMOGRAPHY (2 Units E: LH 30)

LEARNING OUTCOMES
At the end of the course, student should be able to:
1. identify various sources of demographic data for actuarial work;
2. compute some important demographic rates, ratios and indices;
3. explain the fundamental theorem of demography and its application in actuarial work;
4. differentiate between life contingency theory and demographic principles;
5. compare mortality tables in actuarial work and mortality table in demographic analysis;
6. identify the relevance of demography in vital statistics and national planning; and
7. illustrate the features of different life tables.

COURSE CONTENTS
Relationship between actuarial science and demography. Scope, uses and sources of demographic and socio-economic data. Methods of collecting demographic data. Sources and problems of errors in demographic data. Uses of census data. Measures of mortality and fertility. Current and cohort methods of description and analysis. Rates, ratios and indices. Standardization of crude death rates and crude birth rates; construction of national life table from census data. Stationary population theory. Stable population and projection. Components of population change.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.
Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained

ECO 201: INTRODUCTION TO MICROECONOMICS I (2 Units C: LH 30)
LEARNING OUTCOMES
At the end of the course, the students should be able to:
1. discuss the division of economics into microeconomics and macroeconomics;
2. define the laws of individual and market demand and supply and their elasticities; and
3. explain the basic elements in microeconomics in terms of foundation subject matter of household consumer behaviour using indifference curves, elementary production and costs with

isoquant and Isocost, treated in graphical form and simple mathematical capsules.

COURSE CONTENTS

Introduction to microeconomics. The partial equilibrium analysis with respect to basic microeconomic variables like individual and market demand, individual and market supply. Shapes of the demand and supply curves. Price elasticity of demand. Arc and point elasticity of demand. Income and cross elasticities. Consumer behaviour: Cardinalist, Ordinalist using indifference curves, their graphical and mathematical presentation and Partial derivative. Consumer surplus. Income and substitution effects in consumption. Cobweb analysis. Government price control. The production functions. Introductory theories of production and cost: Short- and Long-run production with laws of diminishing returns or variable proportion, marginal rate of technical substitution, Isocost, returns to scale, expansion path and factor substitution. Graphical and mathematical treatments of these.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained

ECO 202: INTRODUCTION TO MICROECONOMICS II (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, the students should be able to:

1. appreciate the basic concepts and the working of market mechanism within the context of theories of demand, supply, consumer behaviour, production in the short and long run;
2. identify distribution of incomes to factors of production, and
3. discuss the structure of markets within the context of perfect and imperfect markets.

COURSE CONTENTS

Recapitulation of the short-run and long-run production periods. The short-run and long-run cost curves. Fixed costs, variable costs, average costs and marginal costs. Cobb Douglass production function. Technical progress and production. Market structure: Axioms and determination of prices and output in perfect markets and imperfect markets, such as perfect competitive market, monopoly and price discrimination, monopolistic competition, oligopoly and different types, duopoly. Pricing of factors of production such as, wages, rent, interest and profit. Elementary price theory and its applications.

ECO 203: INTRODUCTION TO MACROECOOMICS I (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, the students should be able to:

1. distinguish elements of microeconomics which deal individual households, firms and market from elements of macroeconomics that is concerned with aggregates in terms of nature and scope;
2. discuss economic models, general equilibrium and disequilibrium
3. identify the goals of macroeconomics; gross domestic product measurement and components; and

4. explain the basic theories of consumption, savings and investments in economic science as well as
their determinants.

COURSE CONTENTS
Distinction between Microeconomics and Macroeconomics in the context of partial and general equilibrium. The goals of macroeconomics. Nature and Scope of Macroeconomics. Concept of National Income Accounting, Gross Domestic Product (GDP) and its components, its calculation and derivatives. Gross National Product, GDP deflator and its uses, cost of living; consumer price index and National Income with its derivatives. Consumption and Savings theories and their determinants. The Multiplier and the relations between APC, APS, MPC, MPS. Also, the theories of investment and its determinants, as well as introduction to government roles within the context of a 3- Sector model.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained

ECO 204: INTRODUCTION TO MACROECONOMICS II (2 Units E; LH 30)

LEARNING OUTCOMES
At the end of the course, the students should be able to:
1. discuss issues on the aggregate economy in relation to the functions and role of government with its policies;
2. identify the types of inflation and its positive and negative effects;
3. explain the types of unemployment and the Philips curve; and
4. discuss the external sector activities, particularly the international trade and role of exchange rate in payment system.

COURSE CONTENTS
Introduction to government: government spending, taxes, transfers, subsidy and output. Government sector multipliers; multiplier for model with income taxes. Money and its functions, the banking system and functions. Income and Trade policy measures. The External sector: International trade, exchange rate regimes, international reserves and balance of payments.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.
Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained

FIN: 202: PRINCIPLES OF INSURANCE (2 Units E: LH 30)

LEARNING OUTCOMES
By the end of the course students should be able to:
1. explain the nature and functions of insurance;

2. discuss some classifications of risk;
3. identify the insurance market, and intermediaries and their functions;
4. describe major categories of general insurance business;
5. evaluate all types of life assurance contracts their uses and benefits;
6. explain the principle of contract as relate to insurance;
7. define insurance contract and the rules applying to insurance; and
8. discuss the various principles of insurance.

COURSE CONTENTS

Historical evolution of insurance. Its career prospects and institutional structure. The scope and schemes of insurance. Detailed analysis of the various classes of insurance. The insurance contract and analysis. Social insurance and pension schemes. Group insurance. The general principles of insurance. Insurable interest, utmost good faith, indemnity, subrogation, contribution and proximate cause. Insurance and wagering. The insurance markets. Insurers and insured, brokers and agents. Insurance association and organizations. Insurance in practice. Conceptual clarifications of risk and risk management. Peril and hazards. Premiums, renewal, claims and disputes in the practice of insurance. Insurance marketing and history of insurance legislation in Nigeria.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained

FIN 209: ELEMENTS OF BANKING (3 Units E: LH 45)

LEARNING OUTCOMES

After studying this course the student should be able to:

1. describe the concepts of banking and the financial system;
2. explain the principles of banking;
3. elucidate the broad functions of banks;
4. analyze and explain the basic raison d'etre for banks;
5. describe the components of the balance sheets of banks;
6. elucidate the liability and asset portfolio management "problem" of banks; and
7. discuss the roles of the Chartered Institute of Bankers and the CBN.

COURSE CONTENTS

The business of banking. The development of money. Historical development of banking. The Central Bank of Nigeria. The Nigeria banking structure, savings and investment. The Nigerian money market. Bank's balance sheet. Organisational structure of clearing banks, Bills of exchange. Cheques. Methods of payment through the banking system. Bank customers. Bank

accounts services for the exporters and importers. Bank lending. Interpreting the accounts of customers. The Banker's Institute (The Chartered Institute of Bankers of Nigeria (CIBN).

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained

300 Level

NUC 70% CCMAS COURSE CONTENT

GST 312: Peace and Conflict Resolution (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to

1. analyse the concepts of peace, conflict and security;
2. list major forms, types and root causes of conflict and violence;
3. differentiate between conflict and terrorism;
4. enumerate security and peace building strategies; and
5. describe roles of international organisations, media and traditional institutions in peace building.

Course Contents

Concepts of peace, conflict and security in a multi-ethnic nation. Types and theories of conflicts: ethnic, religious, economic and geo-political conflicts. Structural conflict theory, realist theory of conflict and frustration-aggression conflict theory. Root causes of conflict and violence in Africa: indigene and settlers' phenomenon, boundaries/boarder disputes, political disputes, ethnic disputes and rivalries, economic inequalities, social disputes, nationalist movements and agitations, selected conflict case studies – Tiv-Junkun, Zango Kartaf, chieftaincy and land disputes and many more. peace building, management of conflicts and security, peace & human development. Approaches to peace & conflict management: (religious, government, community leaders and many more.). Elements of peace studies and conflict resolution, conflict dynamics assessment scales: constructive & destructive. Justice and legal framework and concepts of social justice. The Nigeria legal system. Insurgency and terrorism. Peace Mediation and Peace Keeping. Peace and security council (international, national and local levels). Agents of conflict resolution: conventions, treaties, community policing, evolution and imperatives. Alternative dispute resolution (ADR): a). dialogue b). arbitration c). negotiation d). collaboration and many more Roles of international organizations in conflict resolution: (a). The United Nations (UN) and its conflict resolution organs. (b). The African Union & Peace Security Council (c). ECOWAS in peace keeping. Media and traditional institutions in peace building. Managing post-conflict situations/crisis: refugees. internally displaced persons (IDPs). The role of NGOs in post-conflict situations/crisis.

ENT 312: Venture Creation (2 Units C: LH 30; PH 45)

Learning Outcomes

At the end of this course, students, through case study and practical approaches, should be able to

1. describe the key steps in venture creation; spot opportunities in problems and in high potential sectors regardless of geographical location;
2. state how original products, ideas, and concepts are developed;
3. develop business concept for further incubation or pitching for funding;
4. identify key sources of entrepreneurial finance;
5. implement the requirements for establishing and managing micro and small enterprises;
6. conduct entrepreneurial marketing and e-commerce;
7. apply a wide variety of emerging technological solutions to entrepreneurship; and
8. appreciate why ventures fail due to lack of planning and poor implementation.

Course Contents

Opportunity identification (sources of business opportunities in Nigeria, environmental scanning, demand and supply gap/unmet needs/market gaps/market research, unutilised resources, social and climate conditions and technology adoption gap). New business development (business planning, market research); entrepreneurial finance (venture capital, equity finance, Micro finance, Personal savings, small business investment organizations and Business plan competition). Entrepreneurial marketing and e-commerce (principles of marketing, customer acquisition & retention, B2B, C2C and B2C models of e-commerce, first mover advantage, E-commerce business models and successful E-commerce companies,). Small business management/family business. Leadership & management. Basic book keeping. Nature of family business and family business growth model. Negotiation and business communication (Strategy and tactics of negotiation/bargaining. Traditional and modern business communication methods). Opportunity discovery demonstrations (business idea generation presentations, business idea contest, brainstorming sessions, idea pitching). Technological Solutions (the concept of market/customer solution, customer solution and emerging technologies. Business applications of new technologies - Artificial Intelligence (AI), Virtual/Mixed Reality (VR), Internet of Things (IoTs), Blockchain, Cloud Computing, Renewable Energy etc. Digital Business and E-Commerce Strategies).

IMT 301: Financial Mathematics (2 Units C: LH 30)

Learning Outcomes

At the end of this course, the students should be able to:

1. demonstrate a firm understanding of modern theory of financial mathematics;
2. explain the basic concepts in Financial Mathematics; and
3. conduct transactions concerning annuities, actuarial, statistics and mortality analysis

Course Contents

Various topics in mathematics of finance are covered, including annuities. Actuarial. statistics and mortality analysis. principles and methods of actuarial treatment of statistical data, including the compilation and other rates. exposure to risk formula selection. multiple decrements. history and distinctive features of the principle's actuarial tables in common use. vital statistics, including censuses of births, deaths, marriages, and migration statistics. forecasting rates of mortality. General Theory of Projection.a short account of the population of Nigeria. construction of national

life table, sickness and other rates. construction of tables. valuation of liabilities under life policies and special topics. multiple decrement (service) and associated single decrement tables. values of and contribution for sickness benefits. pension benefits, disability benefits and widows and orphans benefits.

MTH 302: Ordinary Differential Equations (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. describe line dependence, Wronskian, reduction order, variation of parameters, series solution about ordinary and regular points; and
2. discuss orthogonal polynomials.

Course Contents

Ordinary differential equations: linear dependence, Wronskian, reduction order, variation of parameters, series solution about ordinary and regular points. Special functions: Gamma, Beta, Bessel, Legendre's theorem, hyper geometric. Laplace transform and applications to initial value problems

IMT 303: Mathematical Computing (2 Units C: LH 15; PH 45)

Learning Outcomes

At the end of this course, students should be able:

1. demonstrate knowledge of computer systems and its historical development;
2. give a general overview of computer architecture, hardware, software and liveware;
3. use computers to process information;
4. create documents;
5. demonstrate knowledge in basic information technology;
6. exhibit deep knowledge of principles of application programs;
7. give an overview of MAPLE environment (Document mode and worksheet
8. explain simulating models and MATLAB; Matlab fundamentals
9. write programmes to solve simple real-life problems; and
10. use Latex.

Course Contents

Introduction to computer systems and its historical development. contribution from mathematicians (e.g. Leibnitz, Boole, Pascal, Babbage, Turing, von Neumann). numerical computation and mechanical computing devices. general overview of computer architecture, hardware, software and liveware, programming languages, application packages, the present day use of computers and its future. introduction to Linux Operating System: UNIX commands, directory structure, text editors, user accounts and file permissions, text editors, virtual terminals in text mode. programming with C on Unix system. editing (with emacs), compilation, debugging etc. formatted input-output, control structures, loops, C-functions, pointers, File input/output, command-line arguments (the above shall be discussed with mathematical applications). introduction to X-Windows system;overview of MAPLE environment (document mode and worksheet): basic mathematical operations in maple and their commands, assignment, equation. Boolen equality, lists, sets, expressions, simplify,

factor, expand. solving ordinary differential equations, evaluating and differentiating functions. Writing programmes to solve simple real-life problems. plotting graphs, exporting and printing graphs. simulating models. overview of MATLAB. Matlab fundamentals. Variable and Workspace. operators, expression and statements. vectors and matrices. controlling command window, input and output, executing matrix linear algebra with Matlab. defining Matlab functions and creating scripts, Function M-file. and flow control (Condition and Loop controls).

MTH 305: Complex Analysis II (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. discuss Laurent expansion, Isolated singularities and Residues; and
2. define Residue theorem and Rouche's theorem

Course Contents

Laurent expansions. Isolated singularities and residues. Residue theorem. Calculus of residue, and application to evaluation of integrals and to summation of series. Maximum modulus principle. Argument principle. Rouche's theorem. The fundamental theorem of algebra. Principle of analytic continuation. Multiple valued functions and Riemann surfaces.

IMT 306: Introduction to Operations Research (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. build their own formulations;
2. expand existing formulations;
3. critically evaluate the impact of model assumptions and choose an appropriate solution technique for a given formulation.

Course Contents

Phases of operation research study. classification of operation research models' linear dynamic and integer programming. Decision Theory. Inventory Models, Critical Path Analysis and Project Controls.

IMT 307: Function Analysis (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. define Hilbert spaces, Banach spaces and vector spaces; and
2. describe Banach Algebra.

Course Contents

Hilbert Spaces. bounded linear functionals. operators on Banach spaces. topological vector spaces. and Banach algebra.

MTH 308: Introduction to Mathematical Modelling (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. develop some mathematical models in Biology, Physics, Chemistry, and Social Science;
2. describe flow Diagrams;
3. discuss method of Analysis of models formulated; and
4. describe the method of solutions to the models formulated.

Course Contents

Methodology of model building; identification, formulation and solution of problems, cause-effect diagrams. Equation types: algebraic, ordinary differential, partial differential, difference, integral and functional equations. Applications of mathematical models to physical, biological, social and behavioural sciences.

IMT 312 - Financial Derivatives (2 Units C: LH 30)

Learning Outcomes

Upon the completion of this course, students should be able to:

1. explain the concept of Simple Market Model, Basic Notions and Assumptions, No-Arbitrage Principle, One-Step Binomial Model;
2. handle risk and return, forward and futures contracts and risk-free asset;
3. handle dynamics of stock prices;
4. apply investment strategies to the Binomial Tree Model and other stock models.

Course Contents

Introduction: A Simple Market Model, Basic Notions and Assumptions. No-Arbitrage Principle, One-Step Binomial Model. Risk and Return. Forward and futures Contracts. Options, Risk-Free Asset, Money Market, Zero-Coupon Bonds, Coupon Bonds, Money Market Account. Risky Assets: dynamics of stock prices, return, expected return. Model: Risk-Neutral Probability, Martingale Property, Other Models, Trinomial Tree Model, Stock and Money Market Models. investment strategies, application to the Binomial Tree Model and Fundamental Theorem of Asset Pricing.

IMT 320: Industrial Attachment (12 Weeks) (3 Units C: PH 135)

Students should be attached to some industrial organizations for 12 Weeks at the 300 Level preferably during the long vacation for more industrial experience. Students to be assessed based on seminar presentation, their reports and assessment by supervisors.

INSTITUTIONAL 30% CCMAS

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS

***MAT 316: MECHANICS IV (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Mechanics IV is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

This course contains the law of gravitation, Kepler's Laws. Central orbits; momentum and energy equations, determination of the orbits, apses, critical velocity, stability of circular orbit are some emphasizing topics useful in other disciplines.

The course therefore, makes the beneficiary to appreciate and apply the motion near the earth's surface to solve real life problems. The course is both a preparation for further study and practical use.

OBJECTIVES

The objectives of the course are to:

1. define the law of gravitation;
2. illustrate Kepler's Laws.
3. use motion near the earth's surface to demonstrate real life problem.
4. demonstrate principle of stability of circular orbit
5. illustrate rotating and translating frames.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state the law of gravitation;
2. calculate the momentum and energy equations;
3. describe the motion near the earth's surface;
4. state Coriolis force;
5. describe the Foucault pendulum.

COURSE CONTENTS

The Newton's law of universal gravitation: definition, formula and facts. Kepler's first lawof planetary motion: movement of planets in elliptical orbits with the Sun as a focus. Kepler's second law of planetary motion: a planet covers the same area of space in the same amount of time no matter where it is in its orbit. Kepler's third law of planetary motion: a planet's orbital period is proportional to the size of its orbit (its semi-major axis). Central orbits. Angular momentum about a pole. Energy equations. Determination of the orbits, apses, critical velocity, stability of circular orbit. Moving frames of reference; rotating and translating frames. Coriolis force (movement on

rotating objects). The Coriolis effects. Earth's rotation and its effects on weather. Motion near the earth's surface. The Foucault pendulum. Definition, prediction and proof of Earth's rotation.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 323: VECTORS AND TENSORS ANALYSIS (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE
Graduate who are highly skilled and knowledgeable in teaching Vector and Tensor Analysis is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
In this course, we will study the most fundamental knowledge for understanding Vectors and Tensors were taught in the traditional way. Prior to our applying vector and tensor analysis to our research area of modern continuum mechanics, *vector and tensor* analysis provides a kind of bridge between elementary aspects of linear algebra, geometry and analysis.

Tensor analysis contains tools and definitions used within modeling of continuous media, field equations in physics, electromagnetism, elasticity theory and theory of general relativity. A coordinate-free approach is also given for certain differential operators such as the exterior derivative, the Lie derivative and the covariant derivative. Furthermore, integration theory and Stokes theorem is treated in a general form. Finally, the Riemann curvature tensor and the torsion tensor is also treated.

The brief introduction to tensor analysis in this course offers students knowledge of mathematics to become familiar with the basic concepts of the contemporary theory of intense charged particle beams. The hydrodynamic equations of a dense laminar beam are formulated in tensor form, and a geometrized variant of those equations is given based on the use of a previously unknown coordinate system associated with the flow geometry. Such a flow description is possible only in a non-orthogonal coordinate system. The Cauchy problem for the beam equation in the geometrized statement turns out to be similar to the Cauchy problem in Einstein's general relativity theory..

The course therefore, makes the students to appreciate and apply Tensor algebra, transformation laws and Cartesian tensors to solve real life problems.

OBJECTIVES

The objectives of the course are to:

1. describe Line law of multiple integral;
2. demonstrate Gradient, divergence and curl;
3. define Gauss's theorem;
4. demonstrate Curvilinear coordinates;
5. Apply transformation laws and Cartesian tensors;
6. illustrate Green's Theorem, Stokes's Theorems and Divergence Theorem;
7. demonstrate understanding of the concept and basic skills learned in vector analysis to solve physics problems.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define vector differentiation and vector integration;
2. define gradient, divergence and curl of vectors;
3. state Green's, Stoke's and divergence theorems;
4. solve some problems involving applications of vector differentiation and vector integration;
5. define tensor and Cartesian tensor.
6. apply Stokes's Theorems to solve simple problems.

COURSE CONTENTS

Vector Analysis: Gradient. Divergence and curl of point functions. Vector differentiation and applications. Expansion formulae. Curvilinear coordinates, line, surface and volume integrals. Gauss's, Green's and Stoke's theorems. Proper and improper transformation. Tensor analysis. Tensor algebra. Transformation laws. Cartesian tensors: Summation convention. Transformation equations. Orthogonally conditions. Kronecker tensor and Levi-civita tensor. Tensors of different rank. Inner and outer products. Contraction. Quotient theorems. Symmetric and anti-symmetric tensors. Application to Vector Analysis.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

UNIVERSITY OF LAGOS

FACULTY OF PHYSICAL AND EARTH SCIENCE

DEPARTMENT OF MATHEMATICS

B. Sc. INDUSTRIAL MATHEMATICS

***MAT 325: ANALYTICAL DYNAMICS (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Analytical Dynamics is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Analytical dynamics develops Newtonian mechanics to the stage where powerful mathematical techniques can be used to determine the behaviour of many physical systems. The mathematical framework also plays a role in the formulation of modern quantum and relativity theories.

Analytical Dynamics is designed to teach you how mathematics could be used in solving problems in the contemporary Science/Technology and Engineering world. Therefore, the course is structured to expose you to the skills required to attain a level of proficiency in Analytical Dynamics. The course therefore, makes the beneficiary to appreciate and apply Langrange's equations to Holonomic and Non-Holonomic systems to solve real life problems.

OBJECTIVES

The objectives of the course are to:

1. define the term "constraint"
2. state the types of constraint;
3. state the difference between the various types of constraint;
4. apply D'Alambert's Principle and relevant theorems;
5. apply the Lagrange's equations to find the differential equations;
6. describe the Lagrange function of particle(s) moving in a conservation force field;
7. derive Lagrange's equations for holonomic and non-holonomicconstraint respectively;
8. define and explain simple harmonic motion;
9. state the forces causing simple harmonic motion;
10. explain the suspension by an elastic string;
11. describe the conical pendulum.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define the dynamics of a rigid body;
2. define Langrange's equations for Holonomic system;
3. discuss Kinetic and potential energies;
4. define nodes of oscillation;
5. state the difference between Holonomic and Non-Holonomic systems;
6. define degrees of freedom;
7. define the moment of a Force.

COURSE CONTENTS

Euler's dynamical equation for motion of a rigid body. Angular velocity. Angular momentum. Euler angles for transforming between the body-fixed and inertial frames of reference. The dynamics of rotational motion about a body-fixed point under the action of external force. The symmetrical top precessional motion. Force-free motion of a rigid symmetric top. Motion of spinning symmetric top with magnitudes. Degrees of freedom. Langrange's equations for Holonomic. Derivations and Lagrange multipliers for Holonomic constraints. Non-Holonomic systems. Impulsive Motion. Collation of smooth spheres. Small oscillation. Simple pendulum. Reduction of coplanar forces acting on a rigid body to a force and a couple. Moment of a force. The Hamiltonian. The Calculus of Variation. The Earth's gravitational field. Kinetic energy. Potential energy. Normal coordinates in a system of coupled oscillators and initial values. Nodes of oscillation.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 332: REAL ANALYSIS III (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE
Graduate who are highly skilled and knowledgeable in teaching Real Analysis III is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
Mathematics relies on our ability to be able to solve equations, if not in explicit exact forms, then at least in being able to establish the existence of solutions. To do this requires a knowledge of "Analysis", which in many respects is just Calculus in very general settings. The foundations for this work are commenced in Real Analysis *III*.

This course deals with studies the behavior of improper integrals, their classifications and real life applications. It covers real numbers, sequences and series of real numbers, real functions and their topological concepts. It also covers the concept of open covering, Heine–Borel and Bolzano–Weierstrass theorems, which will seem mysterious at first.
The topics help students to build solid backgrounds in real analysis which are applicable in Economics, Engineering, Physics and Optimization. The course therefore, makes the students to appreciate and apply real analysis involving improper integrals to solve real life problems.

OBJECTIVES
The objectives of the course are to:
1. describe the topology of the real line;
2. illustrate the discontinuity of Improper Integral.
3. state some methods of proof.
4. describe how switching circuits work in logic;
5. describe open and closed sets topologically;
6. state the Weierstrass intermediate theorem.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. demonstrate skills in constructing rigorous mathematical arguments;

2. apply the theory in the course to solve a variety of problems at an appropriate level of difficulty;
3. define the convergence of an improper integral;
4. apply improper integrals to real life problem;
5. state and apply the Bolzano-Weierstrass theorem;
6. state and apply Heine Borel's theorem;
7. define a improper integral of type I;
8. define a metric space.

COURSE CONTENTS

Switching circuits in Logic. Connectives and validity of arguments. Some methods of proof. Extended real number system. Topology of the real line: open and closed sets. Interior, exterior, frontier. Closure of a set. Limit inferior and Limit superior criterion for convergence. Bolzano-Weierstrass theorem. Heine Borel's theorem. Weierstrass intermediate theorem. Enumerable and non-enumerable sets. Fundamental Theorem of Calculus. Improper integral of type I: Infinite limits of integration. Two infinite limits of integration. Discontinuity of Improper Integral. Convergence and divergence of improper integral. Comparison and quotient tests for improper integrals. Improper integral of type II. Real life applications of improper integral. Introduction to metric space topology. Sequence Spaces.

Pre-requisite: MTH 207

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS

***MAT 334: NUMERICAL ANALYSIS II (3 Units C: LH 30, PH 45)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Numerical Analysis II is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Numerical Analysis II is concerned with approximation methods .Polynomials are very useful in the study of Numerical Analysis. Over the years polynomials have been used as approximation to several functions. Although polynomials are sometimes difficult to solve as equations yet they help in appreciating the value of certain functions. Polynomials are of paramount importance when it comes to approximation theory.

Orthogonal polynomials are of fundamental importance in many branches of mathematics in addition to approximation theory and their applications are numerous but we shall be mainly concerned with two special cases, the Legendre polynomials and the Chebyshev polynomials. More general applications are however easily worked out once the general principles have been understood. Legendre polynomial is known to possess some oscillatory property among which makes it of importance in the field of numerical analysis.

For approximation, or interpolation, of a function defined analytically a method due to Hermite is often useful. The method is superficially related to the spline method but in fact the two methods are very different because fitting of spline involves solving a system of linear equations to obtain numerical values for the second derivatives.
The course content is rich in topics useful in other disciplines Thus, the course is beneficial to those who wish to have a solid foundation in numerical analysis. The course is both a preparation for further study and practical use.

OBJECTIVES
The objectives of the course are to:
1. state the difference between discrete data and continuous function
2. define what orthogonal polynomials are.
3. use the least square methods and familiarize themselves with curve fitting;
4. state Simpson and trapezoidal rules and apply the rules to solve real life problems.
5. state the difference between initial value and boundary value problems in ordinary differential equations.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. define simultaneous equations and solve problem using Gaussian elimination method;
2. explain how to fit a curve;
3. define least square method;
4. explain interpolation and approximation;
5. state some numerical differentiation and numerical integration problems; and
6. state some numerical problems in ordinary Differential equations with initial value problems.
7. state the necessary formulae for generating the Legendre polynomials.

COURSE CONTENTS
Solution of simultaneous equations. Direct method. Gaussian elimination method. Gauss Jordan method. L.U. decomposition method. Iterative schemes. Gauss Seidel approximation. Gauss Jacobi approximation. Curve fitting. Least square method. Orthogonal polynomials (Chebyshev and Legendry polynomials). Numerical integration. Simpson rule. Trapezoidal rule. Newton Cote formula. Gaussian quadrature. Initial value problems for ordinary differential equations. Cubic Spine Interpolation. Hermite Approximations.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***CSC 316: INTRODUCTION TO SYSTEMS ANALYSIS AND DESIGN**

(3 Units E: LH 30; PH 45)

SENATE APPROVED RELEVANCE
This course in Computer Science Programme serves as the eye opening to real-life problem solving through problem requirement gathering to deployment of information system. In order to make the student dependable technical analyst, the knowledge of requirement gathering, system design and technical reporting processes are of utmost expedient. Systems analysis and design as a course gives the grace to students to approach intending, future IT-seeking clients with confidence of required steps to IS development. It is then necessary that the University of Lagos situated in a commerce city inculcate into her students, skills to digitalize manual processes for surrounding small scale and large scale industries. The University would not hesitate to continue to produce future information system analysts, systems designers and system developers for the nation as many of her students are out with Google, Microsoft and high-end IBM as contributors and IT specialists.

OVERVIEW
The pedagogy for this course is to take the students through the process of developing information systems or application as a way to automate major business processes. Information systems development life cycle would be itemised and discussed in phases. Business and data modelling approaches would be discussed. The skill of problem requirements (automable functions, data, output reports and even non-functional activities identifications) and documentations would be taught. The roles of a system analyst, system designer, system developer would be enumerated.

The different application development methodologies and reporting would be expatiated. The use of Unified Modelling Language (UML) diagrams would be discussed. Different approaches to acquiring IS would be compared (in-house development, open source or outsourcing to development companies) using relevant yardsticks. Case studies of extent of automation shall be drawn from some SMEs in Lagos or University of Lagos suburbs where the students would visit, document their findings and propose better solution options place of assignment. This knowledge prepares the graduate for the opportunity to work with top guys in the industry.

OBJECTIVES
The objectives of the course are to:
1. Present Information System concepts and developmental rationale for business processes
2. Review importance of Information system in the modern society
3. Study Information system environments

4. Describe the phases in the systems development life cycle in relation to Human-Computer Interaction (HCI) factors.
5. Explore the knowledge behind feasibility report and legal process on IS development.
6. Explore requirement elicitation skill and process
7. Make use of UML for data and process modeling
8. Compare in-house development with IS outsourcing
9. Identify choice of language for choice of problem

LEARNING OUTCOMES

At the end of this course, students should be able to:

1. describe system requirements gathering techniques;
2. explain data modeling technique (entity relationship modeling);
3. explain process modeling technique (data flow diagram);
4. describe system architectural design;
5. describe process and database design and
6. explain user interface design.

COURSE CONTENTS

Structured approach to analysis and design of information systems (IS) for businesses. Software development life cycle. Structured top-down and bottom-up design. Requirement gathering ethics. Feasibility report writing. Information systems development and documentation methodologies. UML diagrams. Data and process modeling with UML. Use case and class diagrams. Dataflow diagram and Entity relationship modeling. Computer aided software engineering. Input and output design diagram. Methods of system change over. IS verification and validation. File and database design. Information systems outsourcing. Open source information system. Choice of programming language on problem solving. Design of user interfaces. Comparison of structured and object-oriented design.

Lab Work: system requirements gathering techniques; data modelling techniques (entity relationship modelling); process modelling techniques (data flow diagram); use of UML diagrams; system architectural design; user interface design.

MINIMUM ACADEMIC STANDARD

Computer Laboratory is provided with computer systems with pre-installed tools and software documentation packages needed for the implementation of some of the notable UML diagrams Two students to one computer system is allowed for group presentation.

UNIVERSITY OF LAGOS

FACULTY OF PHYSICAL AND EARTH SCIENCE

DEPARTMENT OF MATHEMATICS

B. Sc. INDUSTRIAL MATHEMATICS

***CSC 331: REAL-TIME SYSTEMS AND PROGRAMMING (3 Units E: LH 30; PH 45)**

SENATE APPROVED RELEVANCE

The world of embedded and artificial intelligence applications is spreading vast that University of Lagos cannot but expose students to building embedded components and program them to work. Artificial intelligent systems drive most emerging technology in the modern day in the areas

of health, security and agriculture. Arduino , fintech, fishbins and raspberry components for experimentations are within reach with little cost. The University has demonstration laboratories where these devices are taught and deployed for use. The course is therefore appropriate to expose the students to nitty- gritty of building and automating embedded components and expose them to some simulation software. Future graduates with skills to build Internet of Thing (IoT) supported applications that ease out modern day human problem.

OVERVIEW

The ubiquity of real-time systems in recent times made it germane for its exploration in a Computer Science curriculum. Real time systems are systems in which time at which output is produced is significant.

Consequently, the correctness of real-time systems depends not only on the logical result of the computation, but also on the time at which the results are produced. The goal of the course is to provide the underpinning theories and praxis for real-time systems and the programming languages that drives them to undergraduate computer science students.

OBJECTIVES

The objectives of the course are to:

1. define embedded system, real-time applications and real-time operating system.
2. explore the fundamentals of microprocessor based systems.
3. review features and expectations from real time systems and their supporting sosftware.
4. review input-output interfacing technique for real-time systems and real-time devices with the main processor.
5. describe the methodologies for building real-time systems.
6. develop real-time components and automate them to work.
7. examine energy challenges of real-time system and embedded systems.
8. describe scheduling of process in real-Time device environment.
9. formulate real-life problems with real-time App development.
10. implement the problem solution models with any real-time supporting programming language of choice.

LEARNING OUTCOMES

On the completion of the course, students will be able to:

1. Explain the issues and basic concepts of real-time and real-time software development.
2. Apply appropriate software engineering methods and tools in developing embedded real-time software.
3. Develop classroom embedded systems with appropriate simulation and development environment.
4. Analyze the timing performance of a real-time software design using real-time schedulabity analysis tools.
5. Identify challenges of embedded systems and probable solutions.
6. Work in a team to apply real-time software engineering knowledge in developing a medium real-life /embedded program as group mini project.
7. Assess the performance of a real life experimentations.

COURSE CONTENTS

Introduction to Real-Time Systems. Categories of computer systems. Embedded real-time (ERT) systems. Real-time concepts. Embedded real-time development methodologies. Real-time operating systems. Embedded real-time hardware fundamentals. Real-time analysis challenges. Real-Time Software Overview. Challenges in software engineering of ERT software. Software Life Cycle. Software engineering essential for ERT. Requirement Specification and Design. Requirements stage. Prototyping. Design fundamentals. Elements of modular design. Design Models – functional, OOD & data flow. Concurrency. What & why concurrency. Concepts of concurrency. Sequential and Concurrency programming. Real-time & other application areas. Concurrent Programming. Construct. Concurrent C Language. BACI C. Simulator Environment. Real-Time Operating Systems. Basic features. Scheduling concept. Real-Time Operating Systems. Control of share resources. Inter-task communication. Memory management. MicroC/os RTOS. Real-Time Analysis. Schedulability analysis. Scheduling policies. Designing with rate-monotonic analysis. Hardware Fundamental for Software Engineering. Fundamental of microprocessor based systems. Input-output interfacing technique. Embedded real-time robotics case study. Mini Project Presentations and Revision.

Laboratory: The students would be exposed to building real life applications with arduino, Raspberry devices, use of simulators and modeling apps to sense and activate other applications to solve real life problems

MINIMUM ACADEMIC STANDARD

Computer laboratory with computer workstations with pre-installed tools and software packages needed for the implementation of real-time systems in group of two students to one desktop system.

ACS 302: LIFE CONTINGENCIES (3 Units E: LH 45)

LEARNING OUTCOMES

At the end of course, students should be able to:

1. appraise the pricing of life assurance contracts and life annuity contracts;
2. acquire good knowledge of complete future life time and its moments;
3. differentiate between gross premium and net premiums;
4. develop skills for evaluation of mortality functions such as probability of survival and death;
5. identify various life insurance products such as life annuities and insurance;
6. calculate reserves, policy values and mortality profits; and
7. acquire adequate skills for calculations relating to increasing insurances, annuities, with-profit policies, contingent and reversionary benefits.

COURSE CONTENTS

Measurement of mortality. Life annuities. Life insurance. Net annual premiums. Net level premium reserves. Expense factor. Relationships between insurance payable at the moment of death and those payable at the end of year of death. Differential equations for insurances payable at the moment of death. Reserves. Policy values. Mortality profits.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained

ACS 304: FINANCIAL MATHEMATICS FOR ACTUARIES (3 Units E: LH 45)

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. explain in details the use of cash flow models, simple and compound rates of interest and discount;
2. distinguish between nominal and effective rates of interest and discount;
3. identify various types of annuities and perpetuities used to solve financial transaction problems;
4. apply prospective and retrospective methods to determine the outstanding loans;
5. acquire good skills that will enable them derive equations that can yield values for linear interpolation and annuity tables;
6. demonstrate competence in using amortization and sinking fund methods to defray debts.

COURSE CONTENTS
Time value of money. Time preference. Productivity of capital. Uncertain future. Level of interest rates. Actuary's relationship to the Time Value of money. Measurement of interest. Mathematical theory and practical problems in compound and simple interest. Solution of problems in interest. Obtaining numerical results. Basic problems. Equation of value. Unknown time and unknown rate of interest. Elementary annuities: Annuity-immediate, annuity-due on any date. Perpetuities. Fractional terms. Unknown time, unknown rate of interest. Varying interest. Amortization schedules and sinking funds.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.
Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained

ACS 306: LIFE AND HEALTH INSURANCE (2 Units E: LH 30)

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. demonstrate good knowledge and understanding of the wider determinants of health and ill-health;
2. review the roles of people and agencies who undertake work in the promotion of public health;
3. discuss the debates and dilemmas arising from the promotion of public health;
4. apply actuarial principles to life and health insurance business;
5. identify various types of health insurance products; and
6. compute reserves and health insurance contributions.

COURSE CONTENTS
Life insurance. Insurable interest. Types of contract and typical provisions supplementary benefits. With profit policies. Underwriting. Premium calculation. Reserves. Reassurance. Industrial life assurance. Group life assurance. Personal accident and sickness insurance, covers available. Cancellable and non-cancellable contracts. Policy documents. Exclusions, proximate cause. Ratings. Group contracts actuarial principles of premium and reserve. National Health Insurance schemes.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained

ACS 308: RATEMAKING AND INSURANCE PRICING MODELS (2 Units E: LH 30)

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. define various terms in insurance as it relates to ratemaking;
2. estimate written, earned and unearned premiums;
3. identify different types of data used for actuarial analysis;
4. differentiate between aggregated data and segregated data;
5. make adjustment to premiums based on data supplied in book manual;
6. identify qualitative information required for actuarial analysis;
7. solve specialized insurance problems; and
8. identify assumptions underlying different statistical models.

COURSE CONTENTS
Basic insurance terms: Exposure, premium, claim, loss, loss adjustment, expense, underwriting expense, underwriting profit, and goal of ratemaking. Fundamental insurance equation. Ratemaking in perspective. Overall and individual balance. Basic insurance ratios. Ratemaking data. Exposures: Criteria for exposure bases, exposures for large commercial risks, and aggregation of exposures. Premium: Premium aggregation, methods of aggregation for annual terms, adjustments to premium, losses and loss adjustment expenses, other expenses and profit.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained

ACS 310: SURVIVAL MODELS (2 Units E: LH 30)

LEARNING OUTCOMES
At the end of the course**,** students should be able to:
1. demonstrate adequate skills and to competences to prepare mortality table;
2. perform the computation of force of mortality and relate it to force of interest under extra risk

underwriting;
3. describe the procedures required to estimate lifetime distributions such as Kaplan–Meier model;
4. analyze actuarial models and their characteristics; and
5. generate new distributions and modify loss random variable.

COURSE CONTENTS

Revision of probability theory for Actuarial Science. Actuarial survival models: Sources of survival data. Types of variables. Exposure to risk. Probability of living and dying. Mortality tables. Survival function in life contingencies. Time-until-death for a person of age x. Curtate-future-lifetime. Force of mortality. Relation of life table functions to the survival function. Mathematics of survival models. Probability density function. Cumulative Distribution Function (CDF). Hazard rate function; Moment of random variable X. Conditional probabilities of death (failure) and central rate. Truncated distributions. Expectation and variance of future lifetime. Median of future lifetime. Forms of survival models: Lifetimes, curtate future lifetime, expected value and variance. Estimation procedures for lifetime distributions: Kaplan–Meier estimate. Understanding actuarial models. Characteristics of actuarial models. Generating new distribution. discrete distribution. Modification of loss random variable. Aggregate loss models. methods of parameter estimation. Basics stochastic simulation. application of simulation in actuarial modeling.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained

CSC 308: OPERATING SYSTEM (3 Units E: LH 30; PH 45)

LEARNING OUTCOMES

At the end of this course, students should be able to:

1. recognise operating system types and structures;
2. describe OS support for processes and threads;
3. recognise CPU scheduling, synchronisation, and deadlock;
4. resolve OS issues related to synchronisation and failure for distributed systems;
5. explain OS support for virtual memory, disk scheduling, I/O, and file systems;
6. identify security and protection issues in computer systems; and
7. use C and Unix commands, examine behaviour and performance of Linux, and develop various system programmes under Linux to make use of OS concepts related to process synchronisation, shared memory, mailboxes, file systems, etc.

COURSE CONTENTS

Fundamentals of operating systems design and implementation. History and evolution of operating systems. Types of operating systems. Operating system structures. Process management: processes, threads, CPU scheduling, process synchronisation. Memory

management and virtual memory. File systems; I/O systems; Security and protection; Distributed systems; Case studies.

LAB WORK: Practical hands-on engagement to facilitate understanding of the material taught in the course. All the process, memory, file and directory management issues will be demonstrated under the LINUX operating system. Also UNIX commands will be briefly discussed. Alternatively, hands-on exposure may be through the use of operating systems developed for teaching, like TempOS, Nachos, Xinu or MiniOS. Another possibility is through programming exercises that implement and simulate algorithms taught. Simulation of CPU scheduling algorithms, producer-consumer problem, memory allocation algorithms, file organisation techniques, deadlock algorithms and disk scheduling algorithms.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

DTS 304: DATA MANAGEMENT I (3 Units E: LH 30; PH 45)

LEARNING OUTCOMES

At the end of the course the students should be able to:

1. describe the components of a database system and give examples of their use;
2. describe the differences between relational and semi-structured data models;
3. explain and demonstrate the concepts of entity integrity constraint and referential integrity constraint;
4. apply queries, query optimizations and functional dependencies in relational databases;
5. describe properties of normal forms and explain the impact of normalization on the efficiency of database operations;
6. describe database security and integrity issues and their importance in database design; and
7. explain the concepts of concurrency control and recovery mechanisms in databases.

COURSE CONTENTS

Information Management Concepts. Information storage & retrieval. Information management applications. Information capture and representation. Analysis and indexing - search, retrieval, information privacy. Integrity and security. Scalability, Efficiency and Effectiveness. Introduction to database systems. Components of database systems. DBMS functions. Database architecture and data independence. Database query language. Conceptual models. Relational data models. Semi-structured data models. Relational theory and languages. Database Design. Database security and integrity. Introduction to query processing and optimisation. Introduction to concurrency and recovery.

Lab work: Practical exercise on information representation, capture, storage and retrieval. Learn how to analyse data and index for easy searching and indexing. Practical on creating database files and models. How to create and use various database designs. How to query the created database. Methods of concurrency and recovery in database. Learn how to secure the database.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply. This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained.

ECO 306: INTRODUCTORY ECONOMETRICS (3 Units C: LH 30; PH 45)

LEARNING OUTCOMES

At the end of the course, the students should be able to:

1. explain the various concepts used in econometrics;
2. derive simple ordinary least squares estimates and the statistical relationship of data set like the goodness-of-fit, the t-statistics and making inferences from results of simple/multiple regression and interpret estimated models; and
3. carry out data formatting, data mining, data processing and interpretation of estimated models.

COURSE CONTENTS

Beginning with extensions of the two–variable linear model. The course should cover introduction to matrix algebra and algebra of econometrics. The General linear model and generalized lease square procedure, violations of linear model assumptions and the correction of serial correlation (autocorrelation) of the errors, intercorrelation (Multicollinearity) of the variables, stochastic regressions and errors in variables are also discussed. Other topics are identification of problems, meaning, detection and correction of simultaneous models and estimations, K-class estimators and the choice of estimators. Students will be engaged in practical assignments and familiarity with software computer applications for econometrics such as E-views, Stata, SPSS and many others.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained

Laboratory

Equipment and furniture needed in the traditional Accounting Laboratory

1. computers at least 50.

2. hands-on application of accounting software such as Peachtree complete accounting (best software), QuickBooks online (intuit), audit commander SPSS and other statistical software Packages.

FIN 303: PRINCIPLES OF FINANCE (3 Units C: LH 45)

LEARNING OUTCOMES

On the successful completion of this course, students should be able to:

1. explain the concept of financial management and roles of a financial manager;
2. discuss the aims of finance functions;
3. list and discuss the short-term and long-term sources of funds;
4. discuss the methods of issuing the instruments of debt;
5. prepare and explain the 'sources of funds' and 'uses of funds' for the company using information from the balance sheet;
6. explain and apply financial resources management in firms and its challenges;
7. explain the concept of working capital and working capital management;
8. discuss cash management techniques and preparation a cash budget;
9. identify and explain domestic/external sources of capital formation;
10. explain time value of money; and
11. list and explain types of leasing agreements.

COURSE CONTENTS

Financial statements, cash flow and taxes with a look at key financial statements, including the balance sheet, income statements and cash flow statements. The course also deals with capital budgeting, risk and return, cost of capital and financial policy, planning and management. Also examined are the various financial ratios and the type of information that they convey to us with regard to the financial health of a business and the process of looking ahead and anticipating the future. Financial statements analysis – P&L, balance sheet & cash flow. Taxes & depreciation. Basic theories, concepts, techniques and skills of financial management. Tools for measuring and analysing the financial performance of a firm. Financial planning and forecasting. The Financial environment: Markets, institutions, and interest rates. Risk and rates of return, Time value of money. Bonds and their valuation. Stocks and their valuation. Basics of capital budgeting. Introduction to global markets and their impacts on and linkages with the global markets. Application of the finance techniques by doing assignments on various financial topics utilizing MS excel (computer lab) and stock investing.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

Laboratory

Equipment and furniture needed in the traditional Accounting Laboratory

1. computers at least 50.

2. hands-on application of accounting software such as Peachtree complete accounting (best software), QuickBooks online (intuit), audit commander SPSS and other statistical software Packages.

FIN: 313 FINANCIAL MANAGEMENT (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define financial management and its objectives;
2. explain some basic concepts such as right issue, preference share and commercial papers;
3. understand the meaning of opportunity cost of capital and its usefulness;
4. discuss the methods of evaluating capital budgeting;
5. explain the various investment appraisal methods and its associated problems;
6. explain the legal and regulatory considerations for mergers and acquisitions;
7. describe the impact that the issue of dividends may have on a company's share price;
8. explain the various theories of dividend policy and dividend decisions;
9. explain valuation of shares and the various methods of valuing shares;
10. define risk, risk management and the various ways of dealing with financial risk; and
11. explain the different capital structure theories.

COURSE CONTENTS

The nature, scope and purpose of financial management. Management of working capital: cash, inventory, receivables, marketable securities, etc., Analysis and interpretation of basic financial statements. Financial forecasting and planning. Capital budgeting/investment decisions under conditions of certainty and uncertainty. Determinants and implications of dividend policy. Valuation of shares, assets and enterprises. Fixed income securities. Capital structure theories. Sources and cost of capital. Capital rationing. Mergers, acquisition, reconstruction and organization. Risk in finance and methods of avoiding them.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained

INS 301: RISK MANAGEMENT I (3 Units E: LH 45)

LEARNING OUTCOMES

At the end of this course, students should be able to:

1. explain the concept of risk management;
2. identify various classes of risks;
3. recognize risk management process;
4. describe the various risk treatment techniques;
5. apply statistical tools to analyze risks affecting insurance and other business firms; and
6. utilize the concept of loss distribution, pooling arrangement and diversification in making decisions.

COURSE CONTENTS

Definition of some important concepts of risk. Classification of risks. Peril Versus Hazard. Risk and Human Behavior. The cost of risks for households, firms and society. The aim of risk

Management. Corporate Risk Management. Objectives of Business Risk Management. Risk Management Process. Risk Management Vs Insurance Management. The Roles of a Risk Manager. The contributions of Risk Management to a Business Entity. Enterprise Risk Management (ERM). Risk Assessment Fundamentals: Objective of Risk Identification. Risk Identification Techniques. Major Risk Management Techniques. Risk Measurement: Loss Frequency and Loss Severity. Probability Analysis. Probability Distribution. Risk Measures for Random Variables. Pooling Arrangement and Diversification. Theories of accident causation. Effect of Insurance on Loss Control. Risk Management and Shareholder Wealth.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained

INS 302: RISK MANAGEMENT II (3 Units E: LH 45)

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. demonstrate knowledge of the variety of financial and financial related risks facing organizations;
2. explain the approach to risk management through risk identification, risk measurement and risk management (or mitigation);
3. discuss the main types of financial risk – market risk and capital adequacy, credit risk, liquidity risk, operational, legal and compliance risks, reputational risk;
4. explain the methodological principles of value at risk (VAR); and
5. discuss the concept of reputational risk and how to manage it.

COURSE CONTENTS
Nature of risk. Meaning of risk. Classification of risk. Types of financial risk – market risk and capital adequacy. Credit risk. Liquidity risk. Operational, legal and compliance risks. Reputational risk. Benefit of risk. Human perception of risk. Utility theory. Risk assessment; risk identification, risk estimation, risk evaluation. Risk treatment: risk avoidance, risk control, risk transfer, risk financing. Basic statistical concepts relating to insurance and risk management. Risk data, presentation of risk data, statistical measurement, and probability distribution. Risk pricing: underwriting, reinsurance and pricing insurance. Risk-based regulation and capital adequacy. Corporate governance issues of risk. Nigeria's and UK regulations; Capital adequacy. New risk-based approach to capital requirements, maintaining capital adequacy.
MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.
Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained

STA 311: PROBABILITY III (3 Units E: LH 45)

LEARNING OUTCOMES
At the end of the course, students should be able to:

1. define discrete sample spaces and provide rules of probability;
2. explain independence Bayes' theorem and Um models;
3. determine sampling with and without replacement;
4. explain inclusion-exclusion theorem;
5. explain allocation and matching problems;
6. explain probability generating function; and
7. outline the properties of Bernoulli trials, binomial, Poisson, Hypergeometric negative binomial and multinomial distribution, Poisson process.

COURSE CONTENTS

Discrete sample spaces. definitions and rules of probability. Independence Bayes' theorem. Um models; sampling with and without replacement. inclusion-exclusion theorem. allocation and matching problems; probability generating function; Bernoulli trials, binomial, Poisson, Hypergeometric negative binomial and multinomial distribution, Poisson process.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained

STA 312: DISTRIBUTION THEORY I (3 Units E: LH 45)

LEARNING OUTCOMES

At the end of the course, the students should be able to:

1. identify and/or find distributions using any of the transformation techniques;
2. explain cumulants and their generating functions as well as some special univariate distribution;
3. explain the central limit theorem;
4. illustrate bivariate moment generating functions of random variable and bivariate distribution;
5. explain bivariate moment generating functions; and
6. explain bivariate normal distributions associated with the normal, X 2, t and F distribution.

COURSE CONTENTS

Transformation techniques, probability integral transformation, order statistics and their functions. Cumulants and their generating functions. Some special univariate distribution. Central limit theorem. Bivariate moment generating functions of random variable. Bivariate distribution. Bivariate normal distributions. Distribution associated with the normal, X 2, t and F distribution

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained

STA 321: STATISTICS INFERENCE III (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. explain sampling and sampling distribution;
2. differentiate point from interval estimation;
3. outline the principles of hypotheses testing; test hypotheses concerning population means, proportions and variances; of large and small samples, large and small sample cases;
4. conduct a goodness fit tests using the analysis of variance;
5. describe the methods of estimation; using maximum likelihood, least squares and method of moments

COURSE CONTENTS

Estimation methods of estimation (Moments, maximum likelihood, least squares), Properties of Estimators (Consistency, unbiasedness, efficiency, sufficiency, completeness, BLUE, UMVUE). Construction of confidence interval. Tests of Hypothesis, Definition and Fisherian concepts of hypothesis testing. Newman-Pearson Lemma. Likelihood ratio test. Equivalency of confidence interval and hypothesis testing in decision-making under regression correlation analysis.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

STA 322: REGRESSION AND ANALYSIS OF VARIANCE I (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, the students should be able to:

1. explain total, partial and multiple correlation ratio;
2. demonstrate simple and multiple linear regression using the variable selection techniques, stepwise regression;
3. determine the analysis of covariance, influence measures, polynomial regression and orthogonal polynomials;
4. explain simple non-linear way classification;
5. demonstrate two-way classification; three-way classification; balanced and unbalanced two factor nested (hierarchical) classifications;
6. explain multiple comparisons component or variance estimates and tests; and
7. demonstrate these techniques in computing packages.

COURSE CONTENTS

Introduction. Mathematical model in simple linear regression (Y=B0+Bx+e). methods of fitting linear regression; eye-fitting method; method of least squares (LS) Assumptions of LS. Predictor: Y=B0+B1X. model violations and remedies. Residuals. Tests of hypothesis of regression parameters. General theory of least squares. Matrix approach of linear regression. Application of statistical packages. (SPSS and R) to regression introduction to multiple linear regression correlation analysis.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained.

SSC 302: RESEARCH METHOD I (2 Units C: LH 30)

LEARNING OUTCOMES
At the end of the course, the students should be able to:
1. explain the concept and scope of research methods;
2. discuss in detail procedures for conducting a research;
3. discuss the principles guiding research; and
4. discuss types of research – experimental non-experimental and quasi-experimental research.

COURSE CONTENTS
Introduction to Social Science research protocol with particular emphasis on proposal development; literature review (conceptual, empirical and theoretical framework), ethnography of the study area statement of the research problem, sampling procedures and methods of data collection and instruments.

MINIMUM ACADEMIC STANDARDS
The approved staff-student ratio in administration and management sciences is 1:30.
Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained

MTH 399: Industrial Attachment II (24 Weeks) (6 Units C: PH 270)
Students should be attached to some industrial organizations for additional 12 weeks at their 300 Level preferably the whole of second semester and during the long vacation for more industrial experience. Students to be assessed based on seminar presentation, their reports and assessment by supervisors.

400 Level

NUC 70% CCMAS COURSE CONTENT

MTH 401: Theory of Ordinary Differential Equations (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. define existence and uniqueness theorems;
2. describe Volterra and Fredholm types; and
3. solve some problems of reduction of ordinary differential equations to integral equations.

Course Contents
Existence and uniqueness theorems, dependence of solutions on initial data and parameters. Properties of solutions. Sturm comparison and Sonin-Polya theorems. Linear and non-linear systems. Floquet's theory and stability theory. Integral equations: classification, Volterra and

Fredholm types, Neumann series. Fredholm alternative for degenerate Hilbert – Schmidt kernels. Reduction of ordinary differential equations to integral equations. Symmetric kernels, eigenfunction expansion with application.

MTH 402: Theory of Partial Differential Equations (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. define first and second order linear equations;
2. describe Cauchy problems; and
3. solve some problems in parabolic, hyperbolic and elliptic equations.

Course Contents

Theory and solutions of first-order and second order linear equations. Classification, characteristics and canonical forms. Cauchy problems. Elliptic equations: Laplace's and Poisson's formulas, properties of harmonic functions. Hyperbolic equations; wave equations, retarded potential: transmission line equation, Riemann method. Parabolic equation: diffusion equation, singularity function, boundary and initial – value problems.

IMT 404: Optimization Theory (2 Units C: LH 30)

Learning Outcomes

At the end of the of this course, students should be able:

1. develop a fundamental understanding of the role of optimization technique;
2. describe and explain complex systems ; and
3. enumerate the theoretical and computational tools needed to solve equations that arise as models.

Course Contents

Linear programming models. the simplex method, formulation and theory. quality integer programming. transportation problem. Two-person zero-sum games. nonlinear programming. quadratic programming Kuhn-tucker methods. optimality criteria. simple variable optimization. multivariable techniques. gradient methods.

MTH 405: General Topology (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. define the following: topological spaces, continuous functions and topological invariants;
2. describe pointwise and uniform convergence.

Course Contents

Topological spaces, definition, open and closed sets neighbourhoods. Coarser, and finer topologies. Basis and sub- basis. Separation axioms, compactness, local compactness, connectedness. Construction of new topological spaces from given ones. Sub-spaces and quotient spaces. Continuous functions, homeomorphisms and topological invariants. Spaces of continuous functions: Pointwise and uniform convergence.

IMT 411: Mathematical Economics I (2 Units C: LH 30)

Learning Outcomes

At the end of this course, students should be able to:

1. discuss the concepts and theory of microeconomic;
2. explain the mathematical approach to address multi market equilibrium well as Pareto optimality problem;
3. account for the general economic optimization over time, linear models such as Input-Output (I-O) models; and
4. discuss the linear programming concepts and solutions.

Course Contents

Microeconomic theory is treated with a mathematical approach. Topics will include the following: theory of consumer behaviour, constrained optimizing behaviour. the Slutsky equation, construction of utility number. theory of the firm. constrained optimizing behaviour, CES production function, market equilibrium with lagged adjustment and continuous adjustment. multi-market equilibrium. Pareto optimality. general economic optimization over time. linear model. input-output (I-O) models, linear programming concepts and solutions.

IMT 412 - Mathematical Economics II (2 Units C: LH 30)

Learning Outcomes

At the end of this course, the students should be able to:

1. exhibit a sound understanding of mathematical techniques;
2. formulate economic problems in mathematical terms;
3. apply relevant tools for analyzing economic problems; and
4. provide students with the mathematical tools required for economic analysis at the undergraduate level.

Course Contents

Micro-economic theory is treated with a mathematical approach in the following areas: Simple model of income determination. consumption and investment. the IS curve, monetary equilibrium, the LM curve. labour wages and price (inflation) models. full employment equilibrium models of income determination. aggregate demand and supply analysis. balance of trade (payments). model of income determination. stabilization policy. comparative statistics. analysis of monetary fiscal policy. the Harold Domar growth model and the neo-classical growth model.

IMT 499 – Project (6 Units C: LH 270)

Learning Outcomes

This course is the project component of the research method and proposal writing, therefore, the course will:

1. equip students with the skills and knowledge in the analysis and interpretation of data; and
2. expose the students to report writing.

Course Contents
A research project and dissertation to be undertaken on any topic of mathematical interest / its areas of application in the Industrial Mathematics

INSTITUTIONAL 30% CCMAS

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 438: NUMERICAL ANALYSIS III (3 Units C: LH 30, PH 45)**

SENATE APPROVED RELEVANCE
Graduate who are highly skilled and knowledgeable in teaching Numerical Analysis III is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW
Numerical Analysis III is the study of algorithms that use numerical approximation (as opposed to symbolic manipulations) for the problems of mathematical analysis. It is the study of numerical methods that attempt at finding approximate solutions of problems rather than the exact ones.

Numerical analysis finds application in all fields of engineering and the physical sciences, and in the 21st century also the life and social sciences, medicine, business and even the arts. Current growth in computing power has enabled the use of more complex numerical analysis, providing detailed and realistic mathematical models in science and engineering. The course is also concerned with existence of solution which include one step schemes and linear multistep methods, their theory of convergence and stability.

OBJECTIVES
The objectives of the course are to:
1. describe the existence of solutions;
2. illustrate one step schemes;
3. outline the difference between one step and multistep methods;
4. demonstrate extrapolation processes;
5. state integral equation and boundary value problems.

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. define a one step scheme;
2. state the criteria for convergence of one step scheme;
3. state the stability properties of multistep methods.

4. solve integral equation and boundary value problems.
5. apply linear multistep methods.
6. state the formula for Taylor and Runge-Kutta methods.
7. describe problems with singularities.

COURSE CONTENTS

Functions approximations: Tchebyshef polynomials, Rational functions and Pade approximations, least square approximations, min-max approximation. Linear Difference equations with constant coefficients. Numerical solutions of second order linear boundary value problems in ODE (Shooting methods). Approximate Analytic methods for solving BVPs and Integral Equations (Decomposition methods, Variational Iteration Methods, Homotopy Pertubation methods). Solution of Ordinary differential equations. One step methods. Taylor and Runge-Kutta methods, Rooted trees and general order conditions of Runge-Kutta methods, Variable-step size methods. Error bound for Runge-Kutta Methods. Multistep methods. Order conditions. Stability, Local Truncation Error. Error estimates. Stability and Weak-Stability theory. Predictor-Corrector methods and their Error Estimates. Stability theory for Predictor-Corrector algorithms. Methods for special problem: stiffness, Oscillatory, Multirate, Problems with singularities, etc. Extrapolation. Geometric integration of Symplectic (Hamiltonian) Reversible systems. General Linear Methods.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 415: FLUID MECHANICS I (3 Units C: LH 30))**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Fluid Mechanics I is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Fluid Mechanics I is an introductory part of this course. It is concerned with pressure, hydrostatics, and buoyancy; open systems and control volume analysis; mass conservation and momentum conservation for moving fluids; viscous fluid flows, flow through pipes; dimensional analysis; boundary layers, and lift and drag on objects.

Fluid flows are important in many scientific and technological problems including atmospheric and oceanic circulation, energy production by chemical or nuclear combustion in engines and stars, energy utilisation in vehicles, buildings and industrial processes, and biological processes such as the flow of blood. Considerable progress has been made in the mathematical modelling of fluid flows and this has greatly improved our understanding of these problems, but there is still much to discover. This course introduces students to the mathematical description of fluid flows and the solution of some important flow problems. Topics covered are: the mathematical description of fluid flow in terms of Lagrangian and Eulerian coordinates; the derivation of the Navier-Stokes equations from the fundamental physical principles of mass and momentum conservation; use of the stream function, velocity potential and complex potential are introduced to find solutions of the governing equations for inviscid, irrotational flow past bodies and the forces acting on those bodies; analytic and numerical solutions of the Navier-Stokes equation.

OBJECTIVES

The objectives of the course are to:

1. describe Real and Ideal fluid;
2. illustrate motion of fluids particles;
3. solve simplified examples of fluid mechanics,
4. demonstrate Stokes's stream function with applications;
5. apply the acquired knowledge and skills in professional and specialist courses

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define basic terms, values and laws in the areas of fluids properties, statics, kinematics and dynamics of fluids, and hydraulic design of pipes,
2. describe methods of implementing fluid mechanics laws and phenomena while analysing the operational parameters of hydraulic problems, systems and machines,
3. apply tables and diagrams, and equations that define the associated laws,
4. calculate and optimise operational parameters of hydraulic problems, systems and machines.
5. explain the correlation between different operational parameters,
6. select engineering approach to problem solving based on the acquired physics and mathematical knowledge

COURSE CONTENTS

Introduction: Basic concepts of fluid mechanics. Fundamental terms. Physical values. Fluids and their properties. Forces inside fluid. Fluid Statistics: Pascal's law. Euler's equation of fluid statics. Measurement of pressure. Relative statics of fluid constant acceleration, rotation. Forces of hydrostatic pressure. Buoyancy. Flotation. Stability. Fluid Kinematics: Euler and Lagrangian specification of fluid flow. Streamlines. Pathlines. Stream surface. Stream tube. Mass/volume flow. Control volume. Fluid Dynamics: Continuity equation. Basic laws of fluid dynamics conservation of mass, conservation of linear momentum, conservation of energy. Ideal fluid flow. Application of Bernoulli's equation. Real fluid flow. Viscosity. Determination of losses. Reynolds experiment. Laminar and turbulent flow. Boundary layer. Velocity profile. Losses in pipes. Frictional losses. Nikuradse experiments. Moody's diagram. Local losses. Coefficients of resistance. Hydraulic design of pipeline: Different approaches in designing the pipeline – pressure drop, mass/volume low, diameter of pipeline. Graphical view. Energy properties of pumps and

hydraulic machines. Dimensional analysis. Theory of similarity. Flow of fluid in open channels. Non- stationary flow and hydraulic shock.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***MAT 426: NUMERICAL ANALYSIS IV (3 Units E: LH 30, PH 45)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Numerical Analysis IV is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Numerical Analysis IV concerned with the story of how functions, derivatives, integrals, and differential equations are handled as strings of numbers in the computer. At the heart of numerical analysis is an understanding of approximation of the solution of PDE, classification of PDE and approximation of derivatives by finite difference.

Most scientists and engineers are sooner or later faced with computing tasks that require some knowledge of numerical analysis. This course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable majorly in Engineering. Topics like the Crank Nicolson method. Introduction to finite element method, variation formulations are very important in other fields. This course makes the beneficiary to appreciate Engineering problems.

OBJECTIVES

The objectives of the course are to:

1. Learn how to describe solutions of difference equations;
2. illustrate the classification of PDEs;
3. differentiate the Crank Nicolson and Galerkin methods;
4. demonstrate a simple parabolic differential equation;
5. state explicit form of the difference equation and its convergence.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state the classical finite difference methods.;
2. solve problems on high-order compact difference methods;
3. state the explicit form of the difference equation and its convergence;

4. solve problems on finite difference methods for 1-D parabolic, hyperbolic, and elliptic equations;
5. state the classification of PDEs.

COURSE CONTENTS

Brief overview of PDEs and applicable areas. Models and quick review of methods. Finite difference methods for 1-D parabolic, hyperbolic, and elliptic equations. Classical finite difference methods. Applications finite difference schemes such as Crank-Nicholson, Leap-Frog, Dufort-Frankel, Douglas, Newark, Upwind schemes, Lax-Friedrich, Lax-Wendroff, Box, Beam-Warming to PDE model. Method of Lines for time-dependent problems. Analysis of finite difference methods. High-Order compact difference methods for 1-D problems. Finite difference methods for higher dimensional PDEs. ADI and splitting methods. Peaceman-Rachford, Douglas-Rachford. Mitchell-Fair weather schemes. Mixed Equations. Finite element methods. Basic theory. Variational and weak formulation. Weighted residual methods. Rayleigh-Ritz, collocation and Galerkin methods. Engineer's point of view of finite element methods with application to Boundary value problems 1-D PDEs. Finite element methods of Laplace (Poisson) and time-dependent PDEs.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS

***MAT 445: FLUID MECHANICS II (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Fluid Mechanics II is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Fluid Mechanics II is a survey of principal concepts and methods of fluid dynamic. It is a vast area of Mathematics that contains governing equations of viscous flow, exact solutions, low Reynold's number solutions. Boundary layers, Compressible flows.
This course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable majorly in engineering. This course makes the beneficiary to appreciate engineering problems.

OBJECTIVES

The objectives of the course are to:

1. describe the governing equations of viscous flow;
2. define Reynold's number;
3. interpret Reynold's number solutions;
4. demonstrate compressible flows;
5. illustrate boundary layers;
6. describe the differential equations for conservation of mass.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state fluid kinematics,
2. demonstrate an ideal fluid flow,
3. illustrate basic viscous flows,
4. define laminar and turbulent boundary layers,
5. demonstrate momentum-integral methods,
6. demonstrate adiabatic and isentropic compressible flow,
7. describe normal shock waves,
8. state the governing equations of viscous flow,
9. define momentum,
10. state the boundary layer equations.

COURSE CONTENTS

Governing equations of viscous flow. exact solutions. low Reynold's number solutions. Boundary layers. Compressible flows. Review of the kinematics of the flow field. The differential equations for conservation of mass, momentum and energy. The stream function, vorticity and irrotationality. Simple viscous flow fields. The boundary layer equations and the laminar flat plate boundary layer. Momentum-integral estimates. The turbulent boundary layer. Boundary layers with pressure gradient. Experimental external flows. Review of Thermodynamics and the speed of sound. Compressible adiabatic and isentropic steady flow. Isentropic flow with area change. The normal shock wave. Converging and diverging nozzles. Compressible duct flow with friction. Frictionless duct flow with heat transfer.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS

***MAT 446: ELASTICITY AND PLASMA DYNAMICS (2 Units E: LH 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Elasticity and Plasma Dynamics is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in

Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

This course deals with Elasticity and Plasma Dynamics. Elasticity is the tendency of solid objects and materials to return to their original shape after the external forces (load) causing a deformation are removed. Physical reasons for elastic behavior vary among materials and depend on the microscopic structure of the material.

This also course deals with the study of plasmas and particular emphasis on the orbital theory of plasmas and on magneto hydrodynamics (MHD) will be devoted. The course also treats some basic concepts on Elasticity together with some applications to Physics and Engineering.

This course describes the analysis of stress and strain, the elastic solid, Saint-Venant principle, generalized Hooke's law and two-dimensional elasticity. The course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable in other fields.

OBJECTIVES

The objectives of the course are to:

1. Learn how to describe Saint-Venant Principle in elastic solid;
2. illustrate the generalized Hooke's law;
3. state two-dimensional elasticity methods;
4. demonstrate the dynamics of plasma, motion of a particle in a uniform magnetic field;
5. demonstrate physical basis of relativity theory.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. describe the physical basis of relativity theory and its mathematical aids;
2. state Generalized Hooke's law;
3. state Saint-Venant principle;
4. explain the relativity electrodynamics of empty space.
5. state a number of phenomena associated with the behaviour of neutral fluids;
6. describe the behaviour of plasmas and some of the phenomena associated with them.
7. describe tensions and deformations in a solid and to relate them.

COURSE CONTENTS

Analysis of deformations. Displacement vector. Infinitesimal transformations (commutativity). Rotation (antissi metric part of the strain tensor). Values and eigenvectors of the deformation tensor. Relative variation on length, area and volume. Trace and invariants. Saint-Venant pronciple and the strain ellipsoid. Analysis of tensions. Equations of balance and motion. The stress tensor. Values and principal directions of tension. The Mohr diagram and its application. The quadric of tensions. Utility; calculation of stress and its components. Extremes. Hooke's generalized law. The elastic tensor coefficients as a 4th order tensor. Isotropic media. Lamé parameters. Generalized Hooke's law for an isotropic media. Calculation of deformations given the tensions and vice-versa. Relations between Young modules, Poisson's ratio and compressibility and Lamé parameters. Energy density of elastic deformation. elástica. Clapeyron Tehorem. Plasma orbital theory, dynamics of multiple charged particles and processes without

collisions in plasmas. Approaches used in the theory of plasmas Orbits of motion of particles in a plasma. Effect of a perpendicular force, gradient and curvature. Larmor radius and frequency. Magnetic mirrors. Particle acceleration in astrophysics. Van Allen Belt. Cosmic rays. Basic properties of plasmas. Different types of plasma. Electrical neutrality in a plasma. Shielding and the Debye length. The plasma parameter. Oscillations in plasmas. The plasma frequency. Electromagnetic waves in hot and cold plasmas. Magneto hydrodynamics (MHD) Fundamental equations. The plasma equations. The equation of induction and its consequences. Magneto hydrostatics. Applications to Fusion and Tokamaks. Applications to the solar atmosphere. Magneto hydrodynamic waves. Alfvén waves and magneto acoustic waves. Magneto convection and sunspots. The buoyancy of magnetic flux tubes. The Parker instability of the magnetic field. The magnetic field as a carrier of angular momentum, magnetic braking and magnetized winds.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS

***IMT 429: INTRODUCTION TO STOCHASTIC CALCULUS FOR FINANCE (3 Units E: LH 45)**

SENATE APPROVED RELEVANCE
Graduate who are highly skilled and knowledgeable in teaching Introduction to Stochastic Calculus for Finance is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education

OVERVIEW
Stochastic calculus is a branch of mathematics that operates on stochastic processes. It allows a consistent theory of integration to be defined for integrals of stochastic processes with respect to stochastic processes. This field was created and started by the Japanese mathematician Kiyoshi Itô during World War II.

This course deals with processes containing a stochastic component and thus allows the modeling of random systems. Many stochastic processes are based on functions which are continuous, but nowhere differentiable. This rules out differential equations that require the use of derivative terms, since they are unable to be defined on non-smooth functions. Instead, a theory of integration is required where integral equations do not need the direct definition of derivative terms. In quantitative finance, the theory is known as Ito Calculus.

The main use of stochastic calculus in finance is through modeling the random motion of an asset price in the Black-Scholes model. The physical process of Brownian motion (in particular, a geometric Brownian motion) is used as a model of asset prices, through the Weiner Process. This process is represented by a stochastic differential equation, which despite its name is in fact an integral equation.

OBJECTIVES

The objectives of the course are to:

1 learn how to construct Brownian motion as a continuous martingale;
2. state the Martingale convergence theorem;
3. perform Integrals with respect to a random walk;
4. state the multivariate normal distribution;
5. describe limits of sums of independent variables.
6. illustrate multivariate normal distribution.

LEARNING OUTCOMES

At the end of this course, the student should be able to:

1. define a Martingale, its properties and applications.
2. define a Brownian motion, and its properties.
3. construct a Brownian motion and apply it.
4. perform simple integration of simple and continuous processes, and Ito's formula.
5. define the Markov chain.
6. define Riemann integral,
7. state Ito's formula.

COURSE CONTENTS

Martingale conditional expectation. Martingale optional sampling theorem. Martingale convergence theorem. Square integrable martingales. Integrals with respect to a random walk. Brownian motion. Limits of sums of independent variables. Multivariate normal distribution. Limits of random walks. Construction of Brownian motion. Brownian motion as a continuous martingale. Brownian motion as a Markov. Brownian motion as a Gaussian. Computations for Brownian motion. Quadratic variation. Expected value at a future time. Stochastic integration. Understanding the concept of stochastic calculus. Review of Riemann integration. Integration of simple processes. Integration of continuous processes. Ito's formula.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS
***STA 423: STOCHASTIC PROCESSES (3 Units E: LH 30)**
SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching stochastic processes is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of Academy, Industry, community and fostering development in education.

OVERVIEW

A stochastic process is a set of random variables indexed by time or space. Stochastic modelling is an interesting and challenging area of probability and statistics that is widely used in the applied sciences. In this course the students will gain the theoretical knowledge and practical skills necessary for the analysis of stochastic systems. The basic concepts of the theory of stochastic processes and explore different types of stochastic processes including Markov chains, Poisson processes and birth and death processes will be studied.

Stochastic processes are widely used as mathematical models of systems and phenomena that appear to vary in a random manner. Examples include the growth of a bacterial population, an electrical current fluctuating due to thermal noise, or the movement of a gas molecule. Stochastic processes have applications in many disciplines such as Biology, Chemistry, Ecology, neuroscience, physics, computer science, signal processing, image processing and telecommunication. Furthermore, seemingly random changes in financial markets have motivated the extensive use of stochastic processes in finance.

OBJECTIVES

The objectives of the course are to:

1. describe the defining properties of various stochastic process models, especially for Markov processes.
2. explain standard concepts and methods of stochastic modeling
3. Identify the best stochastic process for specific situations
4. apply stochastic analysis to realistic problems
5. recognize Markov chain process and apply it to the Birth and Death processes.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state three properties of various stochastic process models, specifically for Markov processes.
2. compute probabilities of transition between states and return to the initial state after long time intervals in Markov chains.
3. apply the most appropriate stochastic process technique for a given applied problem.
4. solve stochastic process problems mathematically and using software.
5. define Markov chain process and apply it to the Birth and Death processes.

COURSE CONTENTS

Discrete and continuous time Markov chains. General Random Walk. Simple Random Walk. Random walk: Gamblers ruin problem. gambler's ruin and successful runs. Chapman-Kolmogorov Equation. Classification of States. Long run behavior. Limit Theorems, Branching processes. Poisson processes. Counting Process. Inter arrival and Waiting Time Distributions. Conditional Distribution of Arrival Times. Birth-and-death processes. Queuing processes: M/M/I, M/M/s, M/a/I queues and their waiting time distributions. Renewal Process. Some Limits

Theorems. Wald's Equation. Elementary Renewal Theorem, Regenerative Process. Symmetric Random Walk and Arc Sine Laws. Introduction to Markov Chain Monte Carlo (MCMC) methods.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

UNIVERSITY OF LAGOS
FACULTY OF PHYSICAL AND EARTH SCIENCE
DEPARTMENT OF MATHEMATICS
B. Sc. INDUSTRIAL MATHEMATICS

***STA 432: DATA ANALYSIS (3 Units E: LH = 30)**

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching data analysis is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of Academy, Industry, community and fostering development in education.

OVERVIEW

Data Analysis is a process of inspecting, cleaning, transforming and modeling data with the goal of discovering patterns in the raw data and useful information, suggesting conclusions and supporting decision-making. It enables us to discover patterns in the raw data and draw valuable information from them. Data analytics processes and techniques may use applications incorporating machine learning algorithms, simulation, and automated systems. The systems and algorithms work on the unstructured data for human use.

These findings are interpreted and used to help organizations understand their clients better, analyze their promotional campaigns, customize content, creatse content strategies, and develop products. Data analytics help organizations to maximize market efficiency and improve their earnings. Companies may use the information they obtain from data analysis to guide their decisions, leading to improved results. Data analysis removes a lot of guesswork from preparing marketing plans, deciding what material to make, creating goods, and more.

OBJECTIVES

The objectives of the course are to:

1. describe the theory and basis of data analytics;
2. detecting patterns of interest and identifying trends in time and space;
3. demonstrate ways to collect data from multiple sources relevant for Data Analytic;
4. identify data problems, develop and design an analysis plan;
5. use of statistical computing in exploratory data analysis.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. state two theory and basis of data analytics (including computing, statistics and mathematics);
2. evaluate compile and use data and associated materials from three multiple sources relevant for Data Analytics;
3. define two data problems, formulate questions, develop and design an analysis plan, and interpret the results;
4. apply three full range of Data Analytics techniques drawn from mathematics and statistics to address a real-world application problem;
5. implement four statistical packages in exploratory data analysis for real life data from social, biological and physical sciences.

COURSE CONTENTS

A review of statistical methods for data analysis. Regression and ANOVA. Statistical inference. Exploratory data analysis including graphical. Histogram. stem and leaf and box plot. Resistant statistics. Robust methods. Exploratory Data Analysis (EDA). descriptive statistics. tabular numerical and graphical. Numerical. sample quantiles. Location. Spread or dispersion. Skewness. Kurtosis. covariance and correlation. Graphical. frequency plots and histograms. univariate and multivariate box plots. Scatter plot. stem and leaf plot. Confirmatory Data Analysis (CDA). Inferential statistics and interpretation. Resistant statistics and Robust methods. Resistant Measures. Robust Estimators of the Mean. The Trimean. Winsorized Mean. Trimmed. Mean. Jackknife Procedure. Analysis of Categorical Data. The contingency table. Test of independence. Chi-square statistic for 2 by 2 contingency table. Use of statistical packages in data analysis for real life data from social. biological and physical sciences.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained.

UNIVERSITY OF LAGOS

FACULTY OF PHYSICAL AND EARTH SCIENCE

DEPARTMENT OF MATHEMATICS

B. Sc. INDUSTRIAL MATHEMATICS

***CSC522: PRINCIPLE OF PROGRAMMING LANGUAGE (3 Units E: LH 30; PH 45)**

SENATE APPROVED RELEVANCE

This course is a fundamental aspect of our undergraduate computing program as the university of first choice and nations' pride. Our university's core objectives are to produce morally and entrepreneurially sound graduates who are well-rounded and competent in their fields of study. Principle of Programming Language is designed to equip students with foundational knowledge and skills in the use of Contemporary programming languages and platform for effective development of solutions from well-defined requirement gathering. The course aligns with our university's core objectives by providing students with the necessary knowledge and skills to become competent and productive professionals in their chosen fields of study. By taking this

course, students will be able to integrate their moral and ethical values into their use of digital technology while developing entrepreneurial skills.

OVERVIEW

An exposure to important concepts and issues in contemporary programming languages is necessary for a computer science student. Programming languages are one of the most important and direct tools for the construction of a computer system: in modern computer different languages are routinely used for various levels of abstraction.

Programming language is important because it defines the relationship, semantics and grammar which allows the programmers to effectively communicate with the machines that they program.

OBJECTIVES

The objectives of the course are to

1. Improve the background for choosing appropriate programming languages for certain classes of programming problems.
2. Be able to program in an imperative (or procedural), an object-oriented, a functional, and a logical programming language.
3. Be conversant with the the significance of an implementation of a programming language in a compiler or interpreter.
4. Increase the ability to learn new programming languages.
5. Increase the capacity to express programming concepts and choose among alternative ways to express things Simulate useful features in languages that lack them.
6. Be able to design a new programming language.
7. Make good use of debuggers and related tools.
8. Explain why language features are designed the way they are.
9. Explain the language features and paradigms of different programming languages.
10. Describe syntax, semantics, names, scopes, and bindings.
11. Describe control flow, data types and execution of programming languages.
12. Determine appropriate languages for given applications.
13. Apply programming languages effectively.
14. Engage in lifelong learning with new programming languages and paradigms.

LEARNING OUTCOMES

1. Describe the role of theoretical formalisms, such as operational and denotational semantics.
2. Apply these semantics in the context of programming languages.
3. Evaluate differences (advantages/disadvantages) of these theoretical formalisms.
4. Create operational or denotational semantics of simple imperative programs.
5. Analyze the role of types in programming languages.
6. Formalize properties and reason about programs.
7. Apply basic principles for formalizing concurrent programming languages.
8. Specify the concrete and abstract syntax of a programming language, including a precise specification of the binding and scope of declarations.
9. Define the static semantics (compile-time constraints) of a programming language using typing judgments, and how to state and prove that it properly defined.

10. Define the dynamics semantics (run-time behaviour) of a language using operational and denotational methods.
11. Verify rigorously that the statics and dynamics of a language are coherent, a property commonly called type safety.
12. Explain the propositions-as-types principle, which relates programs to proofs and specifications to theorems, and know how to apply it in language design and program verification.
13. Formulate type and assertion languages for specifying the behaviour of a program, and how to verify that a program satisfies such a specification.
14. Specify the cost (sequential and parallel time and space complexity) for a program written in a precisely defined language, and how to verify that a given program meets stated cost bounds.
15. Use software tools to verify both the properties of languages and the specifications of programs written in well-defined languages.
16. Use the static and dynamic semantics of a language to derive a compiler for it that complies with these definitions, and how to use types and verification tools to ensure compiler correctness.
17. Relate a language definition to its implementation, both in terms of the run-time structures required, but also to validate abstract cost measures in an implementation.

COURSE CONTENTS

Introduction and History of programming language. Program structure Data types, abstraction, and polymorphism. Memory Allocation and Deallocation. object-oriented programming. Syntax. Values and types. Names, Scopes, and Bindings. Variables and Constants. Expressions Statements and Control-flow. Data types and Procedures. Parameter Passing. Mutually Recursive Procedures Data Abstraction and Procedural Representation. Runtime Environments. Object-Oriented Programming. Functional Programming. Logic Programming. Programming methodologies using example drawn from a variety of programming languages including LISP, Prolog, ML, Ada, Smalltalk, Icon, APL, JAVA and LUCID, Programming assignment to involve the use of some of these languages.

MINIMUM ACADEMIC STANDARD

Computer Laboratory with different workstations with pre-installed tools and software packages needed for the execution of projects and assignments (One Computer to One Student).

ACS 401: FURTHER LIFE CONTINGENCIES (3 Units E: LH 45)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. differentiate between deterministic and stochastic approaches to annuities and life insurances;
2. apply deterministic and stochastic approaches to evaluate annuities and life insurances;
3. discuss the concept of complete future life time and its moments;
4. differentiate between gross premium and net premiums;
5. compute probabilities of survival and death;
6. compute life annuities, insurance, reserves, policy values and mortality profits; and

7. perform calculations relating to increasing insurances, annuities, with-profit policies and contingent and reversionary benefits.

COURSE CONTENTS

Life annuities. Continuous life annuities. Discrete life annuities. Life annuities with monthly payments. Apportionable annuities-due and complete annuities-immediate. Benefit premiums. Fully continuous premiums. Fully discrete premiums. True monthly payment premiums. Apportion able premiums. Accumulation type benefits. Reserves and policy values.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained

ACS 402: PENSIONS FUNDS AND SOCIAL INSURANCE (3 Units C: LH 45)

LEARNING OUTCOMES

At the end of this course, students should be able to:

1. identify factors influencing choice of benefits design;
2. review the reasons for eligibility criteria a sponsor might set for membership of the scheme;
3. differentiate between defined benefit and defined contribution scheme;
4. identify various features of defined benefit and defined contribution;
5. identify the types of benefits available in pension designed and identify technical issues that may lead to fund adequacy in existing types of pension scheme;
6. demonstrate good knowledge of computation of annuity rates; and
7. demonstrate adequate knowledge of Nigeria pension reform act, 2004 as amended.

COURSE CONTENTS

Pension scheme design. Factors influencing the choice of benefit design. Eligibility criteria for membership of the scheme. Forms of benefits. Main types of benefit scheme. Risk and uncertainty. Main risks for the beneficiaries. Financing benefits scheme. Defined contribution pension. Features and phases of DC. Programmed withdrawal and life annuity computations. Annuity rates computations. Key factors affecting retirement income. Annuity rates determination. Factors affecting development of annuity markets. Construction of rates of retirement, death, withdrawal and other benefits. Alternative funding methods and determination of rates of contribution. Transfer values and optional benefits. Underfunded pension. Employees' benefits and pensions administration. Nigeria Pension Reform Act, 2014 and subsequent amendments.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

ACS 404: ACTUARIAL VALUATION (3 Units E: LH 45)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. identify various funding methods and actuarial pension valuation;
2. compute past and future pension benefits;
3. estimate past earned and future gratuities under defined contribution scheme;
4. develop models for accounting to account for funding gaps under defined contribution scheme;
5. identify the actuarial gains/losses and interpret them accordingly;
6. demonstrate competences on how to treat actuarial surpluses and deficits; and
7. recognize international actuarial standards.

COURSE CONTENTS

Objectives of a valuation. Elements of actuarial valuation. Actuarial valuation methods. Funding methods. Attained age funding method. Entry age funding method. Projected unit funding method. Current unit funding method. Actuarial assumptions. Actuarial valuation of gratuity. Computations of accrued benefits rights. Benefits/liabilities determination. PBO, ABO, VBO. Actuarial value of assets. Actuarial computations. Dealing with surplus and deficit. The actuarial valuation reports. International standards of actuarial practice.

ACS 405: ADVANCED RISK MANAGEMENT (3 Units E: LH 45)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. demonstrate knowledge of the variety of financial and financial related risks facing organizations;
2. explain the approach to risk management through risk identification, risk measurement and risk management (or mitigation);
3. discuss the main types of financial risk – market risk and capital adequacy, credit risk, liquidity risk, operational, legal and compliance risks, reputational risk;
4. explain the methodological principles of value at risk (VAR); and
5. discuss the concept of reputational risk and how to manage it.

COURSE CONTENTS

Nature of risk. Meaning of risk. Classification of risk. Types of financial risk – market risk and capital adequacy. Credit risk. Liquidity risk. Operational, legal and compliance risks. Reputational risk. Benefit of risk. Human perception of risk. Utility theory. Risk assessment: Risk identification, risk estimation, risk evaluation. Risk treatment: Risk avoidance, risk control, risk transfer, and risk financing. Basic statistical concepts relating to insurance and risk management. Risk data, presentation of risk data, statistical measurement, and probability distribution. Risk pricing: Underwriting, reinsurance and pricing insurance. Risk-based regulation and capital adequacy. Corporate governance issues of risk. Nigeria's and UK regulations. Capital adequacy. New risk-based approach to capital requirements, maintaining capital adequacy.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained

CSC 401: ALGORITHMS AND COMPLEXITY ANALYSIS (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. explain the use of big-O, omega, and theta notation to describe the amount of work done by an algorithm,
2. use big-O, omega, and theta notation to give asymptotic upper, lower, and tight bounds on time and space complexity of algorithms,
3. determine the time and space complexity of simple algorithms,
4. deduce recurrence relations that describe the time complexity of recursively defined algorithms,
5. solve elementary recurrence relations,
6. for each of the strategies (brute-force, greedy, divide-and-conquer, recursive backtracking, and dynamic programming), identify a practical example to which it would apply,
7. use pattern matching to analyse substrings, and
8. use numerical approximation to solve mathematical problems, such as finding the roots of a polynomial.

COURSE CONTENTS

Basic algorithmic analysis. Asymptotic analysis of Upper and average complexity bounds. Standard Complexity Classes. Time and space trade-offs in analysis recursive algorithms. Algorithmic Strategies. Fundamental computing algorithms. Numerical algorithms. Sequential and Binary search algorithms. Sorting algorithms, Binary Search trees. Hash tables. Graphs and their representation.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

ECO 401: ADVANCED MICROECONOMICS I (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, the students should be able to:

1. discuss confidently, issues in behavioural/optimization construct in consumer behaviour, investments, production and pricing/distribution under different market or cost structures, maximisation and minimisation framework;
2. explain the issues of equilibrium in single and multimarket situations, multiperiod consumption, investment theory of the firm and the role of time; and

3. discuss welfare economics issues within the context of efficiency or inefficiency in resource use under perfect and imperfect competitive situations; and optimality conditions.

COURSE CONTENTS

Fundamental quantitative relationships, Recapitulation on partial/general equilibrium conditions. Consumer surplus, Slutsky and Hicks income/substitution effects. Consumer behaviour under uncertainty. Dynamic analysis and value theory production functions. Constrained profit maximisation, cost minimisation and profit functions. Duality in production. Production under uncertainty. Factor market equilibrium. The existence and uniqueness of equilibrium as well as the stability of equilibrium. Topics are determination of wages, rent, interest and profit. Optimisation in theories of consumption and production. It also includes the notion of economic efficiency. Efficiency and equity.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

ECO 402: ADVANCED MICROECONOMICS II (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, the students should be able to:

1. advise businesses on optimal combinations of factors of production to achieve optimal output and maximise their profits;
2. advise on management of public resources to achieve citizen's welfare and externalities, including the use of game theory to solve welfare problems in emerging and developing economies; and
3. use game theory to solve welfare problems in emerging and developing economies.

COURSE CONTENTS

Study of externalities as well as Social and private costs and Pareto Optimality Criterion. Social welfare functions are studied using Game theory and with applications relevant to developing economies. Theory of Games: Two-person, Zero-Sum games, Mixed strategies. Linear programming equivalence, cooperative games and the Nash Bargaining solution. Welfare economics: Pareto optimality for consumption, production and in general. Social Welfare Function and its determinants. Social preference and indifference. The Arrow Impossibility Theorem. Income distribution and equity. Relevance of the welfare economics to emerging and developing economies.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space
The NUC standard requirement of 0.65m^2 per full-time student is maintained.

ECO 403: ADVANCED MACROECONOMICS I (2 Units E: LH 30)
LEARNING OUTCOMES
At the end of the course, the students should be able to:
1. present a complete open economy model multiplier, that is, including the household, business unit, the government and the external sector;
2. discuss in details, the theories of consumption and investments and relevant to emerging and developing economies.
3. explain the relevance of the IS-LM framework to macroeconomic policies; the appropriate macroeconomic theories and policies on consumption and investment for emergingand developing economies desirous of sustainable growth and development.

COURSE CONTENTS
Basic model of income determination and extension from a simple closed economy to a four-sector economy with the derivation of the multipliers. Detailed theories of consumption (KIH, RIH, PIH & LCH) and investment (Accelerator theories, MEC, MEI, Profit Theory, Profit & Residual, Cash Flow and many others), (including under conditions of uncertainly) of the classical, Neo-Classical, Keynesian and Post-Keynesian economists; and the relevance of the theories to emerging and developing countries. The advanced treatment of the IS-LM framework within the context of government monetary and fiscal policies and in relation to output, employment, prices and inflation.

ECO 404: ADVANCED MACROECONOMICS II (2 Unit E: LH 30)

LEARNING OUTCOMES
At the end of the course, the students should be able to:
1. discuss growth models from neo-classical to modern theories of economic growth
2. identify employment models from the classical to Keynesian;
3. explain various theories of exchange rate and balance of payments under different exchange rate regimes;
4. prepare government macroeconomic policies and offer sound advice to public and private enterprises or organisation; and
5. carry out relevant research on macroeconomic policies as they affect emerging and developing economies, including Nigeria.

COURSE CONTENTS
Growth Theories: Solow model and endogenous models of economic growth, inclusive growth model. Keynesian models of Income, employment and Prices. Taxation and Laffar curve, Debts and Ricardian equivalence. Capital theory, fiscal and monetary policy theories. Determination of wages, rent, interest and profit. Notion of economic efficiency; Efficiency and equity; the theories

of foreign exchange and balance of payment, required adjustment techniques in balance of payment disequilibrium. The J-Curve and Marshall-Lerner condition for currency adjustments. Theories of Business Cycle. Macroeconomic policies, planning and optimal allocation of resources with particular reference to emerging and developing economies.

MINIMUM ACADEMIC STANDARDS

The approved staff-student ratio in administration and management sciences is 1:30.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained.

ICT 418: DESIGN AND INSTALLATION OF ELECTRICAL & ICT EQUIPMENT
(2 Units E: LH 15; PH 45)

LEARNING OUTCOMES

At the end of this course, students should be able to:

1. describe health and safety standards in electrical installations;
2. explain NCC and FCC codes of practice and standards in Nigeria;
3. implement telecommunication design and installation;
4. configure computer networks; and
5. design and install telephony equipment.

COURSE CONTENTS

Electrical Installation. Introduction to Health and safety at Work Act in Nigeria. Electrical safety. First aid. Electricity supply regulations. Lighting and Illumination: Luminous intensity and flux. Maintenance factor. Coefficient of utilisation. Types of light sources. Calculation of lighting requirements. Glare. Stroboscopic effect. Installation Materials, cables, junction box, terminations, joints. Conduits and conduiting. Trunks and trunking. Electrical Installation design in domestic, commercial, and industry. Alarm and emergency systems. Earthling and Protection. Purposes of earthing. Faraday cage. Rod electrodes. Earth electrode resistance. Earthing system. Earth fault loop impedance. ICT services: NCC and FCC codes of practice and standards. Telecommunication design and installation: Satellite, VSAT, etc. Telephone design and installation. Computer networking design and installation. Wireless LAN design and installation. Preparation of Bill of Engineering Measurement Evaluation. Contract bidding. Consultancy.
Lab Work: Demonstration of simple electrical installations using electrical materials including conduits and trunks. Illustration of ICT installations including wired and wireless computer networks. Demonstration of telecommunications installations

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.
This course should have at least three categories of laboratories: software, network and hardware laboratories. Best practice requires a staff to students' ratio of 1:20 for laboratory practical.

Classroom Space

The NUC standard requirement of 0.65m^2 per full-time student is maintained.

MTH 406: LEBESGUE MEASURE AND INTEGRALS (2 Units E: LH 30)

SENATE APPROVED RELEVANCE

Graduate who are highly skilled and knowledgeable in teaching Lebesgue Measure and Integrals is in accord with UNILAG vision to be centre of academic excellence so as to expand the frontiers of knowledge and provide valued-added Research and Development. Relevance is seen in Mathematics teachers from UNILAG been able to provide a platform for synergy of academy, industry, community and fostering development in education.

OVERVIEW

Lebesgue Measure and Integrals is primarily concerned with Inner and outer measure. measurable and non-measurable sets, measurable functions and Lebesgue integrable functions. This course contains relevant topics which will help the students to develop mastery of the subject area and the topics are applicable in Statistics and other fields. Topics like The dominated convergence theorem for infinite series, Fatou's theorem, Monotone convergence theorem are highly applicable in Optimization.

This course makes the beneficiary to appreciate the geometric interpretation of Lebesgue Integral and the relationship between Riemann and Lebesgue integrals.

OBJECTIVES

The objectives of the course are to:

1. Learn how to describe inner and outer measures;
2. illustrate the length of set and its relation to measure;
3. describe the Cantor set;
4. demonstrate the geometric interpretation of Lebesgue Integral;
5. demonstrate the dominated convergence theorem.

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. define Lebesgue measure;
2. define a Cantor set;
3. state the relationship between Riemann and Lebesgue integrals;
4. state Fatou's theorem;
5. state monotone convergence theorem.

COURSE CONTENTS

Lebesgue measure: Inner and outer measure. Measurable and non-measurable sets, length of set. Measurable subsets of the real line. Borel sets, Vitali's covering theorem, Cantor set. Measurable functions: Egorov's theorem, Baire's class. Lebsgue Integral for bounded, as a limit of sum, non-negative and for unbounded functions, Geometric interpretation of Lebesgue Integral. Relationship of Riemann and Lebesgue integral. The dominated convergence theorem for infinite series, Fatou's theorem, Monotone convergence theorem. The Lebesgue integral for arbitrary function. Introduction to $L^p(E)$- spaces for measurable subsets E of the real line.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

STA 415: REGRESSION AND ANALYSIS OF VARIANCE II (3 Units E: LH 45)

LEARNING OUTCOMES
At the end of the course, students should be able to:
1. explain multicollinearity, autocorrelation and heteroscedasticity;
2. perform the residual analysis;
3. describe the transformations;
4. compare intercepts and slopes;
5. explain simple non – linear regression and Logistic regression;
6. use dummy variables;
7. familiarise with the departures from ANOVA assumptions;
8. estimate missing values; and
9. determine the analysis of covariance in one-way, two-way, three-way and nested (hierarchical) classifications as well as analysis of covariance with two concomitant variables.

COURSE CONTENTS
Multicollinearity, autocorrelation and heteroscedasticity; residual analysis; transformations. comparison of intercepts and slopes; Simple non – linear regression; Logistic regression; Use of dummy variables. Departures from ANOVA assumptions. Transformations. Missing values; analysis of covariance in one-way, two-way, three-way and nested (hierarchical) classifications; analysis of covariance with two concomitant variables.

MINIMUM ACADEMIC STANDARDS
The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space
The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

INS 401 PORJECT MANAGEMENT (2 Units E: LH 30)

LEARNING OUTCOMES
At the end of this course, students should be able to:
1. describe project management planning;
2. describe project scheduling;
3. explain management of project resources;
4. discuss project procurement, monitoring and execution; and

5. explain project communication and time management.

COURSE CONTENTS

Introduction to Project Management. The Project Management Lifecycle: Project management and systems development or acquisition. The project management context. Technology and techniques to support the project management lifecycle, and Project management processes. Managing Project Teams: Project team planning, motivating team members, Leadership, power and conflict in project teams, and managing global project teams. Managing project communication and enhancing team communication. Project Initiation and Planning. Managing Project Scope: Project initiation, how organisations choose projects, Activities, and Developing the project charter. Managing Project Scheduling: Common problems in project scheduling, and Techniques for project scheduling. Managing Project Resources: Types of resources (human, capital, time), and Techniques for managing resources. Project quality and tools to manage project quality. Managing project risk and tools for managing project risk. Managing Project Procurement: Alternatives to systems development, External acquisition, Outsourcing-domestic and offshore. Steps in the procurement process, and managing the procurement process. Project Execution, Control and Closure: Managing project execution, monitoring progress and managing change. Documentation and communication, and Common problems in project execution. Managing Project Control and Closure: Obtaining information, Cost control, Change control, administrative closure, Personnel closure, Contractual closure and Project auditing.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

PHY 401: QUANTUM MECHANICS (2 Units E: LH 30)

LEARNING OUTCOMES

At the end of the course, students should be able to:

1. demonstrate a clear and structured approach for solving problems;
2. compute the angular momentum of a wave function using operators;
3. describe and compute Eigen functions of the angular momentum operators;
4. explain the relationship between quantum spin and angular momentum; and
5. apply spin operators to a simple ½ system to perform calculations.

COURSE CONTENTS

The formulation of quantum mechanics in terms of state vectors and linear operators. Three-dimensional spherically symmetric potentials. The theory of angular momentum and spin. Identical particles and the exclusion principle. Methods of approximation and multi electron atoms.

MINIMUM ACADEMIC STANDARDS

The guidelines on academic staff/student ratio of 1:20 for science programmes shall apply.

Classroom Space

The NUC standard requirement of $0.65m^2$ per full-time student is maintained.

NUC 70% CCMAS

Minimum Academic Standards

Equipment

Acquisition of a sufficient number equipment to enable adequate implementation of the benchmark statements as they relate to programmes to Science / Mathematics Professional practice such as:

Computers
Computer software
Flash drives
Printers
Projectors
Smart board
Smart TV
Video machine
Screen board
Pen drive
Loudspeaker with microphones

Staffing

Academic Staff

1. The established staff/students ratio of 1:20 for Science should be met. There should be a minimum of six academic staff
2. Training and retaining of academic staff and students should be pursued vigorously.
3. All academic staff should have computing skills.
4. 75% of the academic staff should possess PhDs.
5. At least 20% of the academic staff should be Professors/Readers and 35 % Senior Lecturer and 45% others.

Non-Academic Staff /Administrative Staff

The Non-Academic - Academic staff ratio in the University should be 1: 4 maxima.

Technical Support Personnel

The services of technical support staff, which are indispensable in the proper running of laboratories and workshops, are required. It is important to recruit very competent senior technical staff to maintain teaching and research equipment. They are also to undergo regular training to keep them abreast of developments in equipment operation and maintenance.

Qualified Laboratory Technologists – Qualification/Areas of specialization: BSc or HND in Computer / Electronic Engineering, Industrial Mathematics knowledgeable in the use of Maths Application Packages (MatLabs, Mables, LaTex, etc).

Administrative Support Staff

The services of the administrative support staff are indispensable in the proper administration of departments and faculty offices. It is important to recruit very competent, computer literate senior staff.

All Administrative, Secretarial and Clerical Staff should have computing skills.

Library

The University should leverage on available technology to put in place rich databases and other electronic/digital library and information resources. In addition, well stock, current hardcopies of reference/ professional journals and other textual materials, to complement the University central library, should be provided. A well network digital library should serve the entire university community. Availability of wireless facilities (Wifi) with adequate bandwidth should enhance access to these electronic resources.

In any case, there should be internet ready workstations available in the library for least 25% of the total student enrolled in each academic programme. The funding of the library should be in line with NUC guidelines.

Classrooms, Laboratories, Workshops and Office Space

The NUC recommends the following physical space requirements:

Description		**Size m^2**
Professor's Office	-	18.50
Head of Department's Office	-	18.50
Tutorial Teaching Staff's Office	-	13.50
Other Teaching Staff Space	-	7.00
Technical Staff Space	-	7.00
Secretarial Space	-	7.00
Research Laboratory	-	16.50
Seminar Space/per student	-	1.85
Laboratory Space per FTE	-	7.50
Conference Room	-	37.0